PONDICHERRY UNIVERSITY
RAMANUJAN SCHOOL OF MATHEMATICAL SCIENCES
DEPARTMENT OF STATISTICS

M.Sc. QUANTITATIVE FINANCE
MSQF 532: Financial Mathematics
LECTURE NOTES ON

Financial Mathematics

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MSQF 532: FINANCIAL MATHEMATICS CREDITS: 4

Course Objectives:

- *To equip students with a solid understanding of fundamental mathematical concepts and techniques used in financial analysis, valuation of financial instruments, and portfolio management.*
- *To develop the ability to apply quantitative methods for solving practical problems in finance, including risk assessment, derivatives pricing, and financial decision-making.*

Course Outcomes (COs):

CO1: Apply financial mathematics to evaluate investments and corporate finance decisions
CO2: Use quantitative tools for bond, equity, and portfolio valuation
CO3: Analyze risk and return to support financial strategy
CO4: Interpret and construct financial statements for cash flow management
CO5: Apply mathematical models to real-world financial problems

Unit I: Introduction to Financial Mathematics

Role and importance of mathematics in finance — Time value of money concepts — Simple interest and compound interest — Nominal and effective rates — Discounting and present value — Basic annuities: ordinary annuities, annuities due — Growth and decay curves.

Unit II: Valuation of Financial Instruments

Present value and future value of annuities — Perpetuities and growing annuities — Loan amortization and sinking funds — Bond valuation: yield to maturity (YTM), yield to call (YTC) — Equity valuation: dividend discount model (DDDM), price-earnings model.

Unit III: Financial Ratios and Cash Flow Analysis

Key financial ratios: liquidity, profitability, leverage, efficiency — Application of ratios in decision making — Cash flow statements: operating, investing, and financing cash flows — Free cash flow analysis — Break-even and sensitivity analysis.

Unit IV: Risk, Return, and Portfolio Mathematics

Measurement of return: arithmetic vs geometric mean — Standard deviation and variance of returns — Coefficient of variation — Covariance and correlation — Portfolio return and risk — Diversification and optimal portfolio construction — Capital Asset Pricing Model (CAPM).

Unit V: Advanced Topics and Applications

Linear programming in financial decision-making — Introduction to financial derivatives (options, forwards, futures) — Option valuation basics: Binomial and Black-Scholes models — Duration and convexity of bonds — Probability distributions in finance: normal, log-normal — Simulation and scenario analysis in financial planning.

Books for Reference:

1. Bodie, Z., Kane, A., & Marcus, A. J. (2022). *Investments* (12th ed.). McGraw-Hill Education.
2. Tuckman, B., & Serrat, A. (2023). *Fixed Income Securities: Tools for Today's Markets* (4th ed.). Wiley.
3. Fabozzi, F. J. (2023). *Financial Mathematics: A Comprehensive Treatment* (2nd ed.). Wiley.
4. Hull, J. C. (2024). *Options, Futures, and Other Derivatives* (11th ed.). Pearson.
5. Ross, S. A., Westerfield, R. W., Jaffe, J., & Jordan, B. D. (2022). *Corporate Finance* (13th ed.). McGraw-Hill Education.

COs	PO1	PO2	PO3	PO4	PO5
CO1	XXX	XXX	XXX	XX	XX
CO2	XXX	XXX	XXX	XX	XX
CO3	XXX	XX	XXX	XXX	XX
CO4	XX	XX	XX	XXX	XX
CO5	XXX	XXX	XXX	XX	XXX

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Chapter 1

Introduction to Financial Mathematics

1.1 Time Value of Money & Simple Interest Fundamentals

Lecture
13/08/2026

Definition 1.1.1: Simple Interest & Accrued Amount

Let P represent the initial **Principal** invested or borrowed, R the **nominal annual rate of interest** expressed as a percentage ($R\%$ per annum), and T the **time horizon** expressed in years.
The **Simple Interest** (SI) generated is linear in time and principal:

$$\text{SI} = \frac{P \times R \times T}{100} = P \cdot r \cdot T \tag{1.1}$$

where $r = \frac{R}{100}$ is the effective decimal interest rate per year.
The **Total Accrued Amount** (A or FV) at the end of duration T is given by:

$$A = P + \text{SI} = P(1 + rT) \tag{1.2}$$

Time Conversion Rules for Fractional Periods

When the time period is given in units other than full years:

- **Days to Years (Exact Year):** $T = \dfrac{\text{Number of Days}}{365}$
- **Weeks to Years:** $T = \dfrac{\text{Number of Weeks}}{52}$
- **Months to Years:** $T = \dfrac{\text{Number of Months}}{12}$

Additivity of Simple Interest

Under simple interest, the interest earned in each period depends only on the original principal P and not on previously accrued interest (i.e., interest does not compound). Consequently, over multiple disjoint consecutive time periods $[0, t_1], [t_1, t_2], \ldots, [t_{k-1}, t_k]$ with respective rates $R_1, R_2, \ldots, R_k$, the total simple interest is strictly additive:

$$\text{SI}_{\text{total}} = \sum_{j=1}^{k} \frac{P \times R_j \times \Delta t_j}{100} \tag{1.3}$$

Example 1.1.1: Basic Simple Interest and Total Amount

Problem:
Find the simple interest and the total amount to be paid on a principal sum of Rs. 15,000 at an interest rate of 5% per annum after 2 years.

Given Data:

- Principal (P) = Rs. 15,000
- Rate (R) = 5% per annum
- Time (T) = 2 years

Step-by-Step Calculation:

1. **Simple Interest** (SI)**:**

$$\text{SI} = \frac{P \times R \times T}{100} = \frac{15{,}000 \times 5 \times 2}{100} = \text{Rs. } 1{,}500$$

2. **Total Amount** (A)**:**

$$A = P + \text{SI} = 15{,}000 + 1{,}500 = \text{Rs. } 16{,}500$$

Final Result:

$$\text{SI} = \text{Rs. } \mathbf{1{,}500}, \quad A = \text{Rs. } \mathbf{16{,}500}$$

Example 1.1.2: Fractional Period in Days (Exact Day Count)

Problem:
Find the simple interest and accumulated amount on a principal of Rs. 25,000 at an annual rate of 5% for a duration of 219 days.

Given Data:

- Principal (P) = Rs. 25,000
- Rate (R) = 5% p.a.
- Time (T) = 219 days = $\frac{219}{365}$ years = 0.6 years

Calculation:

$$\text{SI} = \frac{P \times R \times T}{100} = \frac{25{,}000 \times 5 \times \frac{219}{365}}{100} = \frac{25{,}000 \times 5 \times 0.6}{100} = \text{Rs. } 750$$

Total Amount:

$$A = P + \text{SI} = 25{,}000 + 750 = \text{Rs. } 25{,}750$$

Final Result:

$$\text{SI} = \text{Rs. } \mathbf{750}, \quad A = \text{Rs. } \mathbf{25{,}750}$$

Example 1.1.3: Fractional Period in Weeks

Problem:
Find the simple interest and accumulated amount on a principal of Rs. 25,000 at an annual rate of 5% for a period of 20 weeks.

Given Data:

- Principal $(P) = \text{Rs. } 25{,}000$
- Rate $(R) = 5\%$ p.a.
- Time $(T) = 20$ weeks $= \dfrac{20}{52}$ years $= \dfrac{5}{13}$ years

Calculation:

$$\text{SI} = \frac{P \times R \times T}{100} = \frac{25{,}000 \times 5 \times \frac{20}{52}}{100} = \frac{1{,}250 \times 20}{52} \approx \text{Rs. } 480.77$$

Total Amount:

$$A = P + \text{SI} = 25{,}000 + 480.77 = \text{Rs. } 25{,}480.77$$

Final Result:

$$\text{SI} = \text{Rs. } \mathbf{480.77}, \quad A = \text{Rs. } \mathbf{25{,}480.77}$$

Example 1.1.4: Year-by-Year Multi-Period Breakdown

Problem:
A principal amount of Rs. 1,00,000 is invested at a rate of 10% simple interest per annum for 3 years. Calculate the simple interest generated individually for the 1st year, 2nd year, and 3rd year, and compute the total accrued amount.

Given Data:

- Principal $(P) = \text{Rs. } 1{,}00{,}000$
- Rate $(R) = 10\%$ p.a.
- Total Horizon $(T) = 3$ years

Year-Wise Computation: Because simple interest is non-compounding, the interest generated in any single 1-year increment $(\Delta t = 1)$ remains invariant:

$$\text{Year 1: } \text{SI}_1 = \frac{1{,}00{,}000 \times 10 \times 1}{100} = \text{Rs. } 10{,}000$$
$$\text{Year 2: } \text{SI}_2 = \frac{1{,}00{,}000 \times 10 \times 1}{100} = \text{Rs. } 10{,}000$$
$$\text{Year 3: } \text{SI}_3 = \frac{1{,}00{,}000 \times 10 \times 1}{100} = \text{Rs. } 10{,}000$$

Aggregate Interest and Terminal Amount:

$$\text{SI}_{\text{total}} = \text{SI}_1 + \text{SI}_2 + \text{SI}_3 = 10{,}000 + 10{,}000 + 10{,}000 = \text{Rs. } 30{,}000$$
$$A = P + \text{SI}_{\text{total}} = 1{,}00{,}000 + 30{,}000 = \text{Rs. } 1{,}30{,}000$$

Final Summary Table:

Period	Principal (Rs.)	Interest (Rs.)	Accrued Amount (Rs.)
Year 1	1,00,000	10,000	1,10,000
Year 2	1,00,000	10,000	1,20,000
Year 3	1,00,000	10,000	1,30,000
Total	—	**30,000**	**1,30,000**

Example 1.1.5: Inverse Problem: Tiered Rate Schedule

Problem:
A total simple interest of Rs. 2,500 is earned by investing an unknown sum of money (P) for a total duration of 13 years under a tiered rate structure:

- 4% p.a. for the first 3 years
- 5% p.a. for the next 4 years
- 8% p.a. for the remaining period beyond 7 years

Determine the initial principal sum invested.

Interval Breakdown: Let P denote the unknown principal sum. The total horizon of $T = 13$ years is decomposed into:

1. **Interval 1:** $T_1 = 3$ years, $R_1 = 4\%$

$$\mathrm{SI}_1 = \frac{P \times 4 \times 3}{100} = 0.12\,P$$

2. **Interval 2:** $T_2 = 4$ years, $R_2 = 5\%$

$$\mathrm{SI}_2 = \frac{P \times 5 \times 4}{100} = 0.20\,P$$

3. **Interval 3:** $T_3 = 13 - (3 + 4) = 6$ years, $R_3 = 8\%$

$$\mathrm{SI}_3 = \frac{P \times 8 \times 6}{100} = 0.48\,P$$

Linear Formulation and Algebraic Solution: The cumulative interest satisfies:

$$\begin{aligned} \mathrm{SI}_{\text{total}} &= \mathrm{SI}_1 + \mathrm{SI}_2 + \mathrm{SI}_3 \\ 2{,}500 &= 0.12\,P + 0.20\,P + 0.48\,P \\ 2{,}500 &= 0.80\,P \\ \implies P &= \frac{2{,}500}{0.80} = \text{Rs. } \mathbf{3{,}125} \end{aligned}$$

Final Result: The initial principal sum invested is Rs. **3,125**.

1.2 Theory of Compound Interest & Growth/Decay Models

Lecture
17/08/2026

Definition 1.2.1: Compound Interest & Terminal Value

Under **Compound Interest**, interest accrued at the end of each compounding period is added to the principal to form the new principal base for subsequent periods (interest on interest).
For a principal P invested at a nominal annual interest rate of $R\%$ ($r = R/100$) for n years compounded annually, the terminal accumulated amount (A or FV) is:

$$A = P\left(1 + \frac{R}{100}\right)^n = P(1+r)^n \tag{1.4}$$

The total **Compound Interest** (CI) generated over the horizon is:

$$\text{CI} = A - P = P\left[\left(1 + \frac{R}{100}\right)^n - 1\right] = P\left[(1+r)^n - 1\right] \tag{1.5}$$

Exponential Growth and Decay / Depreciation Curves

- **Compound Growth / Appreciation:** If a quantity grows at constant annual percentage rate $r = R/100$:
$$A(n) = P(1+r)^n \tag{1.6}$$
- **Compound Decay / Reducing Balance Depreciation:** If an asset or demographic metric depreciates at annual rate $d = R/100$:
$$A(n) = P(1-d)^n \tag{1.7}$$
The **Compound Depreciation** (total value reduction) over n periods is:
$$\text{Total Depreciation} = P - A(n) = P\left[1 - (1-d)^n\right] \tag{1.8}$$
- **Varying Multi-Year Growth/Decay Factors:** For sequential periods with rates $r_1, r_2, \ldots, r_k$:
$$A_k = P \cdot (1 \pm r_1)(1 \pm r_2) \cdots (1 \pm r_k) \tag{1.9}$$

Example 1.2.1: Annual Compounding of Interest

Problem:
Find the compound interest (CI) on a principal sum of Rs. 12,600 for 2 years at an annual interest rate of 10%, compounded annually.

Given Data:

- Principal (P) = Rs. 12,600
- Rate (R) = 10% per annum
- Horizon (n) = 2 years (compounded annually)

Step-by-Step Calculation:

1. **Terminal Accumulated Amount (A):**

$$A = P\left(1 + \frac{R}{100}\right)^n$$
$$A = 12{,}600\left(1 + \frac{10}{100}\right)^2 = 12{,}600(1.10)^2$$
$$A = 12{,}600 \times 1.21 = \text{Rs. } 15{,}246$$

2. **Compound Interest (CI):**

$$\text{CI} = A - P$$
$$\text{CI} = 15{,}246 - 12{,}600 = \text{Rs. } 2{,}646$$

Final Result:

$$\text{Accumulated Amount } A = \text{Rs. } \mathbf{15{,}246}, \quad \text{CI} = \text{Rs. } \mathbf{2{,}646}$$

Example 1.2.2: Compound Depreciation of an Asset (Reducing Balance)

Problem:
A television set was purchased for Rs. 21,100. The asset depreciates in market value at a constant rate of 5% per annum. Determine the residual book value of the TV after 3 years and compute the total compound depreciation incurred.

Given Data:

- Initial Purchase Value (P) = Rs. 21,100
- Annual Rate of Depreciation (d) = 5% per annum
- Horizon (n) = 3 years

Calculation:

1. **Residual Value after 3 Years (A):**

$$A = P\left(1 - \frac{d}{100}\right)^n$$
$$A = 21{,}100\left(1 - \frac{5}{100}\right)^3 = 21{,}100(0.95)^3$$
$$A = 21{,}100 \times 0.857375 = \text{Rs. } 18{,}091.6125 \approx \text{Rs. } 18{,}091.61$$

2. **Total Compound Depreciation:**

$$\text{Total Depreciation} = P - A$$
$$= 21{,}100 - 18{,}091.61 = \text{Rs. } 3{,}008.39$$

Final Result:

$$\text{Residual Value after 3 Years} = \text{Rs. } \mathbf{18{,}091.61}, \quad \text{Compound Depreciation} = \text{Rs. } \mathbf{3{,}008.39}$$

Example 1.2.3: Sub-Annual Compounding: Semi-Annual Conversion

Problem:
Find the compound interest and accumulated amount on a principal of Rs. 48,000 for 1 year at an annual nominal rate of 8%, compounded half-yearly.

Given Data:

- Principal (P) = Rs. 48,000

- Nominal Annual Rate (R) = 8% p.a.
- Horizon (n) = 1 year
- Compounding Frequency (m) = 2 periods per year (half-yearly)

Parameter Conversions:

- Periodic Interest Rate: $i_{\text{periodic}} = \dfrac{R}{m} = \dfrac{8\%}{2} = 4\% = 0.04$
- Total Compounding Cycles: $N = n \times m = 1 \times 2 = 2$ periods

Calculation:

1. **Accumulated Terminal Amount (A):**

$$A = P\left(1 + \frac{R/m}{100}\right)^{n \cdot m}$$
$$A = 48{,}000\left(1 + \frac{4}{100}\right)^{2} = 48{,}000(1.04)^2$$
$$A = 48{,}000 \times 1.0816 = \text{Rs. } 51{,}916.80$$

2. **Compound Interest (CI):**

$$\text{CI} = A - P$$
$$\text{CI} = 51{,}916.80 - 48{,}000 = \text{Rs. } 3{,}916.80$$

Final Result:

$$\text{Terminal Amount } A = \text{Rs. } \mathbf{51{,}916.80}, \quad \text{CI} = \text{Rs. } \mathbf{3{,}916.80}$$

Example 1.2.4: Demographic Modeling: Inverse Tiered Population Decay

Problem:
The population of a town decreases every year due to migration, economic shifts, and unemployment. The present population is 6,31,680. The rate of migration was 4% during the past year and 6% during the year preceding that. Determine what the population was 2 years ago.

Given Data:

- Current Population (A_2) = 6,31,680
- Rate of reduction in Year 1 (two years ago, d_1) = 6%
- Rate of reduction in Year 2 (last year, d_2) = 4%
- Let the initial population 2 years ago = P

Algebraic Formulation & Solution: Applying sequential geometric reduction factors:

$$P \times \left(1 - \frac{d_1}{100}\right) \times \left(1 - \frac{d_2}{100}\right) = A_2$$
$$P \times \left(1 - \frac{6}{100}\right) \times \left(1 - \frac{4}{100}\right) = 6{,}31{,}680$$
$$P \times 0.94 \times 0.96 = 6{,}31{,}680$$
$$P \times 0.9024 = 6{,}31{,}680$$

Solving for initial population P:

$$P = \frac{6{,}31{,}680}{0.9024} = \mathbf{7{,}00{,}000}$$

Final Result: The town population 2 years ago was **7,00,000**.

1.3 Methods of Analysis

Lecture 17/08/2026

There are two primary methods of financial time-value analysis:

1. **Compounding**
2. **Discounting**

1.3.1 Compounding

Compounding over Single and Multiple Conversion Periods

a. Compounding of Interest over n Years

$$\text{Future Value (FV)} = P(1+R)^n \tag{1.10}$$

where:

- P = Principal amount
- R = Annual rate of interest (in decimal $r = R/100$)
- n = Number of years

b. Multiple Compounding Periods

$$\text{Future Value (FV)} = P\left(1+\frac{R}{m}\right)^{n\cdot m} \tag{1.11}$$

where:

- P = Principal amount
- R = Nominal annual rate of interest
- n = Number of years
- m = Frequency of compounding per annum
- $n \cdot m$ = Total number of conversion periods
- $\dfrac{R}{m}$ = Periodic interest rate per conversion period

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Example 1.3.1: Multi-Frequency Compounding Comparison

Problem: Calculate the maturity amount if Rs. 2,00,000 is invested for 2 years at a nominal interest rate of 12% per annum compounded: (a) Annually (b) Semi-annually (c) Quarterly (d) Monthly

Given Data:

- Principal (P) = Rs. 2,00,000
- Nominal Rate (r) = 12% p.a. = 0.12
- Time Horizon (t) = 2 years

General Compounding Formula:

$$A = P\left(1 + \frac{r}{m}\right)^{m \cdot t}$$

Detailed Step-by-Step Calculations:

1. **Compounded Annually ($m = 1$):**

$$A = 2{,}00{,}000 \times \left(1 + \frac{0.12}{1}\right)^{1\times 2} = 2{,}00{,}000 \times (1.12)^2$$
$$= 2{,}00{,}000 \times 1.2544 = \text{Rs. } \mathbf{2{,}50{,}880.00}$$

2. **Compounded Semi-Annually ($m = 2$):**

$$A = 2{,}00{,}000 \times \left(1 + \frac{0.12}{2}\right)^{2\times 2} = 2{,}00{,}000 \times (1.06)^4$$
$$= 2{,}00{,}000 \times 1.26247696 = \text{Rs. } \mathbf{2{,}52{,}495.39}$$

3. **Compounded Quarterly ($m = 4$):**

$$A = 2{,}00{,}000 \times \left(1 + \frac{0.12}{4}\right)^{4\times 2} = 2{,}00{,}000 \times (1.03)^8$$
$$= 2{,}00{,}000 \times 1.26677008 = \text{Rs. } \mathbf{2{,}53{,}354.02}$$

4. **Compounded Monthly ($m = 12$):**

$$A = 2{,}00{,}000 \times \left(1 + \frac{0.12}{12}\right)^{12\times 2} = 2{,}00{,}000 \times (1.01)^{24}$$
$$= 2{,}00{,}000 \times 1.26973465 = \text{Rs. } \mathbf{2{,}53{,}946.93}$$

Summary Comparison Table:

Frequency	Periods (m)	Periodic Rate (r/m)	Cycles ($m \cdot t$)	Maturity Amount (A)
Annually	1	12.00%	2	Rs. **2,50,880.00**
Semi-annually	2	6.00%	4	Rs. **2,52,495.39**
Quarterly	4	3.00%	8	Rs. **2,53,354.02**
Monthly	12	1.00%	24	Rs. **2,53,946.93**

Effective Rate of Interest (ERI)

c. Effective Rate of Interest (ERI) Annually compounded rate of interest that is equivalent to an annual interest rate compounded more than once per year:

$$\text{ERI} = \left(1 + \frac{R}{m}\right)^m - 1 \tag{1.12}$$

where:

- ERI = Effective rate of interest
- R = Nominal rate of interest
- m = Frequency of compounding per year

Doubling Period Formulations

d. Doubling Period [Lecture: 18/08/2026]

The **Doubling Period** is the time required for an investment to double its principal amount at a given rate of interest.

- **Continuous Compounding Basis:** Based on the natural logarithm of 2:

$$\ln(2) \approx 0.693 \quad (69.3\%) \tag{1.13}$$

- **Rule of 72 (Standard Practical Thumb Rule):**

$$\text{Doubling Period} = \frac{72}{\text{Rate of Interest } (R)} \tag{1.14}$$

$$\text{Rate of Interest } (R) = \frac{72}{\text{Doubling Period}} \tag{1.15}$$

- **Rule of 69 (Refined Discrete Compounding Approximation):**

$$\text{Doubling Period} = 0.35 + \frac{69}{R} \tag{1.16}$$

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Example 1.3.2: Determination of Interest Rate using Rule of 72

Problem: Calculate the rate of interest if the doubling period of an investment is: (a) 4 years (b) 6 years

Formula:

$$\text{Rate of Interest } (R) = \frac{72}{\text{Doubling Period}}$$

Calculations:

1. For a doubling period of 4 years:

$$R = \frac{72}{4} = \mathbf{18}\% \text{ p.a.}$$

2. For a doubling period of 6 years:

$$R = \frac{72}{6} = \mathbf{12}\% \text{ p.a.}$$

Lecture 25/08/2026

Example 1.3.3: Determination of Doubling Period using Rule of 69

Problem: Calculate the doubling period using the Rule of 69 if the annual rate of interest is: (a) 6% (b) 9% (c) 12%

Formula:

$$\text{Doubling Period} = 0.35 + \frac{69}{R}$$

Calculations:

1. If interest rate is $R = 6\%$:

$$\text{Doubling Period} = 0.35 + \frac{69}{6} = 0.35 + 11.50 = \mathbf{11.85} \text{ years}$$

2. If interest rate is $R = 9\%$:

$$\text{Doubling Period} = 0.35 + \frac{69}{9} = 0.35 + 7.667 = \mathbf{8.02} \text{ years}$$

3. If interest rate is $R = 12\%$:

$$\text{Doubling Period} = 0.35 + \frac{69}{12} = 0.35 + 5.75 = \mathbf{6.10} \text{ years}$$

Compound Value of a Series of Payments

e. Compound Value of a Series of Payments For an uneven stream of payments P_t invested across horizon n:

$$\text{Compound Value (FV)} = \sum_{t=1}^{n} P_t(1+R)^{n-t} \tag{1.17}$$

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Example 1.3.4: Compound Value of an Uneven Series of Payments

Problem: Calculate the Future Value (FV) of the following series of payments made at the end of each year over a 4-year investment horizon at an interest rate of $R = 9\%$ per annum:

- $P_1 = \text{Rs. } 1{,}500$ at the end of the 1st year
- $P_2 = \text{Rs. } 3{,}000$ at the end of the 2nd year
- $P_3 = \text{Rs. } 4{,}500$ at the end of the 3rd year
- $P_4 = \text{Rs. } 6{,}000$ at the end of the 4th year

Analytical Formulation: Payments deposited at the end of year t earn compound interest for $(n-t)$ remaining years until the terminal horizon $n = 4$:

$$\text{FV} = \sum_{t=1}^{4} P_t(1+R)^{4-t} = P_1(1+R)^3 + P_2(1+R)^2 + P_3(1+R)^1 + P_4(1+R)^0$$

Step-by-Step Payment Accumulation:

- **1st Payment (Rs. 1,500):** Invested for $4-1=3$ years:

$$\text{FV}_1 = 1{,}500 \times (1+0.09)^3 = 1{,}500 \times 1.295029 = \text{Rs. } \mathbf{1{,}942.55}$$

- **2nd Payment (Rs. 3,000):** Invested for $4-2=2$ years:

$$\text{FV}_2 = 3{,}000 \times (1+0.09)^2 = 3{,}000 \times 1.188100 = \text{Rs. } \mathbf{3{,}564.30}$$

- **3rd Payment (Rs. 4,500):** Invested for $4-3=1$ year:

$$\text{FV}_3 = 4{,}500 \times (1+0.09)^1 = 4{,}500 \times 1.090000 = \text{Rs. } \mathbf{4{,}905.00}$$

- **4th Payment (Rs. 6,000):** Invested for $4-4=0$ years (at terminal date):

$$\text{FV}_4 = 6{,}000 \times (1+0.09)^0 = 6{,}000 \times 1.000000 = \text{Rs. } \mathbf{6{,}000.00}$$

Compound Value Factor (CVAF) Summary Table:

Year (t)	**Payment** (P_t)	**Horizon** ($4-t$)	**Factor** $(1.09)^{4-t}$	**Compound Value** (FV_t)
1	Rs. 1,500	3 yrs	1.295029	Rs. 1,942.55
2	Rs. 3,000	2 yrs	1.188100	Rs. 3,564.30
3	Rs. 4,500	1 yr	1.090000	Rs. 4,905.00
4	Rs. 6,000	0 yrs	1.000000	Rs. 6,000.00
Total Terminal Future Value (FV):				Rs. **16,411.85**

Final Result: The aggregate compound value of the series of payments at the end of 4 years is Rs. **16,411.85**.

Compound Value of an Annuity (Future Value)

f. Compound Value of an Annuity (Ordinary Annuity vs. Annuity Due) For a sequence of equal constant periodic cash flows A invested at periodic interest rate r over horizon n:

1. Analytical Closed-Form Formulations:

- **Ordinary Annuity (Period-End Payments):**

$$\text{FV}_{\text{ord}} = A \times \left[\frac{(1+r)^n - 1}{r}\right] = A \cdot \text{CVAF}(r,n) \tag{1.18}$$

- **Annuity Due (Period-Beginning Payments):**

$$\text{FV}_{\text{due}} = \text{FV}_{\text{ord}} \times (1+r) = A \times \left[\frac{(1+r)^n - 1}{r}\right](1+r) \tag{1.19}$$

2. Year-by-Year Compound Accumulation Method ($\sum$):

- **Ordinary Annuity Future Value:** Each payment compounds forward to terminal year n:

$$\text{FV}_{\text{ord}} = \sum_{t=1}^{n} A \times (1+r)^{n-t} \tag{1.20}$$

Compounding Logic: Payment 1 (end of Yr 1) compounds for $n-1$ years: $A(1+r)^{n-1}$; Payment t compounds for $n-t$ years; and the final payment n (end of Yr n) compounds for 0 years: $A(1+r)^0 = A$.

- **Annuity Due Future Value:** Because payments occur at the start of each period, each payment earns 1 extra year of compounding:

$$\text{FV}_{\text{due}} = \sum_{t=0}^{n-1} A \times (1+r)^{n-t} \tag{1.21}$$

Compounding Logic: Payment 1 (start of Yr 1, $t=0$) compounds for n full years: $A(1+r)^n$; Payment n (start of Yr n, $t=n-1$) compounds for 1 year: $A(1+r)^1$.

Parameter Definitions:

- A = Constant periodic cash payment or deposit
- r = Periodic interest rate in decimal ($R/100$)
- n = Total number of compounding annuity conversion periods
- t = Period index
- $\text{CVAF}(r,n)$ = Compound Value Annuity Factor = $\dfrac{(1+r)^n - 1}{r}$

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Example 1.3.5: Compound Value of an Annual Annuity

Problem: Calculate the maturity value (Future Value) of an annuity if Rs. 20,000 is paid annually for 7 years at an interest rate of 12% per annum compounded annually.

Given Data:

- Periodic Annuity Flow (A) = Rs. 20,000
- Duration (n) = 7 years
- Interest Rate (R) = 12% p.a. = 0.12

Analytical Solution:

$$\text{FV} = A \times \left[\frac{(1+R)^n - 1}{R}\right] = 20{,}000 \times \left[\frac{(1+0.12)^7 - 1}{0.12}\right]$$

Evaluating the compounding power factor $(1.12)^7 \approx 2.2106814$:

$$\begin{aligned}\text{FV} &= 20{,}000 \times \left[\frac{2.2106814 - 1}{0.12}\right] = 20{,}000 \times \left[\frac{1.2106814}{0.12}\right] \\ &= 20{,}000 \times 10.089012 = \text{Rs. } \mathbf{2{,}01{,}780.24}\end{aligned}$$

Table Approximation Note: Using standard 3-decimal compound annuity factor tables ($\text{CVAF}(12\%, 7) \approx 10.089$ or rounded value 10.090):

$$\text{FV} \approx 20{,}000 \times 10.090 = \text{Rs. } \mathbf{2{,}01{,}800.00}$$

Final Result: The maturity accumulated value of the annuity after 7 years is Rs. **2,01,780.24** (approx. Rs. **2,01,800.00**).

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Example 1.3.6: Maturity Value of Monthly Recurring Deposit Annuity

Problem: Bhuvanesh pays a monthly installment of Rs. 2,000 towards a recurring deposit with a financier. The rate of interest is 6% per annum compounded monthly. What is the maturity value of this recurring deposit after 10 months?

Given Data:

- Monthly Deposit (A) = Rs. 2,000
- Nominal Annual Rate (R) = 6% p.a.
- Periodic Monthly Rate (i) = $\frac{6\%}{12} = 0.5\% = 0.005$
- Total Number of Monthly Instalments (n) = 10 months

Analytical Solution:

$$\text{FV} = A \times \left[\frac{(1+i)^n - 1}{i}\right] = 2{,}000 \times \left[\frac{(1+0.005)^{10} - 1}{0.005}\right]$$

Evaluating $(1.005)^{10} \approx 1.05114013$:

$$\begin{aligned}\text{FV} &= 2{,}000 \times \left[\frac{1.05114013 - 1}{0.005}\right] = 2{,}000 \times \left[\frac{0.05114013}{0.005}\right] \\ &= 2{,}000 \times 10.228026 = \text{Rs. } \mathbf{20{,}456.05}\end{aligned}$$

Approximation Note: Using standard recurring deposit financial tables ($\text{CVAF}(0.5\%, 10) \approx 10.22$):

$$\text{FV} \approx 2{,}000 \times 10.22 = \text{Rs. } \mathbf{20{,}440.00}$$

Final Result: The maturity value of the recurring deposit after 10 months is Rs. **20,456.05** (approx. Rs. **20,440.00**).

1.3.2 Discounting & Present Value Techniques

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Present Value of Lump Sum & Series of Cash Flows

a. Present Value of a Lump Sum The present value (PV) of a single future cash flow FV discounted across n periods at rate R:

$$\text{PV} = \text{FV} \times \frac{1}{(1+R)^n} = \text{FV} \cdot (1+R)^{-n} = \text{FV} \cdot \text{PVF}(R, n) \tag{1.22}$$

b. Present Value of a Series of Cash Flows For an uneven sequence of future cash inflows CF_t received across periods $t = 1, 2, \ldots, n$:

$$\text{PV} = \sum_{t=1}^{n} \frac{\text{CF}_t}{(1+R)^t} = \sum_{t=1}^{n} \text{CF}_t \times (1+R)^{-t} \tag{1.23}$$

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Example 1.3.7: Present Value of a Single Future Lump Sum

Problem: Find the present value of Rs. 10,000 receivable after 4 years at an interest rate of 12% per annum. (The PV factor in the standard table for 12% and 4 years is 0.6355).

Given Data:

- Future Value (FV) = Rs. 10,000
- Duration (n) = 4 years
- Discount Rate (R) = 12% p.a. = 0.12

Calculation:

$$\text{PV} = \text{FV} \times \frac{1}{(1+R)^n} = 10{,}000 \times \frac{1}{(1+0.12)^4} = 10{,}000 \times \frac{1}{1.573519}$$

$$\text{PV} = 10{,}000 \times 0.635518 = \text{Rs. } \mathbf{6{,}355.18}$$

Using 3-Decimal Table Factor (0.635):

$$\text{PV} = 10{,}000 \times 0.635 = \text{Rs. } \mathbf{6{,}350.00}$$

Final Result: The present value of Rs. 10,000 receivable after 4 years is Rs. **6,355.18** (or Rs. **6,350.00** using table factor).

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Example 1.3.8: Construction of Present Value Factor Schedule (12% p.a.)

Problem: Calculate the Present Value Factors (PVF) at an interest rate of 12% per annum for a period of 5 years.

Formula:

$$\text{PVF}(12\%, n) = \frac{1}{(1+0.12)^n} = (1.12)^{-n}$$

Schedule of Discount Factors:

Year (n)	Mathematical Computation	Present Value Factor (PVF)
1	$\frac{1}{(1.12)^1} = \frac{1}{1.120000}$	**0.8929**
2	$\frac{1}{(1.12)^2} = \frac{1}{1.254400}$	**0.7972**
3	$\frac{1}{(1.12)^3} = \frac{1}{1.404928}$	**0.7118**
4	$\frac{1}{(1.12)^4} = \frac{1}{1.573519}$	**0.6355**
5	$\frac{1}{(1.12)^5} = \frac{1}{1.762342}$	**0.5674**

Present Value of Annuities (Ordinary Annuity vs. Annuity Due)

c. Present Value of an Annuity (Ordinary Annuity – Period-End Payments)
1. Analytical Closed-Form Formulation:

$$\text{PV} = A \times \left[\frac{1-(1+r)^{-n}}{r}\right] = A \times \text{PVAF}(r,n) \tag{1.24}$$

2. Year-by-Year Discounting Summation Method ($\sum$):

$$\text{PV} = \sum_{t=1}^{n} \frac{A}{(1+r)^t} = \frac{A}{(1+r)^1} + \frac{A}{(1+r)^2} + \cdots + \frac{A}{(1+r)^n} \tag{1.25}$$

Discounting Logic: Calculate the present value of each individual payment by discounting it back to the present ($t=0$) at periodic rate r, and then sum them up.

d. Present Value of Annuity Due (Period-Beginning Payments) 1. Analytical Closed-Form Formulation:

$$\text{PV}_{\text{due}} = A \times \left[\frac{1-(1+r)^{-n}}{r}\right] \times (1+r) = A \times \text{PVAF}(r,n) \times (1+r) \tag{1.26}$$

2. Year-by-Year Discounting Summation Method ($\sum$):

$$\text{PV}_{\text{due}} = \sum_{t=0}^{n-1} \frac{A}{(1+r)^t} = A + \frac{A}{(1+r)^1} + \frac{A}{(1+r)^2} + \cdots + \frac{A}{(1+r)^{n-1}} \tag{1.27}$$

Discounting Logic: Because payments occur at the start of each period, every cash flow is shifted one period earlier compared to an ordinary annuity. The first payment at $t=0$ requires no discounting (A), while the final payment at $t=n-1$ is discounted for $n-1$ periods: $\dfrac{A}{(1+r)^{n-1}}$.

Parameter Definitions:

- A = Constant periodic annuity cash flow
- r = Periodic discount rate in decimal ($R/100$)
- n = Total number of payment periods / installments
- t = Period index
- $\text{PVAF}(r,n)$ = Present Value Annuity Factor = $\dfrac{1-(1+r)^{-n}}{r}$
- $(1+r)$ = Advance receipt compounding factor for annuity due

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Example 1.3.9: Present Value of an Ordinary Annuity Stream

Problem: Raja has entered into an agreement that will fetch him Rs. 60,000 per annum for the next 4 years. Calculate the present value of these future cash inflows at a discount rate of 20% per annum.

Given Data:

- Annual Cash Inflow (A) = Rs. 60,000
- Horizon (n) = 4 years
- Discount Rate (R) = 20% p.a. = 0.20

Method 1: Year-by-Year Cash Flow Discounting Table

Year (t)	Cash Inflow (CF_t)	PV Factor at 20%: $\frac{1}{(1.20)^t}$	Present Value (PV_t)
1	Rs. 60,000	$\frac{1}{(1.20)^1} = 0.833333$	Rs. 50,000.00
2	Rs. 60,000	$\frac{1}{(1.20)^2} = 0.694444$	Rs. 41,666.67
3	Rs. 60,000	$\frac{1}{(1.20)^3} = 0.578704$	Rs. 34,722.22
4	Rs. 60,000	$\frac{1}{(1.20)^4} = 0.482253$	Rs. 28,935.19
Total Aggregate Present Value (PV):			Rs. **1,55,324.08**

Method 2: Analytical PVAF Closed-Form Formula

$$\begin{aligned} \text{PV} &= A \times \left[\frac{1-(1+R)^{-n}}{R}\right] = 60{,}000 \times \left[\frac{1-(1.20)^{-4}}{0.20}\right] \\ &= 60{,}000 \times \left[\frac{1-0.482253}{0.20}\right] = 60{,}000 \times \left[\frac{0.517747}{0.20}\right] \\ &= 60{,}000 \times 2.588735 = \text{Rs. } \mathbf{1{,}55{,}324.08} \end{aligned}$$

Table Factor Approximation Note: Using standard Present Value Annuity Table factor for 4 years at 20% ($\text{PVAF}(20\%, 4) \approx 2.589$):

$$\text{PV} \approx 60{,}000 \times 2.589 = \text{Rs. } \mathbf{1{,}55{,}340.00}$$

Final Result: The present value of Raja's annuity agreement is Rs. **1,55,324.08** (or Rs. **1,55,340.00** using 3-decimal table factor).

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Example 1.3.10: Present Value of an Annuity Due

Problem: Mr. Thanay expects to receive Rs. 20,000 at the **beginning** of each year for 4 years. Calculate the present value of this annuity due assuming an annual interest rate of 9%.

Given Data:

- Annual Instalment (A) = Rs. 20,000
- Horizon (n) = 4 payments made at $t = 0, 1, 2, 3$
- Discount Rate (R) = 9% p.a. = 0.09

Method 1: Year-by-Year Discounting of Beginning-of-Year Receipts Because payments occur at the beginning of each year, the first payment is received at $t = 0$ (no discounting):

- **Beginning of Year 1** ($t = 0$): $\text{PV}_0 = \text{Rs. } 20{,}000$
- **Beginning of Year 2** ($t = 1$): $\text{PV}_1 = \frac{20{,}000}{(1.09)^1} = \text{Rs. } 18{,}348.62$
- **Beginning of Year 3** ($t = 2$): $\text{PV}_2 = \frac{20{,}000}{(1.09)^2} = \text{Rs. } 16{,}833.60$

- **Beginning of Year 4 ($t = 3$):** $\text{PV}_3 = \dfrac{20{,}000}{(1.09)^3} = \text{Rs. } 15{,}443.67$

Summing all individual present values:

$$\text{PV}_{\text{due}} = 20{,}000.00 + 18{,}348.62 + 16{,}833.60 + 15{,}443.67 = \text{Rs. } \mathbf{70{,}625.89}$$

Method 2: Annuity Due Analytical Formula

$$\begin{aligned}
\text{PV}_{\text{due}} &= A \times \left[\frac{1-(1+R)^{-n}}{R}\right] \times (1+R) \\
&= 20{,}000 \times \left[\frac{1-(1.09)^{-4}}{0.09}\right] \times 1.09 \\
&= 20{,}000 \times \left[\frac{1-0.708425}{0.09}\right] \times 1.09 \\
&= 20{,}000 \times 3.239720 \times 1.09 = \text{Rs. } \mathbf{70{,}625.89} \approx \text{Rs. } \mathbf{70{,}626.00}
\end{aligned}$$

Final Result: The present value of Mr. Thanay's annuity due is Rs. **70,625.89** (approx. Rs. **70,626.00**).

Present Value of a Constant Perpetuity

e. Present Value of a Perpetuity (Constant Perpetuity – Infinite Horizon) A **Perpetuity** is an annuity whose periodic cash flows continue indefinitely. The present value of a constant perpetual stream of cash flows C discounted at rate R per period is:

$$\text{PV of Constant Perpetuity} = \frac{C}{R} \tag{1.28}$$

where:

- C = Constant periodic cash inflow (e.g. annual coupon, preferred dividend)
- R = Discount / capitalization rate per payment period

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Example 1.3.11: Valuation of Preference Shares as a Constant Perpetuity

Problem: A company has issued 10% preference shares with a face value of Rs. 100 each. What would be the fair market price (Present Value) of the preference share if the required rate of return for the investor is 8% per annum?

Given Data:

- Face Value of Preference Share = Rs. 100
- Annual Preference Dividend Rate = 10%
- Constant Annual Dividend (C) = 10% × Rs. 100 = Rs. 10.00
- Required Rate of Return / Discount Rate (R) = 8% = 0.08

Analytical Solution: Since preference shares pay a constant periodic dividend indefinitely without a specified maturity date, their value is given by the constant perpetuity formula:

$$\text{PV} = \frac{C}{R} = \frac{10}{0.08} = \text{Rs. } \mathbf{125.00}$$

Conclusion: The investor will be willing to pay up to Rs. **125.00** to purchase the preference share to earn a 10% dividend on its face value at their required 8% yield.

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Example 1.3.12: Present Value of a Perpetual Cash Flow Stream

Problem: Let us assume that an investor expects a perpetual annual sum of Rs. 500 from an investment. What is the present value of this perpetuity if the benchmark interest rate is 10% per annum?

Given Data:

- Perpetual Annual Cash Inflow (C) = Rs. 500
- Discount Rate (R) = 10% = 0.10

Analytical Calculation:

$$\text{PV} = \frac{C}{R} = \frac{500}{0.10} = \text{Rs. } \mathbf{5{,}000.00}$$

Final Result: The present value of the perpetual Rs. 500 annual cash stream at 10% discount rate is Rs. **5,000.00**.

Present Value of a Growing Perpetuity

f. Present Value of a Growing Perpetuity A **Growing Perpetuity** is an infinite stream of periodic cash flows that grow at a constant compound rate G per period. When the discount rate exceeds the growth rate ($R > G$):

$$\text{PV of Growing Perpetuity} = \frac{C_1}{R - G} \tag{1.29}$$

where:

- C_1 = Expected cash inflow at the end of the first period
- R = Discount rate / required rate of return per period
- G = Constant compound growth rate of cash flows per period
- **Condition for Convergence:** $R > G$ (if $R \leq G$, the present value diverges to infinity).

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Example 1.3.13: Valuation of a Growing Dividend Stream (Gordon Growth Perpetuity)

Problem: A company is expected to declare a dividend of Rs. 4.00 at the end of the 1st year from now (C_1), and this dividend is projected to grow at a constant annual rate of 5% indefinitely. What is the present value of this perpetual dividend stream if the investor's required rate of return is 10% per annum?

Given Data:

- First-Year Dividend (C_1) = Rs. 4.00

- Constant Annual Dividend Growth Rate (G) $= 5\% = 0.05$
- Required Rate of Return / Discount Rate (R) $= 10\% = 0.10$

Convergence Check: Since $R = 0.10 > G = 0.05$, the finite valuation condition ($R > G$) is satisfied.
Analytical Solution: Applying the growing perpetuity (Gordon growth) formula:

$$\begin{aligned} \text{PV} &= \frac{C_1}{R-G} = \frac{4}{0.10-0.05} \\ &= \frac{4}{0.05} = \text{Rs. } \mathbf{80.00} \end{aligned}$$

Final Result: The present value (fair market price) of the perpetual growing dividend stream is Rs. **80.00**.

Present Value of a Growing Annuity

g. Present Value of a Growing Annuity (Finite Horizon n) For a stream of n periodic cash flows growing at rate g per period across a finite horizon (with $r \neq g$):

1. Analytical Closed-Form Formulation:

$$\text{PV} = \frac{C_1}{r-g}\left[1 - \left(\frac{1+g}{1+r}\right)^n\right] \tag{1.30}$$

2. Year-by-Year Summation Formulation ($\sum$):

$$\text{PV} = \sum_{t=1}^{n} \frac{C_1 \times (1+g)^{t-1}}{(1+r)^t} \tag{1.31}$$

Two-Step Year-by-Year Valuation Process:
1. **Step 1: Calculate the growing cash flow for year t:**

$$C_t = C_1(1+g)^{t-1}$$

2. **Step 2: Discount each individual cash flow back to time $t = 0$:**

$$\text{PV}(C_t) = \frac{C_t}{(1+r)^t} = \frac{C_1(1+g)^{t-1}}{(1+r)^t}$$

Finally, sum all n discounted values: $\text{PV} = \sum_{t=1}^{n} \text{PV}(C_t)$.

Parameter Definitions:

- C_1 = Cash flow realized at the end of Period 1
- C_t = Growing cash flow realized at the end of Period t
- r = Periodic discount rate / required rate of return in decimal ($R/100$)
- g = Constant compound growth rate per period in decimal ($G/100$)
- n = Total duration / number of periods
- t = Period index ($t = 1, 2, \ldots, n$)

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Example 1.3.14: Present Value of a Growing Annuity Stream

Problem: Mr. Deva has rented out a commercial shop for 4 years at an initial annual rent of Rs. 12,000. The tenant has agreed to an escalation clause where the rent increases by 10% every year. If the required rate of return (discount rate) is 8%, find the present value of the expected series of rental payments.

Given Data:

- Initial First-Year Rent (C_1) = Rs. 12,000
- Lease Duration (n) = 4 years
- Annual Rent Escalation Rate (G) = 10% = 0.10
- Discount Rate (R) = 8% = 0.08

Method 1: Year-by-Year Escalated Cash Flow Discounting Table

- **Year 1:** Cash Flow C_1 = Rs. 12,000

$$\text{PV}_1 = \frac{12{,}000}{(1.08)^1} = \frac{12{,}000}{1.08} = \text{Rs. } \mathbf{11{,}111.11}$$

- **Year 2:** Cash Flow $C_2 = 12{,}000 \times (1 + 0.10) = \text{Rs. } 13{,}200$

$$\text{PV}_2 = \frac{13{,}200}{(1.08)^2} = \frac{13{,}200}{1.1664} = \text{Rs. } \mathbf{11{,}316.87}$$

- **Year 3:** Cash Flow $C_3 = 13{,}200 \times (1 + 0.10) = \text{Rs. } 14{,}520$

$$\text{PV}_3 = \frac{14{,}520}{(1.08)^3} = \frac{14{,}520}{1.259712} = \text{Rs. } \mathbf{11{,}526.54}$$

- **Year 4:** Cash Flow $C_4 = 14{,}520 \times (1 + 0.10) = \text{Rs. } 15{,}972$

$$\text{PV}_4 = \frac{15{,}972}{(1.08)^4} = \frac{15{,}972}{1.360489} = \text{Rs. } \mathbf{11{,}740.16}$$

Summing all discounted cash inflows:

$$\text{PV} = 11{,}111.11 + 11{,}316.87 + 11{,}526.54 + 11{,}740.16 = \text{Rs. } \mathbf{45{,}694.68}$$

Method 2: Growing Annuity Analytical Closed-Form Formula When growth rate $G > R$:

$$\begin{aligned}\text{PV} &= \frac{C_1}{G - R}\left[\left(\frac{1+G}{1+R}\right)^n - 1\right] = \frac{12{,}000}{0.10 - 0.08}\left[\left(\frac{1.10}{1.08}\right)^4 - 1\right] \\ &= \frac{12{,}000}{0.02}\left[(1.0185185)^4 - 1\right] = 6{,}00{,}000 \times [1.0761578 - 1] \\ &= 6{,}00{,}000 \times 0.0761578 = \text{Rs. } \mathbf{45{,}694.68}\end{aligned}$$

Final Result: The present value of the 4-year escalating rental stream is Rs. **45,694.68**.

Sinking Fund (S.F.) Accumulation & Periodic Instalment

h. Sinking Fund (S.F.) Accumulation A **Sinking Fund** is a dedicated reserve established by investing equal periodic installments at compound interest to accumulate a targeted future capital sum (e.g. to redeem debentures or replace capital equipment):

$$\text{FV} = A \times \left[\frac{(1+R)^n - 1}{R}\right] = A \cdot \text{CVAF}(R, n) \tag{1.32}$$

The required periodic installment A is given by:

$$A = \frac{\text{FV} \times R}{(1+R)^n - 1} = \frac{\text{FV}}{\text{CVAF}(R, n)} \tag{1.33}$$

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Example 1.3.15: Determination of Annual Sinking Fund Instalment

Problem: An amount of Rs. 2,00,000 is required at the end of 5 years from now to repay a debenture liability. What amount should be accumulated (invested) at the end of each year at an annual interest rate of 12% so that the fund reaches Rs. 2,00,000 after 5 years?

Given Data:

- Target Future Liability (FV) = Rs. 2,00,000
- Accumulation Horizon (n) = 5 years
- Annual Rate of Interest (R) = 12% p.a. = 0.12

Analytical Solution: Let A denote the required annual sinking fund installment deposited at the end of each year:

$$\text{FV} = A \times \left[\frac{(1+R)^n - 1}{R}\right] \implies A = \frac{\text{FV} \times R}{(1+R)^n - 1}$$

Evaluating the compound annuity accumulation factor $\text{CVAF}(12\%, 5)$:

$$(1.12)^5 \approx 1.76234168$$

$$\text{CVAF}(12\%, 5) = \frac{1.76234168 - 1}{0.12} = \frac{0.76234168}{0.12} \approx 6.352847$$

Solving for the annual sinking fund installment A:

$$A = \frac{2{,}00{,}000}{6.352847} = \text{Rs. } \mathbf{31{,}481.95}$$

Table Approximation Notes:

- Using standard 3-decimal Compound Value Annuity Factor tables ($\text{CVAF}(12\%, 5) \approx 6.353$):

$$A \approx \frac{2{,}00{,}000}{6.353} = \text{Rs. } \mathbf{31{,}481.19}$$

- Using standard Sinking Fund Factor tables ($\text{SFF}(12\%, 5) = \dfrac{1}{\text{CVAF}} \approx 0.15741$):

$$A = 2{,}00{,}000 \times 0.15741 = \text{Rs. } \mathbf{31{,}482.00}$$

Final Result: The required annual sinking fund installment to be invested each year is Rs. **31,481.95** (approx. Rs. **31,482.00**).

Example 1.3.16: Annuity Due Sinking Fund (T Limited)

Problem: T Limited is creating a sinking fund to redeem its debentures of Rs. 40,00,000 issued on April 5, 2008 and maturing on April 4, 2020. The first annual payment to the fund will be made on April 5, 2008 (at the beginning of each operating year). The company will make equal annual payments and expects that the fund will earn an interest rate of 14% per annum. How much will be the amount of the annual sinking fund payment?

Given Data:

- Target Sinking Fund Liability (FV) = Rs. 40,00,000
- Debenture Issue Date: April 5, 2008 Maturity Date: April 4, 2020
- Horizon (n) = 2020 − 2008 = 12 years
- Timing: Beginning of each year (Annuity Due)
- Annual Interest Rate (R) = 14% p.a. = 0.14

Formula:

$$\text{FV} = \text{Instalment } (A) \times \left[\frac{(1+R)^n - 1}{R}\right] \times (1+R)$$

Substitution & Calculation:

$$40{,}00{,}000 = A \times \left[\frac{(1+0.14)^{12} - 1}{0.14}\right] \times (1+0.14)$$

$$40{,}00{,}000 = A \times \left[\frac{4.817906 - 1}{0.14}\right] \times 1.14$$

$$40{,}00{,}000 = A \times \left[\frac{3.817906}{0.14}\right] \times 1.14$$

$$40{,}00{,}000 = A \times 27.270756 \times 1.14$$

$$40{,}00{,}000 = A \times 31.088662$$

$$A = \frac{40{,}00{,}000}{31.088662} = \text{Rs. } \mathbf{1{,}28{,}664.31}$$

Final Result: The required annual sinking fund instalment is Rs. **1,28,664.31**.

1.4 Growth and Decay Curves

Lecture
03/09/2026

Definition 1.4.1: Exponential Growth and Decay Models

Let t denote time ($t \geq 0$) and $n(t)$ denote the modeled quantity at time t. The dynamics of continuous exponential growth and decay are characterized by:

- **Exponential Growth Model:** $n(t) = n_0\, e^{kt}, \quad k > 0$ (initial quantity $n_0 = n(0)$, growth constant k)
- **Exponential Decay Model:** $n(t) = n_0\, e^{-kt}, \quad k > 0$ (initial quantity $n_0 = n(0)$, decay constant k)

For scaling over time interval $[0, T]$ between boundary values $[n_{\min}, n_{\max}]$:

$$k = \frac{1}{T} \ln\left(\frac{n_{\max}}{n_{\min}}\right) = \frac{1}{5} \ln\left(\frac{70}{10}\right) = \frac{\ln 7}{5} \approx 0.389182 \tag{1.34}$$

- **Growth Curve:** $n(t) = 10\, e^{0.389182\, t}$, increasing monotonically from 10 to 70.
- **Decay Curve:** $n(t) = 70\, e^{-0.389182\, t}$, decreasing monotonically from 70 to 10.

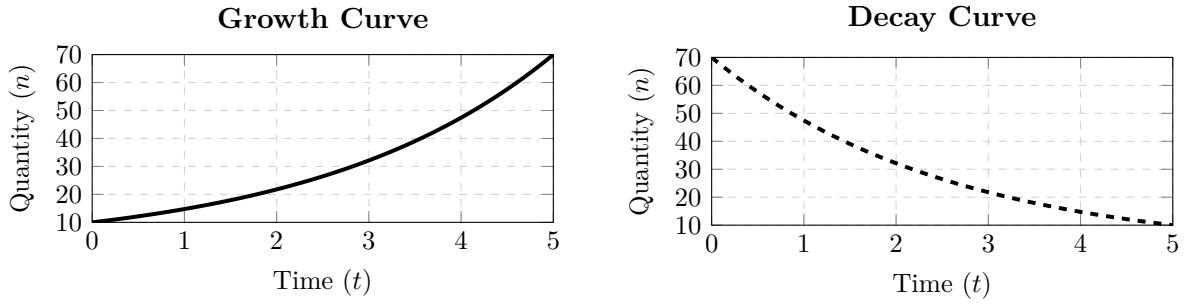


Figure 1.1: Research-Style Greyscale Exponential Growth (Solid) and Decay (Dashed) Curves

Decay Curves in Financial Trading: Price Impact Analysis

Market Microstructure Interpretation of the Decay Curve: In financial markets and quantitative execution, a **decay curve** models how a trade's impact on asset price evolves over time:

- **X-axis:** Time elapsed following trade execution.
- **Y-axis:** Price change / price impact.

Institutional Users (Large Banks): Decay curves are systematically employed by large investment banks and market-making desks to:

1. **Monitor trade impacts:** Measuring how individual block orders or algorithmic parent orders move the bid-ask quote and midpoint over time.
2. **Detect toxic behavior:** Identifying informed order flow, adverse selection, and toxic flow that degrades fill quality and market liquidity.

Core Purpose:

1. **Understand long-term trade impact:** Disentangling temporary market impact (liquidity consumption that decays back to fundamental value) from permanent price impact (information revelation).
2. **Ensure market stability:** Managing institutional execution schedules (e.g. TWAP/VWAP/POV) to prevent excessive volatility and preserve market stability.

Chapter 2

Valuation of Financial Instruments

2.1 Foundations of Valuation and Yield Concepts

Lecture
07/09/2026

Definition 2.1.1: Foundational Valuation Concepts

In corporate finance and financial mathematics, asset valuation depends fundamentally on the perspective and purpose of evaluation:

1. **Book Value:** The accounting net worth of an asset or firm recorded on the balance sheet (historic cost minus accumulated depreciation).
2. **Market Value:** The actual price at which an asset or financial security can be bought or sold in an active, competitive market.
3. **Going Concern Value:** The economic value a prospective purchaser is prepared to pay for an operating business with established operations and recurring earnings power.
4. **Liquidating Value (Breakup Value):** The net residual sum realized if all assets of the enterprise are sold piecemeal and all external liabilities are settled.
5. **Capitalised Value:** The present value of all anticipated future cash inflows generated by a fixed asset or project, discounted at the firm's required cost of capital.

2.1.1 Yield Fundamentals

Market Yield and Current Yield

The **Current Yield** measures the annual income return generated by a fixed-income security or preference share relative to its current market price:

$$\text{Current Yield} = \frac{\text{Annual Dividend / Coupon}}{\text{Current Market Price}} \times 100 \tag{2.1}$$

The **Dividend Yield** (dY) of an equity or preference share is defined as:

$$dY = \frac{\text{DPS}}{P_0} \times 100 \tag{2.2}$$

where DPS is the annual dividend per share and P_0 is the current market price per share.

Example 2.1.1: Current Yield of Preference Shares

An 8% preference share of ABC Limited with face value Rs. 100 is selling in the market for Rs. 64. Determine the current yield of the preference share.

Formula:

$$\text{Current Yield} = \frac{\text{Annual Dividend}}{P_0} \times 100$$

Substitution:

$$\text{Current Yield} = \frac{8}{64} \times 100 = 0.125 \times 100 = 12.5\%$$

Value: 12.5%

Example 2.1.2: Expected Return on XYZ Limited Preference Shares

The preference shares of XYZ Ltd are selling for Rs. 50 per share and pay an annual dividend of Rs. 3. What is the expected rate of return if the security is purchased at the market price?

Formula:

$$\text{Expected Return} = \frac{\text{DPS}}{P_0} \times 100$$

Substitution:

$$\text{Expected Return} = \frac{3}{50} \times 100 = 6\%$$

Value: 6%

Example 2.1.3: Valuation and Investment Decision (Bhuvanesh / PQR Ltd)

Bhuvanesh owns 200 preference shares of PQR Ltd, currently selling at Rs. 40 per share and paying an annual dividend of Rs. 7 per share.

1. What is the expected return (dividend yield)?
2. If Bhuvanesh requires a 12% return, should he sell or buy more preference shares?

1. Expected Return:

$$\text{Expected Return} = \frac{7}{40} \times 100 = 17.5\%$$

2. Intrinsic Valuation and Decision Rule:

$$P^* = \frac{D}{K} = \frac{7}{0.12} = \text{Rs. } 58.33$$

Since the expected rate of return (17.5%) exceeds the required rate of return (12%) and the market price ($P_0 = \text{Rs. } 40$) is below intrinsic worth ($P^* = \text{Rs. } 58.33$), the security is undervalued. Hence, Bhuvanesh should **buy more preference shares**.

2.2 Yield to Maturity, Holding Period Return, and Equity Valuation

Lecture
08/09/2026

Definition 2.2.1: Yield Measures and Holding Period Return

Yield to Maturity (YTM / Redemption Yield) The overall annualized rate of return earned by an investor who holds a fixed-income bond or redeemable preference share to its maturity date. YTM reflects:
1. Recurring annual coupon/dividend cash flows.
2. Periodic capital appreciation or depreciation over the holding span (difference between face value and purchase price).

Holding Period Return (HPR) The total return achieved over a planned holding duration:

$$\text{HPR} = \frac{D_t + \Delta P}{P_1} = \frac{D_t + (P_t - P_1)}{P_1} \tag{2.3}$$

where D_t is the dividend received during the period, P_1 is the purchase price, P_t is the terminal selling price, and $\Delta P = P_t - P_1$ represents capital gain/loss.

Earnings Yield The proportion of accounting net earnings generated per share relative to market price:

$$\text{Earnings Yield} = \frac{\text{EPS}}{P_0} \tag{2.4}$$

- **Gross Yield:** Used by corporate treasurers to estimate the effective cost of equity capital.
- **Net Yield:** Used by equity investors to evaluate the true underlying investment return after taxes.

2.2.1 Approaches to Equity Share Valuation

Methods of Equity Valuation

Equity valuation methodologies fall into four structured frameworks:

1. **Accounting-Based Models:** Book Value (balance sheet net worth) and Liquidation Value.
2. **Dividend Discount Models (DDM):** Present value of all projected future dividend streams.
3. **Discounted Cash Flow (DCF) Models:** Operating cash flows (FCFF) and Free Cash Flows to Equity (FCFE).
4. **Relative Valuation Metrics:** Price-to-Earnings (P/E), Price-to-Book (P/B), and Price-to-Sales (P/S).

2.2.2 The General Dividend Discount Model (DDM)

Definition 2.2.2: General DDM Assumptions

Under fundamental dividend discount theory:

1. Dividends are paid annually at discrete intervals.
2. The first dividend D_1 is received exactly one year after acquisition.
3. Equity liquidation occurs at year-end on ex-dividend terms.

Finite and Infinite DDM Formulations

For a **Finite Holding Period (n years)** with terminal market price P_n:

$$P_0 = \sum_{t=1}^{n} \frac{D_t}{(1+K_e)^t} + \frac{P_n}{(1+K_e)^n} \tag{2.5}$$

Specifically, for a **Two-Year Holding Horizon ($n = 2$)**:

$$P_0 = \frac{D_1}{1+K_e} + \frac{D_2 + P_2}{(1+K_e)^2} \tag{2.6}$$

For an **Infinite Holding Horizon ($n \to \infty$)**, share price equals the discounted sum of all future perpetual dividends:

$$P_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+K_e)^t} \tag{2.7}$$

where P_0 is intrinsic share price, D_t is expected dividend at year t, and K_e is the required equity return.

2.3 Dividend Valuation Models: Constant Dividend and Gordon Basics

2.3.1 Model 1: Zero Growth (Constant Perpetual Dividend)

Lecture
10/09/2026

Zero Growth Valuation Formula

When a firm pays an invariant perpetual dividend ($D_t = D$ for all t):

$$P_0 = \frac{D}{K_e} \tag{2.8}$$

Example 2.3.1: Zero-Growth Equity Valuation

A firm pays an annual dividend of 20% on equity shares of face value Rs. 100 each. Find the intrinsic value of the share given that dividends remain constant perpetually and the required rate of return is 15%.

Formula:

$$P_0 = \frac{D}{K_e} = \frac{\text{Face Value} \times \text{Dividend Rate}}{K_e}$$

Substitution:

$$D = 100 \times 20\% = \text{Rs. } 20, \quad K_e = 0.15$$

$$P_0 = \frac{20}{0.15} = \text{Rs. } 133.33$$

Value: Rs. 133.33

2.3.2 Model 2: Constant Growth Dividend Model

Gordon Constant Growth Formula

Assuming dividends expand at a steady compound annual growth rate g ($g < K_e$):

$$P_0 = \frac{D_1}{K_e - g} = \frac{D_0(1+g)}{K_e - g} \tag{2.9}$$

Example 2.3.2: Constant Growth Share Valuation

A share of face value Rs. 100 pays an expected dividend of 12% at the end of Year 1 (D_1). Dividend growth is estimated at 3% per annum (g). If the investor's required rate of return is 16% (K_e), calculate the intrinsic equity value.

Formula:

$$P_0 = \frac{D_1}{K_e - g}$$

Substitution:

$$D_1 = 100 \times 12\% = \text{Rs. } 12, \quad K_e = 0.16, \quad g = 0.03$$

$$P_0 = \frac{12}{0.16 - 0.03} = \frac{12}{0.13} = \text{Rs. } 92.31$$

Value: Rs. 92.31

2.3.3 Specific Dividend Valuation Models: Gordon's Model Fundamentals

Definition 2.3.1: Gordon Model Assumptions and Growth Formulation

Under Myron J. Gordon's dividend capitalization framework:

1. Earnings are either distributed as cash dividends or retained for internal reinvestment.
2. The retention ratio b and rate of return on reinvested capital r remain strictly constant.
3. Growth rate in future earnings and dividends is endogenous:

$$g = b \times r \tag{2.10}$$

4. The firm is an all-equity enterprise with infinite life, and $K_e > br$.

Gordon's Valuation Formula

$$P_0 = \frac{\text{EPS}_1(1-b)}{K_e - b \cdot r} = \frac{D_1}{K_e - g} \tag{2.11}$$

where EPS_1 is expected earnings per share, b is the retention ratio, $(1-b)$ is the dividend payout ratio, K_e is the cost of equity, and r is the internal rate of return.

Example 2.3.3: Earnings Reinvestment vs. Full Payout

A firm generates $\text{EPS}_1 = \text{Rs. } 9$ and faces a cost of capital $K_e = 15\%$. Compare the expected dividend D_1 under:

1. Case 1: Retention ratio $b = 0$ (entire earnings distributed as dividend).
2. Case 2: Payout ratio $1 - b = 60\%$ (retention ratio $b = 40\%$).

Case 1 ($b = 0$):

$$D_1 = \text{EPS}_1 \times (1-b) = 9 \times (1-0) = \text{Rs. } 9.00$$

Case 2 ($b = 0.40$):

$$D_1 = \text{EPS}_1 \times (1-b) = 9 \times 0.60 = \text{Rs. } 5.40$$

Value: Case 1: Rs. 9.00 | Case 2: Rs. 5.40

2.4 Gordon's Model: Growth, Normal, and Declining Firm Dynamics

Lecture
15/09/2026

Firm Classification and Payout Policy Implications

Gordon's model establishes three distinct economic regimes based on the relationship between return on investment (r) and capitalization rate (K_e):

1. **Growth Firm ($r > K_e$):** The firm earns more on retained capital than equity holders could earn externally. The optimal dividend payout ratio is **NIL (0%)**, and retention should be maximized ($b = 100\%$).
2. **Normal Firm ($r = K_e$):** Internal returns match investors' required returns. The dividend payout ratio is **irrelevant**; market price remains identical across all payout levels.
3. **Declining Firm ($r < K_e$):** Internal returns are lower than the opportunity cost of capital. The optimal dividend payout ratio is 100%, and retention should be zero ($b = 0\%$).

Example 2.4.1: Gordon's Model for a Growth Firm (XY Ltd)

XY Ltd is a growth firm with $\text{EPS} = \text{Rs. } 14$, capitalization rate $K_e = 15\%$, and internal rate of return $r = 20\%$. Determine the market price per share under Gordon's model for retention ratios of: (a) 40%, (b) 60%, (c) 20%.

General Formula:

$$P_0 = \frac{\text{EPS} \times (1-b)}{K_e - b \cdot r} = \frac{D}{K_e - g}$$

(a) Retention $b = 40\%$ (Payout $= 60\%$):

$$D = 14 \times 0.60 = \text{Rs. } 8.40, \quad g = 0.40 \times 0.20 = 0.08$$
$$P_0 = \frac{8.40}{0.15 - 0.08} = \frac{8.40}{0.07} = \text{Rs. } 120.00$$

(b) Retention $b = 60\%$ (Payout $= 40\%$):

$$D = 14 \times 0.40 = \text{Rs. } 5.60, \quad g = 0.60 \times 0.20 = 0.12$$
$$P_0 = \frac{5.60}{0.15 - 0.12} = \frac{5.60}{0.03} = \text{Rs. } 186.67$$

(c) Retention $b = 20\%$ (Payout $= 80\%$):

$$D = 14 \times 0.80 = \text{Rs. } 11.20, \quad g = 0.20 \times 0.20 = 0.04$$
$$P_0 = \frac{11.20}{0.15 - 0.04} = \frac{11.20}{0.11} = \text{Rs. } 101.82$$

Conclusion: Because $r > K_e$ ($20\% > 15\%$), market price per share increases as the retention ratio increases. The growth firm maximizes shareholder wealth by retaining maximum earnings.

Example 2.4.2: Gordon's Model for a Normal Firm (MN Ltd)

MN Ltd is a normal firm with EPS = Rs. 45, cost of capital $K_e = 18\%$, and return on investment $r = 18\%$. Ascertain the market value per share using Gordon's model if dividend payout is: (a) 30%, (b) 60%, (c) 90%.

General Formula:

$$P_0 = \frac{\text{EPS} \times (1 - b)}{K_e - b \cdot r}$$

(a) Payout $= 30\%$ (Retention $b = 70\%$):

$$D = 45 \times 0.30 = \text{Rs. } 13.50, \quad g = 0.70 \times 0.18 = 0.126$$
$$P_0 = \frac{13.50}{0.18 - 0.126} = \frac{13.50}{0.054} = \text{Rs. } 250.00$$

(b) Payout $= 60\%$ (Retention $b = 40\%$):

$$D = 45 \times 0.60 = \text{Rs. } 27.00, \quad g = 0.40 \times 0.18 = 0.072$$
$$P_0 = \frac{27.00}{0.18 - 0.072} = \frac{27.00}{0.108} = \text{Rs. } 250.00$$

(c) Payout $= 90\%$ (Retention $b = 10\%$):

$$D = 45 \times 0.90 = \text{Rs. } 40.50, \quad g = 0.10 \times 0.18 = 0.018$$
$$P_0 = \frac{40.50}{0.18 - 0.018} = \frac{40.50}{0.162} = \text{Rs. } 250.00$$

Conclusion: For a normal firm ($r = K_e = 18\%$), the share price is invariant to payout choice ($P_0 = \text{Rs. } 250.00$). Dividend policy is completely neutral.

Example 2.4.3: Gordon's Model for a Declining Firm (ABC Ltd)

ABC Ltd is a declining firm with EPS = Rs. 35, capitalization rate $K_e = 15\%$, and productivity of retained earnings $r = 10\%$. Using Gordon's model, determine the market price per share if payout is: (a) 20%, (b) 40%, (c) 70%.

General Formula:

$$P_0 = \frac{\text{EPS} \times (1-b)}{K_e - b \cdot r}$$

(a) Payout $= 20\%$ **(Retention** $b = 80\%$**):**

$$D = 35 \times 0.20 = \text{Rs. } 7.00, \quad g = 0.80 \times 0.10 = 0.08$$

$$P_0 = \frac{7.00}{0.15 - 0.08} = \frac{7.00}{0.07} = \text{Rs. } 100.00$$

(b) Payout $= 40\%$ **(Retention** $b = 60\%$**):**

$$D = 35 \times 0.40 = \text{Rs. } 14.00, \quad g = 0.60 \times 0.10 = 0.06$$

$$P_0 = \frac{14.00}{0.15 - 0.06} = \frac{14.00}{0.09} = \text{Rs. } 155.56$$

(c) Payout $= 70\%$ **(Retention** $b = 30\%$**):**

$$D = 35 \times 0.70 = \text{Rs. } 24.50, \quad g = 0.30 \times 0.10 = 0.03$$

$$P_0 = \frac{24.50}{0.15 - 0.03} = \frac{24.50}{0.12} = \text{Rs. } 204.17$$

Conclusion: Because $r < K_e$ ($10\% < 15\%$), market price per share increases monotonically as the dividend payout ratio increases. The optimal policy for a declining firm is 100% payout.

2.5 Walter's Dividend Valuation Model and Policy Decisions

2.5.1 Theories of Dividend Policy

Lecture
16/09/2026

Definition 2.5.1: Dividend Relevance vs. Irrelevance

- **Theories of Relevance:** Maintain that dividend payouts actively dictate share valuation and firm wealth. Key formulations: *Gordon's Model* and *Walter's Model.*
- **Theories of Irrelevance:** Posit that dividend distributions do not affect shareholder wealth or the cost of capital in perfect markets. Key formulation: *Modigliani-Miller (MM) Hypothesis.*

2.5.2 Prof. E. James Walter's Model

Definition 2.5.2: Walter's Model Foundations and Assumptions

Prof. E. James Walter argued that dividend policy is an inseparable investment decision. The relationship between the firm's internal rate of return (r) and its capitalization rate (k) determines value:

1. **Internal Equity Financing:** Retained earnings represent the sole financing source; no new equity or debt is issued.
2. **Constant Returns:** Return on investment r and cost of capital k remain constant over time.
3. **Infinite Horizon:** The firm has an infinite economic life.
4. **Constant Earnings and Dividends:** EPS and DPS are invariant in assessing baseline values.

Walter's Valuation Formula

The theoretical market price per share (P) under Walter's model is given by:

$$P = \frac{D + \frac{r}{k}(E - D)}{k} \tag{2.12}$$

where:

- P = Theoretical market price per share (MPS)
- D = Dividend per share (DPS)
- E = Earnings per share (EPS)
- r = Firm's internal rate of return on investment
- k = Cost of capital / capitalization rate
- $(E - D)$ = Retained earnings per share
- $\frac{r}{k}(E - D)$ = Capitalized market worth of retained earnings

Total Value of the Firm (V) with N outstanding equity shares is:

$$V = N \times P \tag{2.13}$$

Example 2.5.1: Walter's Model for a Growth Firm (Nirmal Ltd)

Nirmal Ltd has a cost of capital $k = 10\%$ and rate of return on investment $r = 18\%$. The company has 500,000 outstanding equity shares and an EPS = Rs. 20. Compute the market price per share and total value of the firm under Walter's model for: (a) No retention (0%), (b) 40% retention, (c) 80% retention.

Formula:

$$P = \frac{D + \frac{r}{k}(E - D)}{k}, \quad V = N \times P$$

Here $E = \text{Rs. } 20$, $r = 0.18$, $k = 0.10$, so $\frac{r}{k} = \frac{0.18}{0.10} = 1.80$.

(a) Retention 0% **(Payout** $100\%, D = 20$**):**

$$P = \frac{20 + 1.80(20 - 20)}{0.10} = \frac{20 + 0}{0.10} = \text{Rs. } 200.00$$

$$V = 500{,}000 \times 200 = \text{Rs. } 10{,}00{,}00{,}000 \quad (10 \text{ Crore})$$

(b) Retention 40% **(Payout** 60%, $D = 20 \times 0.60 = 12$**):**

$$P = \frac{12 + 1.80(20 - 12)}{0.10} = \frac{12 + 14.40}{0.10} = \frac{26.40}{0.10} = \text{Rs. } 264.00$$
$$V = 500{,}000 \times 264 = \text{Rs. } 13{,}20{,}00{,}000 \quad (13.2 \text{ Crore})$$

(c) Retention 80% **(Payout** 20%, $D = 20 \times 0.20 = 4$**):**

$$P = \frac{4 + 1.80(20 - 4)}{0.10} = \frac{4 + 28.80}{0.10} = \frac{32.80}{0.10} = \text{Rs. } 328.00$$
$$V = 500{,}000 \times 328 = \text{Rs. } 16{,}40{,}00{,}000 \quad (16.4 \text{ Crore})$$

Conclusion: Because $r > k$ (18% > 10%), Nirmal Ltd is a growth firm. As dividend payout increases, firm value diminishes. The optimal payout ratio is 0% (full retention).

Example 2.5.2: Walter's Model for a Normal Firm

The EPS of a company is Rs. 15, its rate of capitalization $k = 12\%$, and return on investment $r = 12\%$. Compute the market value per share under Walter's model for payout ratios of: (a) 20%, (b) 50%, (c) 70%. Which is the optimum payout ratio?

Formula:

$$P = \frac{D + \frac{r}{k}(E - D)}{k}$$

Since $r = k = 12\%$, the ratio $\frac{r}{k} = \frac{0.12}{0.12} = 1.00$. Hence:

$$P = \frac{D + 1(15 - D)}{0.12} = \frac{15}{0.12} = \text{Rs. } 125.00$$

(a) Payout 20% **(**$D = 15 \times 0.20 = $ **Rs.** 3.00**):** $P = \text{Rs. } 125.00$
(b) Payout 50% **(**$D = 15 \times 0.50 = $ **Rs.** 7.50**):** $P = \text{Rs. } 125.00$
(c) Payout 70% **(**$D = 15 \times 0.70 = $ **Rs.** 10.50**):** $P = \text{Rs. } 125.00$
Conclusion: Because $r = k$, share price remains constant at Rs. 125.00 across all payout ratios. Dividend policy is neutral; there is no unique optimum payout ratio.

Example 2.5.3: Walter's Model for a Declining Firm

The EPS of a firm is Rs. 12, its capitalization rate is $k = 15\%$, and rate of return on investment is $r = 9\%$. Compute the market price per share using Walter's model for dividend payout ratios of: (a) 25%, (b) 50%, (c) 100%. Which is the ideal payout ratio?

Formula:

$$P = \frac{D + \frac{r}{k}(E - D)}{k}$$

Given $E = \text{Rs. } 12$, $k = 0.15$, $r = 0.09$, so $\frac{r}{k} = \frac{0.09}{0.15} = 0.60$.
(a) Payout 25% **(**$D = 12 \times 0.25 = $ **Rs.** 3.00**):**

$$P = \frac{3 + 0.60(12 - 3)}{0.15} = \frac{3 + 5.40}{0.15} = \frac{8.40}{0.15} = \text{Rs. } 56.00$$

(b) Payout 50% ($D = 12 \times 0.50 =$ **Rs.** 6.00)**:**

$$P = \frac{6 + 0.60(12 - 6)}{0.15} = \frac{6 + 3.60}{0.15} = \frac{9.60}{0.15} = \text{Rs. } 64.00$$

(c) Payout 100% ($D = 12 \times 1.00 =$ **Rs.** 12.00)**:**

$$P = \frac{12 + 0.60(12 - 12)}{0.15} = \frac{12 + 0}{0.15} = \text{Rs. } 80.00$$

2.6 Modigliani–Miller (MM) Dividend Irrelevance Theory and Capital Budgeting Dynamics

Lecture 17/09/2026

Franco Modigliani and Merton Miller (1961) presented a seminal theoretical framework establishing that under conditions of perfect capital markets, a firm's dividend policy has **no effect** on its market valuation or share price. The market value of the firm is governed exclusively by its fundamental earning power and real investment policy, rather than how net earnings are split between distributed dividends and retained earnings.

2.6.1 Theoretical Foundations and Valuation Equations

The MM approach assumes that investors are rational and indifferent between current dividends and capital appreciation in a frictionless market. The current market price of an equity share (P_0) represents the discounted present value of the dividend received at the end of the operating period (D_1) plus the market price of the share prevailing at that time (P_1):

$$P_0 = \frac{D_1 + P_1}{1 + K_e} \tag{2.14}$$

Rearranging Equation (2.14) yields the market price of the share at the end of the period (P_1):

$$P_1 = P_0(1 + K_e) - D_1 \tag{2.15}$$

Where:

- P_0 = Current market price per share at the beginning of the period ($t = 0$).
- P_1 = Market price per share at the end of the period ($t = 1$).
- D_1 = Dividend per share declared/distributed at the end of the period.
- K_e = Rate of capitalization appropriate to the risk class of the firm.

2.6.2 Financing Investment Requirements and New Share Issuance

When a firm undertakes capital investment outlays (I), the required funding is financed first through available retained earnings, with any deficit financed via the issuance of new equity shares:

$$\text{Retained Earnings Available} = E - nD_1 \tag{2.16}$$

$$\text{Capital Deficit to be Raised } (\Delta M) = I - (E - nD_1) \tag{2.17}$$

$$\text{Number of New Shares Issued } (m = \Delta n) = \frac{\Delta M}{P_1} = \frac{I - (E - nD_1)}{P_1} \tag{2.18}$$

Where:

- I = Total capital investment budget required for the period.
- E = Total net income of the firm for the operating period.
- n = Number of existing shares outstanding at the beginning of the period.
- nD_1 = Aggregate dividend distribution to existing shareholders.
- ΔM = Additional equity capital required to be raised externally.
- $m = \Delta n$ = Number of newly issued shares.
- P_1 = Market issue price per share at the end of the period.

Accounting Statement Format for External Equity Financing

In corporate financial practice, the capital requirement and new equity issuance are determined using the standard accounting statement format shown in Table 2.1.

Particulars	**Details (Rs.)**	**Amount (Rs.)**
Investment Proposal Outlay (I)		xxx
Less: Retained Earnings Available for Investment:		
Total Net Income for the Period (E)	xxx	
Less: Total Dividend Distribution ($n \cdot D_1$)	(xxx)	
Retained Earnings Available ($E - n \cdot D_1$)		(xxx)
Amount to be Raised by Issue of New Shares (ΔM)		**xxx**
Number of New Shares to be Issued ($m = \Delta n$): $m = \Delta n = \frac{\text{Amount to be Raised by Issue of New Shares}}{\text{Issue Price per Share}} = \frac{\Delta M}{P_1} = \frac{I - (E - n \cdot D_1)}{P_1}$		

Table 2.1: Accounting Statement Layout for MM External Equity Financing

2.6.3 Proof of Total Market Value Invariance

The aggregate market value of the firm at the end of the period (V_1) is given by multiplying the total post-issue shares by the year-end market price:

$$V_1 = (n + m)P_1 \tag{2.19}$$

Substituting $mP_1 = I - E + nD_1$ from Equation (2.17) into Equation (2.19):

$$\begin{aligned} V_1 &= nP_1 + mP_1 \\ &= nP_1 + I - E + nD_1 \\ &= n(P_1 + D_1) + I - E \end{aligned} \tag{2.20}$$

From Equation (2.14), $P_1 + D_1 = P_0(1 + K_e)$. Substituting this gives:

$$V_1 = nP_0(1 + K_e) + I - E \tag{2.21}$$

Equation (2.21) contains neither D_1 nor the payout ratio. Consequently, total market value is invariant to dividend distributions: dividend payouts reduce internal funds, requiring compensatory new share issuance which dilutes future claims by exactly the dividend amount.

Example 2.6.1: Stewart Ltd: MM Dividend Irrelevance and Valuation

Stewart Ltd has 40,000 shares outstanding currently trading at Rs. 15 each. The board of directors has recommended Rs. 2 per share as dividend. The rate of capitalization appropriate to the firm's risk class is 20%.

1. Based on the MM approach, calculate the share price at the end of the year if dividend is: (a) distributed, (b) not declared.
2. Determine the number of new shares to be issued at year-end assuming net income is Rs. 1, 20, 000 and the investment budget is Rs. 2, 80, 000 under both cases.
3. Show that total market value of shares at year-end remains identical in both cases.
4. Is the MM approach realistic? What factors mar its empirical validity?

Given Parameters:

$$n = 40,000 \text{ shares}, \quad P_0 = \text{Rs. } 15, \quad K_e = 20\% = 0.20 \implies 1 + K_e = 1.20$$
$$E = \text{Rs. } 1,20,000, \quad I = \text{Rs. } 2,80,000$$

(i) Market Price of the Share at Year End (P_1):

$$P_0 = \frac{D_1 + P_1}{1 + K_e} \implies P_1 = P_0(1 + K_e) - D_1$$

- **(a) When Dividend is Distributed ($D_1 = \textbf{Rs. } 2$):**

$$15 = \frac{2 + P_1}{1.20} \implies 15 \times 1.20 = 2 + P_1 \implies 18 = 2 + P_1 \implies \mathbf{P_1} = \text{Rs. } \mathbf{16.00}$$

- **(b) When Dividend is Not Declared ($D_1 = \textbf{Rs. } 0$):**

$$15 = \frac{0 + P_1}{1.20} \implies 15 \times 1.20 = P_1 \implies \mathbf{P_1} = \text{Rs. } \mathbf{18.00}$$

(ii) Amount to be Raised and Number of New Shares Issued:

$$\text{Deficit } (\Delta M) = I - (E - nD_1), \qquad m = \frac{\Delta M}{P_1}$$

Particulars	**Dividend Distributed** ($D_1 = \text{Rs. } 2$)	**Dividend Not Declared** ($D_1 = \text{Rs. } 0$)
Capital Investment Outlay (I)	Rs. 2,80,000	Rs. 2,80,000
Less: Retained Earnings Available:		
Total Net Income (E)	Rs. 1,20,000	Rs. 1,20,000
Less: Total Dividend Paid (nD_1)	(Rs. 80,000)	(Rs. 0)
Retained Earnings Available for Investment	Rs. 40,000	Rs. 1,20,000
Amount to be Raised by New Issue (ΔM)	**Rs. 2,40,000**	**Rs. 1,60,000**
Issue Price per Share (P_1)	Rs. 16.00	Rs. 18.00
Number of New Shares Issued (m)	**15,000 shares** ($2,40,000/16$)	**8,888.89 shares** ($1,60,000/18 \approx 8,889$)

(iii) Proof of Total Market Value Invariance at Year End (V_1):

$$V_1 = (n + m) \times P_1$$

Particulars	Dividend Distributed	Dividend Not Declared
Existing Shares Outstanding (n)	40,000	40,000
(+) New Shares Issued (m)	15,000	8,888.89
Total Shares at Year End ($n + m$)	**55,000**	**48,888.89**
Market Price per Share (P_1)	Rs. 16.00	Rs. 18.00
Total Market Value of Shares (V_1)	**Rs. 8,80,000** $(55,000 \times \text{Rs. } 16)$	**Rs. 8,80,000** $(48,888.89 \times \text{Rs. } 18)$

Conclusion: Market value of equity at year-end remains exactly **Rs. 8,80,000** under both distribution policies, demonstrating complete dividend irrelevance.

(iv) Empirical Validity and Critical Limitations of the MM Approach:

The MM hypothesis is theoretical. In real-world capital markets, its validity is compromised by severe market frictions:

1. **Absence of Taxes:** MM assumes zero tax or identical taxation. In practice, personal income tax rates on dividends often exceed effective capital gains tax rates, creating tax-induced clienteles.
2. **Flotation and Transaction Costs:** Issuing new equity entails significant underwriting, legal, and administrative expenses. Thus, external equity is more costly than internal retained cash flows.
3. **Informational Asymmetry & Signaling:** In reality, dividend increases signal managerial confidence regarding sustainable future profitability, causing positive market price reactions.
4. **Financing and Investment Separation:** MM assumes real investment policy is predetermined and invariant to dividend choices, whereas capital rationing often forces firms to curb investments if cash is paid out as dividends.

2.7 Bond Valuation & Yield to Call (YTC)

Lecture 22/09/2026

Definition 2.7.1: Callable Bond & Yield to Call (YTC)

A **Callable Bond** grants the issuing corporation the contractual right (option) to repurchase (call back) and retire the debt obligation prior to its contractual maturity date at a predetermined call price (CP).

The **Yield to Call** (YTC) measures the annualized rate of return realized by an investor who purchases a callable bond at its current market price (P_0) and holds it until the company exercises its call option, receiving contractual periodic coupon payments plus a terminal call price that typically includes a call penalty fee.

Yield to Call (YTC) Formulations

a. Approximate Yield to Call Formula For rapid practical analysis, the annualized YTC is estimated using the standard average yield approximation:

$$\text{Approximate YTC} = \frac{C + \dfrac{CP - P_0}{N_c}}{\dfrac{CP + P_0}{2}} \tag{2.22}$$

where:

- C = Annual coupon payment (Coupon Rate × Par Value)
- CP = Call price paid at redemption = Par Value + Call Penalty Fee
- P_0 = Current purchase / market price of the bond
- N_c = Number of years remaining from purchase until the call date

b. Exact Discounted Cash Flow Equation The exact YTC is the internal rate of return equating the present value of coupon and call cash flows to the purchase price:

$$P_0 = \sum_{t=1}^{N_c} \frac{C}{(1+\text{YTC})^t} + \frac{CP}{(1+\text{YTC})^{N_c}} \tag{2.23}$$

2.7.1 Comparative Analysis: Yield to Maturity (YTM) vs. Yield to Call (YTC)

The primary distinction between YTM and YTC centers on the expected holding horizon and the terminal payoff structure:

Dimension	**Yield to Maturity** (YTM)	**Yield to Call** (YTC)
Holding Horizon	Investor holds the bond until official maturity date (N).	Shorter holding period terminating at designated call date ($N_c < N$).
Terminal Payout	Exact par face value (Par).	Elevated call price with call penalty fee ($CP >$ Par).
Discount Bond ($P_0 <$ Par)	Lower capital appreciation rate amortized across full horizon N.	YTC $\geq$ YTM — Capital discount is recovered over a shorter horizon N_c alongside penalty.
Premium Bond ($P_0 >$ Par)	Higher annualized yield distributed across longer tenure.	YTC $\leq$ YTM — Purchase premium is written off rapidly across shorter horizon N_c.

Table 2.2: Comparative Matrix: Yield to Maturity (YTM) vs. Yield to Call (YTC)

Lecture
22/09/2026

Example 2.7.1: Determination of Yield to Call (YTC) vs. Yield to Maturity (YTM)

Problem:
A bond with a par value of $1,000 and an annual coupon rate of 5% is currently trading at a 10% discount to its par value. The bond was purchased on 31/12/2023 with a contractual maturity date of 31/12/2033 (10-year maturity period).

The issuing corporation retains a call option to retire the bond early on 31/12/2027 (4 years after purchase) subject to a call penalty fee of 3%.

(a) Determine the current market purchase price (P_0) and the call price (CP).
(b) Calculate the approximate Yield to Call (YTC).
(c) Calculate the approximate Yield to Maturity (YTM) and verify the yield relationship.

Given Data:

- Par Face Value (Par) = $\$\,1{,}000$
- Current Market Price (P_0) = $\$\,1{,}000 \times (1 - 0.10) = \$\,900$
- Annual Coupon Rate = $5\% \implies C = 5\% \times \$\,1{,}000 = \$\,50$
- Maturity Tenure (N) = 10 years (from 31/12/2023 to 31/12/2033)
- Call Horizon (N_c) = 4 years (from 31/12/2023 to 31/12/2027)
- Call Penalty Fee = $3\% \times \$\,1{,}000 = \$\,30$
- Call Redemption Price (CP) = $\$\,1{,}000 + \$\,30 = \$\,1{,}030$

Calculations:

1. **Approximate Yield to Call (**YTC**):**

$$\text{Approximate YTC} = \frac{C + \dfrac{CP - P_0}{N_c}}{\dfrac{CP + P_0}{2}} = \frac{50 + \dfrac{1{,}030 - 900}{4}}{\dfrac{1{,}030 + 900}{2}}$$

$$= \frac{50 + \dfrac{130}{4}}{\dfrac{1{,}930}{2}} = \frac{50 + 32.50}{965} = \frac{82.50}{965} \approx \mathbf{0.08549} \quad (\mathbf{8.55}\%)$$

2. **Approximate Yield to Maturity (**YTM**):**

$$\text{Approximate YTM} = \frac{C + \dfrac{\text{Par} - P_0}{N}}{\dfrac{\text{Par} + P_0}{2}} = \frac{50 + \dfrac{1{,}000 - 900}{10}}{\dfrac{1{,}000 + 900}{2}}$$

$$= \frac{50 + \dfrac{100}{10}}{\dfrac{1{,}900}{2}} = \frac{50 + 10}{950} = \frac{60}{950} \approx \mathbf{0.06316} \quad (\mathbf{6.32}\%)$$

Conclusion & Analytical Interpretation:
Because the bond trades at a 10% discount to par value ($P_0 = \$\,900 < \$\,1{,}000$), the investor benefits from accelerated capital appreciation over the compressed 4-year call period plus the 3% penalty premium, yielding:

$$\text{YTC } (8.55\%) > \text{YTM } (6.32\%)$$

This validates the fundamental pricing theorem: *for a discount bond, the Yield to Call strictly exceeds the Yield to Maturity (*$\text{YTC} \geq \text{YTM}$*).*

2.8 Loan Amortization Models

Lecture
23/09/2026

Definition 2.8.1: Loan Amortization

Loan Amortization is the quantitative process of extinguishing a financial debt obligation over a predetermined horizon through a schedule of uniform periodic installments (A). Each periodic payment covers: (i) the interest accrued on the outstanding principal balance, and (ii) a partial repayment of the principal sum borrowed.

Loan Amortization Mathematical Mechanics

- **Equated Periodic Installment (A):** Derived from the present value of an ordinary annuity:
$$A = \frac{P_0}{\text{PVAF}(r,n)} = P_0 \left[\frac{r}{1-(1+r)^{-n}}\right] \tag{2.24}$$
where P_0 is the initial loan principal sum, r is the periodic interest rate, and n is the total number of repayment periods.
- **Periodic Interest Component (I_t):** Interest due in period t based on unamortized balance B_{t-1}:
$$I_t = B_{t-1} \times r \tag{2.25}$$
Because the loan balance B_{t-1} declines with each repayment, I_t **strictly decreases** monotonically over time.
- **Periodic Principal Repayment Component (ΔB_t):**
$$\Delta B_t = A - I_t = A - (B_{t-1} \times r) \tag{2.26}$$
Because total installment A is constant and I_t declines, ΔB_t **strictly increases** monotonically over time.
- **Remaining Unamortized Debt Balance (B_t):**
$$B_t = B_{t-1} - \Delta B_t = B_{t-1}(1+r) - A \implies B_n = 0 \tag{2.27}$$

2.8.1 Taxonomy of Amortization Loans

Corporate and retail credit structures employ four primary amortization archetypes:

1. **Fixed-Rate Fully Amortized Loans:** The contractual interest rate and periodic debt-service payment (A) remain constant across the entire horizon. Early payments consist primarily of interest, while later payments consist overwhelmingly of principal liquidation.

2. **Variable / Adjustable-Rate Loans (ARM):** The interest rate resets periodically based on benchmark interbank references (e.g. SOFR, Treasury yield). Periodic payment amounts adjust at specified intervals to preserve full amortization by maturity.

3. **Bullet / Balloon Loans:** Equal periodic payments cover interest obligations only (or minimal amortizing principal), leaving a substantial terminal lump sum (the "balloon balance") payable upon final contractual maturity.

4. **Negative Amortization Loans:** Periodic installment payments are set below the accrued periodic interest ($A < I_t$). The unserviced interest is capitalized into the loan principal base, causing total outstanding debt to grow ($B_t > B_{t-1}$) rather than amortize.

2.8.2 Strategic Importance & Corporate Advantages of Amortization

- **Predictable Payment Schedule:** Facilitates long-term corporate liquidity planning, cash flow forecasting, and working capital stability.
- **Reduced Financial Burden over Time:** Steady principal liquidation continuously curtails the interest-bearing debt base, reducing aggregate interest expense.
- **Credit Score & Debt Capacity Enhancement:** Consistent, timely amortization establishes verifiable creditworthiness and builds corporate borrowing headroom.
- **Debt Accumulation Avoidance:** Enforces disciplined capital retirement, preventing perpetual refinancing traps and liquidity crises at maturity.

2.9 Price-Earnings Valuation Model & Earnings Per Share

Lecture
08/10/2026

Definition 2.9.1: Price-Earnings Ratio (P/E Ratio) & EPS

The **Price-Earnings Ratio (P/E Ratio)** is a market-based equity valuation multiple expressing the proportional relationship between the current market price of an equity share and its underlying earnings per share (EPS). It reflects the valuation multiplier the capital market assigns to each rupee of earnings generated by the firm.

P/E Ratio and Earnings Per Share Formulations

a. Price-Earnings Ratio (P/E)

$$\text{P/E Ratio} = \frac{\text{Market Price per Equity Share (MPS)}}{\text{Earnings per Equity Share (EPS)}} \tag{2.28}$$

b. Earnings per Share (EPS) For an enterprise with ordinary equity capital:

$$\text{EPS} = \frac{\text{Net Income Available to Equity Holders}}{\text{Total Number of Outstanding Equity Shares}} \tag{2.29}$$

In corporate capital structures with preference share capital:

$$\text{EPS} = \frac{\text{Net Profit After Tax (NPAT)} - \text{Preference Dividend}}{\text{Number of Equity Shares Outstanding}} \tag{2.30}$$

2.9.1 Managerial Utility & Investor Interpretation of the P/E Ratio

- **Investment Decision Support:** Serves as a primary screening metric for individual and institutional investors to gauge whether an equity security is trading at an attractive earnings valuation relative to industry peers.
- **Comprehensive Success Metric:** The EPS reflects owner-centric profitability, while the P/E ratio synthesizes market perception of future growth prospects, business quality, and operational risk.
- **Valuation Multiple Interpretation:** A P/E ratio of k signifies that investors in the secondary market are willing to pay Rs. k in capital outlay for every Rs. 1 of current annual net earnings produced by the firm.

Lecture
08/10/2026

Example 2.9.1: Calculation and Economic Interpretation of P/E Ratio

Problem:
A company's equity shares are currently trading in the secondary market at Rs. 100 per share. The company reports an Earnings Per Share (EPS) of Rs. 5.00 for the financial year.

(a) Compute the company's Price-Earnings (P/E) ratio.
(b) State the economic meaning and practical conclusion of the computed multiple from an investor's perspective.

Given Data:

- Current Market Price per Share (MPS) = Rs. 100
- Earnings per Share (EPS) = Rs. 5

Calculation:

$$\text{P/E Ratio} = \frac{\text{MPS}}{\text{EPS}} = \frac{100}{5} = \mathbf{20}$$

Conclusion & Practical Interpretation:
The firm's Price-Earnings ratio is **20**. This indicates that investors are willing to pay Rs. **20** in market price for every Rs. **1** of annual net profit earned by the company. Alternatively, it reflects an underlying earnings capitalization yield of $\frac{1}{\text{P/E}} = \frac{1}{20} = 5\%$.

Chapter 3

Financial Ratios and Cash Flow Analysis

3.1 Fundamentals of Ratio Analysis & Classification

Lecture
12/10/2026

Definition 3.1.1: Financial Ratio & Ratio Analysis

A **Financial Ratio** is a quantitative mathematical expression establishing a proportional, fractional, or percentage relationship between two or more interrelated accounting figures extracted from corporate balance sheets, income statements, or cash flow reports.
Ratio Analysis is the diagnostic technique of standardizing and interpreting these numerical proportions to assess the operational efficiency, liquidity, leverage, solvency, and profitability of a business concern.

3.1.1 Managerial Purposes & Core Advantages of Ratio Analysis

Financial statement ratios serve seven strategic corporate functions:

1. **Financial Forecasting & Budgeting:** Historical ratio trajectories reveal structural growth rates, operating expense ratios, and asset utilization rhythms, facilitating predictive multi-year pro forma cash flow modeling.
2. **Managerial Control:** Acts as a continuous diagnostic monitoring instrument by highlighting negative variances between actual operational performance and benchmark targets.
3. **Facilitating Corporate Communication:** Distills complex, voluminous accounting transactions into standardized, intuitive indicator scores readily comprehended by board members, institutional investors, and lenders.
4. **Measuring Operational Efficiency:** Quantifies the rotation velocity and conversion efficiency with which assets and working capital are transformed into revenues.
5. **Investment Decision Support:** Equips equity analysts and shareholders with normalized valuation multiples (e.g. P/E, dividend yields, return on capital) to identify mispriced securities.
6. **Evaluating Financial Solvency:** Simultaneously examines short-term debt-servicing capacity (liquidity) and long-term capital structure stability (financial gearing and leverage).
7. **Inter-Firm & Intra-Firm Benchmarking:** Facilitates cross-sectional peer comparison against industry leaders as well as longitudinal trend analysis across consecutive operating cycles.

3.1.2 Functional Taxonomy: Classification of Ratios by Purpose

In corporate accounting, ratios are classified into three primary functional families based on the economic dimensions they diagnose:

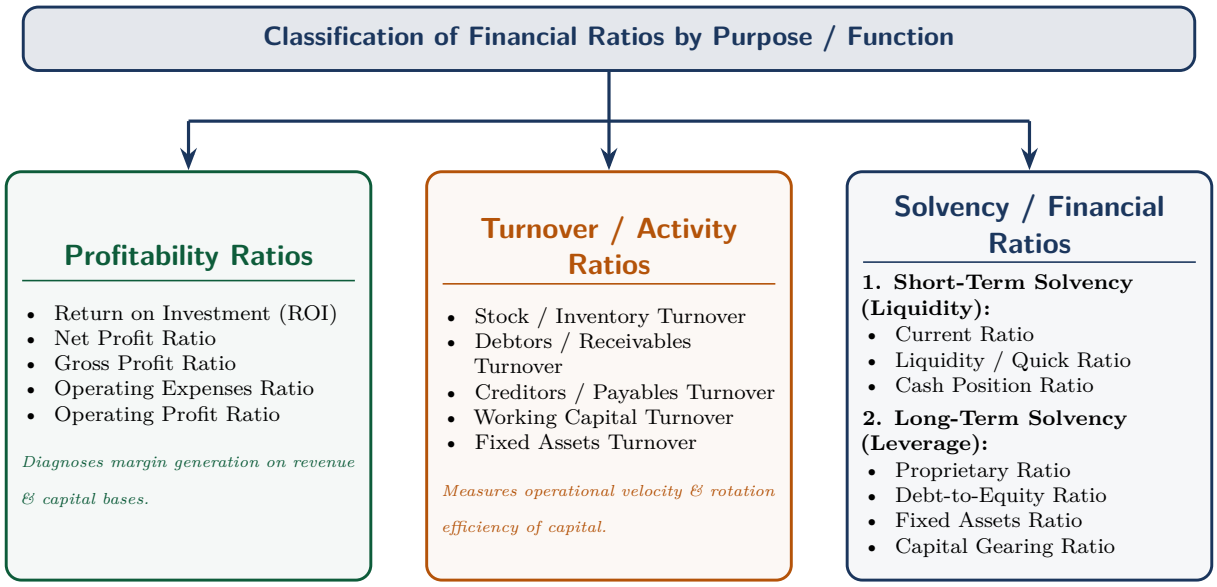


Figure 3.1: Taxonomy of Financial Ratios by Managerial Purpose and Analytical Function

3.1.3 Profitability Ratios: Formulation & Operational Margin Identities

Profitability Ratios measure an enterprise's ability to generate operational surplus, gross margin, and net earnings relative to revenue, total assets, or equity investment:

Profitability Ratios Formulations

- **Gross Profit Ratio:** Reflects margin after direct production costs (COGS):

$$\text{Gross Profit Ratio} = \frac{\text{Gross Profit}}{\text{Net Sales}} \times 100 \tag{3.1}$$

  where Net Sales = Total Sales − Sales Returns, and Gross Profit = Net Sales − COGS.
- **Operating Profit Ratio:** Measures operational profit after administrative, selling, and depreciation expenses:

$$\text{Operating Profit Ratio} = \frac{\text{Operating Profit}}{\text{Net Sales}} \times 100 \tag{3.2}$$

  where Operating Profit = Gross Profit − Operating Expenses = Net Operating Income.
- **Operating Ratio:** Measures aggregate cost efficiency (direct + operating expenses per rupee of sales):

$$\text{Operating Ratio} = \frac{\text{COGS} + \text{Operating Expenses}}{\text{Net Sales}} \times 100 \tag{3.3}$$

- **Fundamental Invariance Identity:**

$$\text{Operating Ratio} + \text{Operating Profit Ratio} = 100\% \tag{3.4}$$

- **Net Profit Ratio:** Final margin available to equity holders after non-operating expenses and taxes:

$$\text{Net Profit Ratio} = \frac{\text{Net Profit After Tax (NPAT)}}{\text{Net Sales}} \times 100 \tag{3.5}$$

Example 3.1.1: Comprehensive Evaluation of Profitability Ratios

Problem:
The following financial statement items relate to the trading and operating activities of a company for the year ended 31/12/1987:

Particulars	**Amount (Rs.)**	**Particulars**	**Amount (Rs.)**
Sales	1,00,000	Salary of Salesmen	1,800
Purchases	70,000	Advertising	700
Closing Stock	14,000	Travelling Expenses	500
Sales Returns	4,000	Salaries (Office)	3,000
Dividend Received	1,200	Rent	6,000
Profit on Sale of Fixed Assets	600	Stationery	200
Loss on Sale of Shares	300	Depreciation	1,000
Opening Stock	11,000	Other Expenses	2,000
		Provision for Tax	7,000

You are required to prepare the formal Trading and Operating Profit & Loss Account for the period, and compute: (i) Gross Profit Ratio, (ii) Operating Profit Ratio, (iii) Operating Ratio, and (iv) Net Profit Ratio.

Trading, Operating, and Profit & Loss Account for the Year Ended 31/12/1987:

Dr. Particulars	Amount (Rs.)	Cr. Particulars	Inner (Rs.)	Amount (Rs.)
To Opening Stock	11,000	By Sales	1,00,000	
To Purchases	70,000	*Less:* Sales Returns	(4,000)	96,000
To Gross Profit c/d	29,000	By Closing Stock		14,000
Total	**1,10,000**	**Total**		**1,10,000**
To Office Salaries	3,000	By Gross Profit b/d		29,000
To Rent	6,000			
To Stationery	200			
To Depreciation	1,000			
To Other Operating Expenses	2,000			
To Salesmen Salary	1,800			
To Advertising	700			
To Travelling Expenses	500			
To Operating Profit c/d	**13,800**			
Total	**29,000**	**Total**		**29,000**
To Loss on Sale of Shares	300	By Operating Profit b/d		13,800
To Provision for Tax	7,000	By Dividend Received		1,200
To Net Profit transferred to Equity	**8,300**	By Profit on Sale of Assets		600
Total	**15,600**	**Total**		**15,600**

Calculations of Profitability Ratios:

1. **Gross Profit Ratio:**

$$\text{Gross Profit Ratio} = \frac{\text{Gross Profit}}{\text{Net Sales}} \times 100 = \frac{29{,}000}{96{,}000} \times 100 = \mathbf{30.21}\%$$

Interpretation: The enterprise retains Rs. 30.21 from every Rs. 100 of net revenue after absorbing direct manufacturing and procurement costs (COGS).

2. **Operating Profit Ratio:**

$$\text{Operating Profit Ratio} = \frac{\text{Operating Profit}}{\text{Net Sales}} \times 100 = \frac{13{,}800}{96{,}000} \times 100 = \mathbf{14.38}\%$$

Interpretation: Pure core operational business activities yield an operating margin of 14.38% before considering capital gains, dividend windfalls, or income taxes.

3. **Operating Ratio:**

$$\begin{aligned}\text{COGS} &= \text{Opening Stock} + \text{Purchases} - \text{Closing Stock}\\ &= 11{,}000 + 70{,}000 - 14{,}000 = \text{Rs. } 67{,}000\\ \text{Operating Expenses} &= 3{,}000 + 6{,}000 + 200 + 1{,}000 + 2{,}000 + 1{,}800 + 700 + 500\\ &= \text{Rs. } 15{,}200\\ \text{Total Operating Cost} &= \text{COGS} + \text{Operating Expenses} = 67{,}000 + 15{,}200 = \text{Rs. } 82{,}200\end{aligned}$$

$$\text{Operating Ratio} = \frac{\text{Total Operating Cost}}{\text{Net Sales}} \times 100 = \frac{82{,}200}{96{,}000} \times 100 = \mathbf{85.63}\%$$

Verification of Invariance Identity:

$$\text{Operating Ratio} + \text{Operating Profit Ratio} = 85.63\% + 14.38\% = \mathbf{100.01}\% \approx \mathbf{100}\%$$

4. **Net Profit Ratio:**

$$\text{Net Profit Ratio} = \frac{\text{Net Profit After Tax}}{\text{Net Sales}} \times 100 = \frac{8{,}300}{96{,}000} \times 100 = \mathbf{8.65}\%$$

(If computed prior to non-operating adjustments as pure net operating income after tax: $\frac{13{,}800-7{,}000}{96{,}000} \times 100 = \frac{6{,}800}{96{,}000} \times 100 = 7.08\%$).

Final Results Summary:

- Gross Profit Ratio = **30.21**%
- Operating Profit Ratio = **14.38**%
- Operating Ratio = **85.63**%
- Net Profit Ratio = **8.65**% (or **7.81**% based on standard pre-dividend adjusted net line)

3.2 Turnover / Activity Ratios & Operational Velocity

Lecture
13/10/2026

Definition 3.2.1: Turnover Ratio & Operational Velocity

Turnover Ratios (also termed **Activity Ratios** or **Performance Ratios**) measure the operational efficiency and rotation velocity with which an enterprise converts its balance sheet assets and capital resources into revenue or cost of sales.
Unlike profitability metrics expressed as percentages, turnover ratios are mathematically expressed in **number of times** (rotation frequency per annum). A higher multiple generally signals superior managerial agility, efficient working capital deployment, and minimized idle capital.

Comprehensive Activity and Velocity Formulations

- **Inventory / Stock Turnover Ratio (ITR):**

$$\text{ITR} = \frac{\text{Cost of Goods Sold (COGS)}}{\text{Average Inventory}} \quad \text{or} \quad \frac{\text{Net Sales}}{\text{Average Inventory}} \tag{3.6}$$

where Average Inventory $= \dfrac{\text{Opening Stock} + \text{Closing Stock}}{2}$.

$$\text{Inventory Holding Period (Velocity)} = \frac{365}{\text{ITR}} \text{ days} = \frac{12}{\text{ITR}} \text{ months} \tag{3.7}$$

- **Debtors / Accounts Receivable Turnover Ratio (DTR):**

$$\text{DTR} = \frac{\text{Net Credit Sales}}{\text{Average Accounts Receivable}} \tag{3.8}$$

where Net Credit Sales = Credit Sales − Sales Returns, and Average Receivables = Debtors + Bills Receivable.

$$\text{Debtors Collection Period (Velocity)} = \frac{365}{\text{DTR}} \text{ days} = \frac{12}{\text{DTR}} \text{ months} \tag{3.9}$$

- **Creditors / Accounts Payable Turnover Ratio (CTR):**

$$\text{CTR} = \frac{\text{Net Credit Purchases}}{\text{Average Accounts Payable}} \tag{3.10}$$

where Net Credit Purchases = Credit Purchases − Purchase Returns, and Average Payables = Creditors + Bills Payable.

$$\text{Creditors Payment Period (Velocity)} = \frac{365}{\text{CTR}} \text{ days} = \frac{12}{\text{CTR}} \text{ months} \tag{3.11}$$

- **Working Capital Turnover Ratio:**

$$\text{Working Capital Turnover} = \frac{\text{Net Sales (or Cost of Sales)}}{\text{Net Working Capital}} \tag{3.12}$$

where Net Working Capital = Current Assets − Current Liabilities.

- **Fixed Assets Turnover Ratio:**

$$\text{Fixed Assets Turnover} = \frac{\text{COGS}}{\text{Net Fixed Assets}} \quad \text{or} \quad \frac{\text{Net Sales}}{\text{Net Fixed Assets}} \tag{3.13}$$

- **Capital Turnover Ratio:**

$$\text{Capital Turnover} = \frac{\text{Cost of Sales (or Net Sales)}}{\text{Capital Employed}} \tag{3.14}$$

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Example 3.2.1: Comprehensive Calculation of Turnover Ratios and Velocity Periods

Problem:
The accounting records of a manufacturing corporation disclose the following operating data for the financial year:

Particulars	Amount (Rs.)	Particulars	Amount (Rs.)
Total Sales	5,00,000	Cash Purchases	50,000
Cash Sales	40,000	Purchase Returns	10,000
Sales Returns	20,000	Creditors at the end	60,000
Debtors at the end	80,000	Bills Payable at the end	20,000
Bills Receivable at the end	20,000	Reserve on Discount on Creditors	2,000
Reserve for Doubtful Debts	1,000	Opening Stock	50,000
Total Purchases	3,00,000	Closing Stock	40,000
Gross Profit	1,00,000	Fixed Assets	10,00,000

From the above information, compute: (a) Stock Turnover Ratio, (b) Fixed Asset Turnover Ratio (under both Cost of Sales and Net Sales bases), (c) Debtors Turnover Ratio and Debtors Collection Period (in months and days), (d) Creditors Turnover Ratio and Creditors Payment Period (in months and days).

Reconciliation of Intermediate Financial Aggregates:

1. **Net Sales:**

$$\text{Net Sales} = \text{Total Sales} - \text{Sales Returns} = 5{,}00{,}000 - 20{,}000 = \text{Rs. } \mathbf{4{,}80{,}000}$$

2. **Cost of Goods Sold (Cost of Sales):**

$$\text{COGS} = \text{Net Sales} - \text{Gross Profit} = 4{,}80{,}000 - 1{,}00{,}000 = \text{Rs. } \mathbf{3{,}80{,}000}$$

3. **Average Stock:**

$$\text{Average Stock} = \frac{\text{Opening Stock} + \text{Closing Stock}}{2} = \frac{50{,}000 + 40{,}000}{2} = \text{Rs. } \mathbf{45{,}000}$$

4. **Net Credit Sales:**

$$\begin{aligned}\text{Net Credit Sales} &= \text{Total Sales} - \text{Cash Sales} - \text{Sales Returns} \\ &= 5{,}00{,}000 - 40{,}000 - 20{,}000 = \text{Rs. } \mathbf{4{,}40{,}000}\end{aligned}$$

5. **Average Accounts Receivable:** (Note: Reserves for doubtful debts are provisions and are not deducted from gross receivables in turnover velocity analysis):

$$\text{Accounts Receivable} = \text{Debtors} + \text{Bills Receivable} = 80{,}000 + 20{,}000 = \text{Rs. } \mathbf{1{,}00{,}000}$$

6. **Net Credit Purchases:**

$$\begin{aligned}\text{Net Credit Purchases} &= \text{Total Purchases} - \text{Cash Purchases} - \text{Purchase Returns} \\ &= 3{,}00{,}000 - 50{,}000 - 10{,}000 = \text{Rs. } \mathbf{2{,}40{,}000}\end{aligned}$$

7. **Average Accounts Payable:**

$$\text{Accounts Payable} = \text{Creditors} + \text{Bills Payable} = 60{,}000 + 20{,}000 = \text{Rs. } \mathbf{80{,}000}$$

Step-by-Step Ratio Computations:

1. **Stock Turnover Ratio:**

$$\text{Stock Turnover Ratio} = \frac{\text{COGS}}{\text{Average Stock}} = \frac{3{,}80{,}000}{45{,}000} = \mathbf{8.44} \text{ times}$$

(If evaluated on Net Sales basis: $\frac{4{,}80{,}000}{45{,}000} = 10.67$ times).

2. **Fixed Asset Turnover Ratio:**

- *Based on Cost of Sales:*

$$\text{Fixed Asset Turnover} = \frac{\text{COGS}}{\text{Fixed Assets}} = \frac{3{,}80{,}000}{10{,}00{,}000} = \mathbf{0.38} \text{ times}$$

- *Based on Net Sales:*

$$\text{Fixed Asset Turnover} = \frac{\text{Net Sales}}{\text{Fixed Assets}} = \frac{4{,}80{,}000}{10{,}00{,}000} = \mathbf{0.48} \text{ times}$$

3. **Debtors Turnover Ratio and Debtors Collection Velocity:**

$$\text{Debtors Turnover Ratio} = \frac{\text{Net Credit Sales}}{\text{Accounts Receivable}} = \frac{4{,}40{,}000}{1{,}00{,}000} = \mathbf{4.40} \text{ times}$$

$$\text{Collection Period (Months)} = \frac{12}{\text{Debtors Turnover}} = \frac{12}{4.40} = \mathbf{2.73} \text{ months}$$

$$\text{Collection Period (Days)} = \frac{365}{\text{Debtors Turnover}} = \frac{365}{4.40} = \mathbf{82.95} \approx \mathbf{83} \text{ days}$$

4. **Creditors Turnover Ratio and Creditors Payment Velocity:**

$$\text{Creditors Turnover Ratio} = \frac{\text{Net Credit Purchases}}{\text{Accounts Payable}} = \frac{2{,}40{,}000}{80{,}000} = \mathbf{3.00} \text{ times}$$

$$\text{Payment Period (Months)} = \frac{12}{\text{Creditors Turnover}} = \frac{12}{3.00} = \mathbf{4.00} \text{ months}$$

$$\text{Payment Period (Days)} = \frac{365}{\text{Creditors Turnover}} = \frac{365}{3.00} = \mathbf{121.67} \approx \mathbf{122} \text{ days}$$

Final Results Summary:

- Stock Turnover Ratio = **8.44** times
- Fixed Asset Turnover Ratio = **0.38** times (Cost of Sales) / **0.48** times (Net Sales)
- Debtors Turnover Ratio = **4.40** times (**2.73** months or **83** days)
- Creditors Turnover Ratio = **3.00** times (**4.00** months or **122** days)

3.3 Solvency Ratios: Liquidity & Financial Gearing

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Definition 3.3.1: Solvency and Liquidity Diagnostic Framework

Solvency denotes the overarching capacity of a business enterprise to fulfill its financial debt commitments as they mature. Solvency diagnostics operate across two interrelated temporal horizons:

1. **Short-Term Solvency (Liquidity):** Assesses the immediate and near-term ability of a firm to settle its current debt commitments, trade obligations, and operating expenses without impairing ongoing business operations.
2. **Long-Term Solvency (Financial Leverage):** Examines the structural adequacy of the firm's capital architecture to service long-term funded debt, periodic debenture interest charges, and preference dividend commitments over a multi-year horizon.

Short-Term Liquidity Ratios & Regulatory Norms

- **Current Ratio (Working Capital Ratio):**

$$\text{Current Ratio} = \frac{\text{Current Assets (CA)}}{\text{Current Liabilities (CL)}} \tag{3.15}$$

where CA denotes short-term resources (Inventories, Debtors, Cash, Bank, Bills Receivable, Prepaids, Marketable Securities), and CL denotes short-term obligations (Creditors, Bills Payable, Bank Overdraft, Outstanding Expenses, Short-Term Provisions).

Conventional Ideal Benchmark Norm: $2:1$

- **Quick Ratio / Liquid Ratio / Acid-Test Ratio:** Reflects the capacity to meet instant obligations without relying on inventory liquidations:

$$\text{Quick Ratio} = \frac{\text{Quick Assets (QA)}}{\text{CL}} = \frac{\text{CA} - \text{Inventories} - \text{Prepaid Expenses}}{\text{Current Liabilities}} \tag{3.16}$$

Conventional Ideal Benchmark Norm: $1:1$

- **Cash Position Ratio (Absolute Liquid Ratio):** The most rigorous measure of immediate cash liquidity, considering exclusively cash and near-cash balances:

$$\text{Cash Position Ratio} = \frac{\text{Cash} + \text{Bank} + \text{Current Marketable Investments}}{\text{Current Liabilities}} \tag{3.17}$$

Conventional Ideal Benchmark Norm: $0.75:1 \quad (\text{or } 0.5:1)$

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Example 3.3.1: Comprehensive Evaluation of Liquidity and Short-Term Solvency

Problem:
The Balance Sheet and Profit and Loss Account of a manufacturing corporation as on 31/12/1997 disclose the following financial records:

Balance Sheet as on 31/12/1997			
Liabilities	**Amount (Rs.)**	**Assets**	**Amount (Rs.)**
Issued Capital (40,000 shares of Rs. 100 each)	40,00,000	Land and Building	30,00,000
Reserves	18,00,000	Plant and Machinery	16,00,000
Profit and Loss Account	6,00,000	Stock	29,60,000
6% Debentures	6,00,000	Debtors	14,20,000
Creditors	26,00,000	Cash at Bank	6,20,000
Total	**96,00,000**	**Total**	**96,00,000**

You are required to evaluate: (a) Current Ratio, (b) Quick Ratio (Acid-Test Ratio), and (c) Cash Position Ratio. Interpret the liquidity standing of the company against standard corporate norms.

Step 1: Itemized Aggregation of Balance Sheet Components:

1. **Current Assets (CA):**

$$\begin{aligned}\text{Current Assets} &= \text{Stock} + \text{Debtors} + \text{Cash at Bank} \\ &= 29{,}60{,}000 + 14{,}20{,}000 + 6{,}20{,}000 = \text{Rs. } \mathbf{50{,}00{,}000}\end{aligned}$$

2. **Current Liabilities (CL):**

$$\text{Current Liabilities} = \text{Creditors} = \text{Rs. } \mathbf{26{,}00{,}000}$$

3. **Quick Assets (QA):**

$$\begin{aligned}\text{Quick Assets} &= \text{Current Assets} - \text{Stock} \\ &= 50{,}00{,}000 - 29{,}60{,}000 = \text{Rs. } \mathbf{20{,}40{,}000} \\ \text{Alternatively:} \quad \text{QA} &= \text{Debtors} + \text{Cash at Bank} \\ &= 14{,}20{,}000 + 6{,}20{,}000 = \text{Rs. } \mathbf{20{,}40{,}000}\end{aligned}$$

4. **Absolute Cash Assets:**

$$\text{Cash Assets} = \text{Cash at Bank} = \text{Rs. } \mathbf{6{,}20{,}000}$$

Step 2: Computation of Liquidity Ratios:

1. **Current Ratio:**

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}} = \frac{50{,}00{,}000}{26{,}00{,}000} = \mathbf{1.92 : 1}$$

2. **Quick Ratio (Acid-Test Ratio):**

$$\text{Quick Ratio} = \frac{\text{Quick Assets}}{\text{Current Liabilities}} = \frac{20{,}40{,}000}{26{,}00{,}000} = \mathbf{0.78 : 1}$$

3. **Cash Position Ratio:**

$$\text{Cash Position Ratio} = \frac{\text{Cash at Bank}}{\text{Current Liabilities}} = \frac{6{,}20{,}000}{26{,}00{,}000} = \mathbf{0.24 : 1}$$

Managerial Interpretation & Diagnostic Summary:

- **Current Ratio** ($1.92:1$ **vs Norm** $2.00:1$)**:** The enterprise maintains Rs. 1.92 of short-term assets for every Rs. 1.00 of current debt. While superficially close to the traditional $2:1$ rule of thumb, it conceals underlying compositional risk.
- **Quick Ratio** ($0.78:1$ **vs Norm** $1.00:1$)**:** Liquid asset cover falls significantly below the $1:1$ benchmark. Specifically, inventories account for $\frac{29,60,000}{50,00,000} = 59.2\%$ of total current assets.
- **Cash Position Ratio** ($0.24:1$ **vs Norm** $0.75:1$)**:** Immediate cash coverage of 0.24 signals vulnerability to sudden supplier payment calls or working capital shocks if stock realization slows down.

3.4 Fund Flow & Cash Flow Statement Analysis

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Definition 3.4.1: Financial Flow Statements: Funds Flow vs. Cash Flow

Financial statement flow analysis examines the dynamic transmission of financial resources across consecutive accounting periods:

1. **Funds Flow Statement:** A diagnostic statement portraying movements in the firm's **Net Working Capital** (NWC = Current Assets − Current Liabilities). It explains how long-term resources (capital, debt borrowings, operational profits) were generated (Sources) and deployed into fixed capital or debt retirement (Applications).
2. **Cash Flow Statement:** A specialized report portraying the gross inflows and outflows of **Cash and Cash Equivalents** (currency, demand deposits, short-term marketable instruments) arising from Operating, Investing, and Financing activities.

3.4.1 Principles Governing Working Capital and Cash Movements

The operational mechanics of working capital and cash adjustments follow complementary accounting principles:

Accounting Transaction / Movement	Impact on Working Capital	Impact on Cash Position
Increase in Current Asset (↑ CA)	Increase in Working Capital (+)	Cash Outflow / Applied (−)
Decrease in Current Asset (↓ CA)	Decrease in Working Capital (−)	Cash Inflow / Released (+)
Increase in Current Liability (↑ CL)	Decrease in Working Capital (−)	Cash Inflow / Source (+)
Decrease in Current Liability (↓ CL)	Increase in Working Capital (+)	Cash Outflow / Applied (−)

Operational Cash Flow & Funds Flow Identities

- **Funds from Operations (FFO):**

$$\begin{aligned}\text{FFO} = \text{Net Profit} + \text{Non-Cash Expenses (Depreciation, Amortization)} \\ + \text{Non-Operating Losses} - \text{Non-Operating Incomes}\end{aligned} \tag{3.18}$$

- **Cash Flow from Operations (CFO):**

$$\text{CFO} = \text{FFO} + \sum(\uparrow \text{CL}) + \sum(\downarrow \text{CA}) - \sum(\downarrow \text{CL}) - \sum(\uparrow \text{CA}) \tag{3.19}$$

- **Net Working Capital Identity:**

$$\Delta\,\text{NWC} = \Delta\text{Current Assets} - \Delta\text{Current Liabilities} \tag{3.20}$$

- **Funds Flow Fundamental Balance Equation:**

$$\text{Total Sources of Funds} - \text{Total Applications of Funds} = \Delta\,\text{NWC} \tag{3.21}$$

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Example 3.4.1: Computation of Cash Flow from Operations

Problem:
The records of an industrial enterprise disclose that the **Funds from Operations** for the financial year ended 31/12/1998 amounted to Rs. 84,000. The balances of Current Assets and Current Liabilities on 01/01/1998 and 31/12/1998 were as follows:

Balance Sheet Item	**01/01/1998 (Rs.)**	**31/12/1998 (Rs.)**
Trade Creditors	1,82,000	1,94,000
Trade Debtors	2,75,000	3,15,000
Bills Receivable	40,000	35,000
Bills Payable	27,000	31,000
Inventories	1,85,000	1,70,000
Trade Investments (Short-term)	40,000	70,000
Outstanding Expenses	20,000	25,000
Prepaid Expenses	5,000	8,000

You are required to prepare a formal Statement of Cash Flow from Operating Activities.

Statement of Cash from Operating Activities for the Year Ended 31/12/1998:

Particulars	Inner (Rs.)	Amount (Rs.)
Funds from Operations (as given)		**84,000**
Add: Increases in Current Liabilities & Decreases in Current Assets:		
Increase in Trade Creditors (1,94,000 − 1,82,000)	12,000	
Increase in Bills Payable (31,000 − 27,000)	4,000	
Increase in Outstanding Expenses (25,000 − 20,000)	5,000	
Decrease in Bills Receivable (40,000 − 35,000)	5,000	
Decrease in Inventories (1,85,000 − 1,70,000)	15,000	41,000
Subtotal		**1,25,000**
Less: Decreases in Current Liabilities & Increases in Current Assets:		
Increase in Trade Debtors (3,15,000 − 2,75,000)	(40,000)	
Increase in Trade Investments (70,000 − 40,000)	(30,000)	
Increase in Prepaid Expenses (8,000 − 5,000)	(3,000)	(73,000)
Net Cash Flow Generated from Operations (CFO)		**Rs. 52,000**

Analytical Interpretation:
Although the enterprise generated Rs. 84,000 of operational surplus before working capital movements, Rs. 73,000 of liquidity was absorbed into expanded customer receivables, short-term trade investments, and prepaid costs. Offsetting inflows of Rs. 41,000 from supplier credit expansion and inventory liquidation resulted in a net operational cash generation of Rs. **52,000**.

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Example 3.4.2: Comprehensive Funds Flow Statement Formulation

Problem:
The following comparative Balance Sheets relate to the business concern of Mr. Rajan (Sole Trader) for the years 1997 and 1998:

Liabilities	1997 (Rs.)	1998 (Rs.)	Assets	1997 (Rs.)	1998 (Rs.)
Creditors	18,000	20,500	Cash	2,000	1,800
Bank Loan (Long-term)	15,000	22,500	Debtors	17,500	19,200
Rajan's Capital	74,000	74,500	Stock	12,500	11,000
			Land	10,000	15,000
			Building	25,000	27,500
			Machinery	40,000	43,000
Total	**1,07,000**	**1,17,500**	**Total**	**1,07,000**	**1,17,500**

Additional Financial Information for 1998: (i) Drawings by the proprietor Mr. Rajan for personal use amounted to Rs. 13,000. (ii) Provision for depreciation on machinery stood at Rs. 13,500 on 01/01/1998 and at Rs. 18,000 on 31/12/1998. You are required to prepare: (a) Schedule of Changes in Working Capital, (b) Supporting Working Notes, and (c) Funds Flow Statement for 1998.

Step 1: Statement of Changes in Working Capital:

Particulars	1997 (Rs.)	1998 (Rs.)	Increase in WC	Decrease in WC
Current Assets (CA):				
Cash	2,000	1,800	—	200
Debtors	17,500	19,200	1,700	—
Stock	12,500	11,000	—	1,500
Total Current Assets (A)	**32,000**	**32,000**		
Current Liabilities (CL):				
Creditors	18,000	20,500	—	2,500
Total Current Liabilities (B)	**18,000**	**20,500**		
Net Working Capital (A − B)	**14,000**	**11,500**		
Net Decrease in Working Capital		**2,500**	2,500	—
Total	**14,000**	**14,000**	**4,200**	**4,200**

Step 2: Supporting Accounting Working Notes:

1. **Working Note 1: Computation of Net Profit & Funds from Operations (FFO):**

Closing Capital (31/12/1998)	= Rs. 74,500
Add: Drawings during the year	= Rs. 13,000
Adjusted Capital Total	= Rs. 87,500
Less: Opening Capital (01/01/1998)	= (Rs. 74,000)
Net Profit for the Year	= Rs. **13,500**
Add: Non-Cash Depreciation on Machinery (18,000 − 13,500)	= Rs. 4,500
Funds from Operations (FFO)	= Rs. **18,000**

2. **Working Note 2: Machinery Ledger Account (at Cost):**

Dr. Particulars	**Amount (Rs.)**	**Cr. Particulars**	**Amount (Rs.)**
To Balance b/d	40,000	By Depreciation for year	4,500
To Bank (Purchases balancing figure)	**7,500**	By Balance c/d	43,000
Total	**47,500**	**Total**	**47,500**

Step 3: Statement of Sources and Applications of Funds (Funds Flow Statement):

Sources of Funds	**Amount (Rs.)**	**Applications of Funds**	**Amount (Rs.)**
Funds from Operations (WN 1)	18,000	Purchase of Land (15,000 − 10,000)	5,000
Long-term Bank Loan Raised	7,500	Purchase of Building (27,500 − 25,000)	2,500
(22,500 − 15,000)		Purchase of Machinery (WN 2)	7,500
		Drawings of Proprietor	13,000
Total Sources (A)	**25,500**	**Total Applications (B)**	**28,000**
Net Decrease in Working Capital (B − A)			**Rs. 2,500**
Reconciled Total	**28,000**	**Total**	**28,000**

3.5 Cost-Volume-Profit (CVP) and Break-Even Analysis

Lecture
26/10/2026

Definition 3.5.1: Cost-Volume-Profit Analysis & Marginal Costing

Cost-Volume-Profit (CVP) Analysis is a quantitative managerial economics technique that evaluates how dynamic variations in operating costs (both fixed and variable components) and production/sales volume interact to determine operational profit and economic feasibility.

3.5.1 Cost Architecture: Fixed vs. Variable Elements

- **Fixed Costs (Period Costs / Time Costs):** Operating expenditures that remain structurally invariant in total sum regardless of production volume fluctuations within the relevant operating capacity (e.g. factory rent, executive salaries, property taxes, straight-line depreciation). On a per-unit basis, fixed cost declines asymptotically as volume increases.
- **Variable Costs (Product Costs / Marginal Costs):** Operational expenditures that vary in direct linear proportion to aggregate output volume (e.g. direct raw materials, piece-rate direct labor, sales commissions, direct power). On a per-unit basis, variable cost remains constant.

3.5.2 Fundamental CVP Formulations & Identity Mechanics

Comprehensive CVP and Break-Even Formulas

- **Contribution Margin (C):**

$$C = \text{Sales Revenue } (S) - \text{Variable Cost } (V) \tag{3.22}$$

$$C = \text{Fixed Cost } (F) + \text{Profit } (P) \tag{3.23}$$

$$C_{\text{unit}} = s - v = \text{Selling Price per Unit} - \text{Variable Cost per Unit} \tag{3.24}$$

- **Profit-Volume Ratio (P/V Ratio):** Expresses the percentage of sales revenue remaining after variable cost coverage:

$$\text{P/V Ratio} = \frac{C}{S} \times 100 = \frac{S - V}{S} \times 100 = \frac{F + P}{S} \times 100 = \frac{\Delta\text{Profit}}{\Delta\text{Sales}} \times 100 \tag{3.25}$$

- **Break-Even Point (BEP):** The unique operational equilibrium volume where total sales revenue equals total operating costs (Total Profit = 0):

$$\text{BEP (in Output Units)} = \frac{\text{Fixed Cost } (F)}{\text{Contribution per Unit } (s - v)} = \frac{F}{C_{\text{unit}}} \tag{3.26}$$

$$\text{BEP (in Sales Value)} = \frac{\text{Fixed Cost } (F)}{\text{P/V Ratio}} = \text{BEP (Units)} \times s \tag{3.27}$$

- **Break-Even Ratio:**

$$\text{Break-Even Ratio} = \frac{\text{Break-Even Sales}}{\text{Actual / Budgeted Sales}} \times 100 \tag{3.28}$$

- **Margin of Safety (MOS):** The cushion of actual sales volume exceeding the break-even threshold:

$$\text{MOS} = \text{Actual Sales} - \text{Break-Even Sales} = \frac{\text{Profit}}{\text{P/V Ratio}} \tag{3.29}$$

$$\text{MOS Ratio} = \frac{\text{MOS}}{\text{Actual Sales}} \times 100 = 100\% - \text{Break-Even Ratio} \tag{3.30}$$

- **Angle of Incidence (θ):** The geometric angle formed at the intersection of the Total Revenue line ($y = s \cdot x$) and Total Cost line ($y = F + v \cdot x$) on the Break-Even Chart. A wider angle indicates a superior P/V ratio and accelerated profit growth beyond the BEP.

3.5.3 Managerial Advantages and Analytical Limitations

Strategic Advantages of Break-Even Analysis

1. **Cost Structure Delineation:** Precisely delineates total operating expenditure into fixed overheads and marginal components.
2. **Threshold Volume Ascertainment:** Identifies the critical minimum sales output (in units and rupees) required to avert operational losses.
3. **Decision-Making Optimization:** Facilitates pricing strategy, make-or-buy evaluations, and tender quoting through CVP simulation.
4. **Inter-Firm Competitiveness Benchmarking:** Evaluates operating leverage and resilience against peer entities across business cycles.
5. **Profit Forecasting & Financial Planning:** Models target sales volume needed to generate pre-determined profit objectives: $S_{\text{target}} = \frac{F + P_{\text{target}}}{\text{P/V Ratio}}$.
6. **Product Portfolio Optimization:** Evaluates multi-division product mixes to maximize aggregate contribution margins.
7. **Cost Control & Budgetary Benchmarking:** Establishes rigorous cost-absorption variance thresholds.
8. **Corporate Surplus Aggregation:** Synthesizes divisional contributions to establish corporate earnings after central fixed overhead deduction.

Critical Theoretical & Practical Limitations

1. **Imperfect Cost Segregation:** In practice, semi-variable and stepped-fixed costs fluctuate non-linearly, particularly beyond 100% operating capacity.
2. **Price Elasticity Violations:** Assumes a constant unit selling price regardless of volume, ignoring commercial price discounts and downward-sloping demand curves.
3. **Detailed Cost Masking:** Aggregated break-even models obscure granular departmental efficiency variations.
4. **Inventory Valuation Distortion:** Marginal costing ignores the absorption of fixed overheads into opening and closing stock reserves.
5. **Multi-Product Mix Instability:** Assumes a rigid, constant product sales mix; shifts in relative consumer demand distort composite break-even coordinates.
6. **Exclusion of Capital Structure Dynamics:** Fails to account for return on invested capital, debt leverage charges, and working capital carrying costs.
7. **Static Environmental Assumptions:** Becomes highly unreliable under volatile macroeconomic conditions, input inflation, or technological disruption.

Example 3.5.1: Break-Even Formulation and Visual Graphic Modeling for A Ltd

Problem:
A Ltd. has furnished the following operating data:

- Fixed Cost (F) = Rs. 80,000
- Variable Cost per unit (v) = Rs. 4
- Estimated Sales Value (S) = Rs. 2,00,000
- Selling Price per unit (s) = Rs. 20

You are required to: (a) Determine Contribution per unit and total, (b) Calculate the P/V Ratio, (c) Determine Break-Even Point in output units and monetary sales value, (d) Calculate the Margin of Safety, (e) Analyze the Angle of Incidence, and (f) Construct the formal Break-Even Chart.

Step 1: Intermediate Operational Calculations:

1. **Estimated Output Volume:**

$$\text{Output Units } (Q) = \frac{\text{Estimated Sales Value}}{\text{Selling Price per Unit}} = \frac{2{,}00{,}000}{20} = \mathbf{10{,}000} \text{ units}$$

2. **Contribution Margin:**

$$C_{\text{unit}} = s - v = 20 - 4 = \text{Rs. } \mathbf{16} \text{ per unit}$$
$$\text{Total Contribution} = Q \times C_{\text{unit}} = 10{,}000 \times 16 = \text{Rs. } \mathbf{1{,}60{,}000}$$

3. **Profit-Volume Ratio (P/V Ratio):**

$$\text{P/V Ratio} = \frac{C_{\text{unit}}}{s} \times 100 = \frac{16}{20} \times 100 = \mathbf{80}\%$$

4. **Break-Even Point (BEP):**

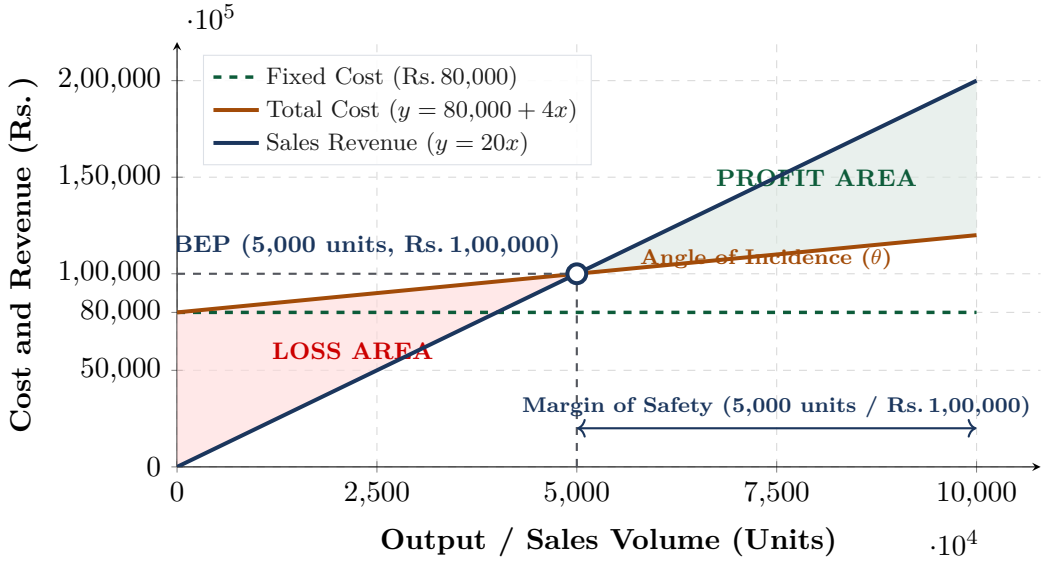
$$\text{BEP (in Units)} = \frac{\text{Fixed Cost}}{C_{\text{unit}}} = \frac{80{,}000}{16} = \mathbf{5{,}000} \text{ units}$$
$$\text{BEP (in Sales Value)} = \frac{\text{Fixed Cost}}{\text{P/V Ratio}} = \frac{80{,}000}{0.80} = \text{Rs. } \mathbf{1{,}00{,}000}$$
$$\text{Alternatively:} \quad \text{BEP (Value)} = 5{,}000 \times 20 = \text{Rs. } \mathbf{1{,}00{,}000}$$

5. **Margin of Safety (MOS):**

$$\begin{aligned}\text{MOS(Value)} &= \text{Actual Sales} - \text{BEP (Sales)} \\ &= 2{,}00{,}000 - 1{,}00{,}000 = \text{Rs. } \mathbf{1{,}00{,}000}\end{aligned}$$
$$\text{MOS(Units)} = 10{,}000 - 5{,}000 = \mathbf{5{,}000} \text{ units}$$
$$\text{MOS Ratio} = \frac{1{,}00{,}000}{2{,}00{,}000} \times 100 = \mathbf{50}\%$$

6. **Angle of Incidence Analysis:** At the BEP (5,000 units, Rs. 1,00,000), the sales revenue curve ($m_S = 20$) intersects the total cost curve ($m_{\text{TC}} = 4$). The slope differential $\Delta m = 20 - 4 = 16$ creates a steep angle of incidence ($\theta = \arctan(16)$), indicating aggressive profit expansion once operations cross 5,000 units.

Step 2: Graphic Vector Break-Even Chart Representation:



Step 3: Managerial Synthesis and Operational Diagnostics:

1. **High Contribution Velocity (80% P/V Ratio):** With a variable cost ratio of only 20% ($v/s = 4/20$), every incremental rupee of sales beyond the break-even threshold contributes Rs. 0.80 directly toward operating net profit, reflecting powerful operating leverage.

2. **Robust Safety Buffer (50% Margin of Safety):** The enterprise can absorb up to a 50% contraction in sales demand (from 10,000 units down to 5,000 units, or a decline of Rs. 1,00,000 in turnover) before incurring an operational net loss.

3. **Steep Angle of Incidence (θ):** The wide slope divergence between the sales revenue curve ($y = 20x$) and the total cost curve ($y = 80{,}000 + 4x$) confirms that profitability accelerates aggressively once fixed overheads (Rs. 80,000) are fully amortized.

4. **Downside Risk Mitigation:** Because fixed overhead represents only 40% of budgeted revenue (80,000/2,00,000), the enterprise maintains substantial operational agility during cyclical market downturns.

Example 3.5.2: Contribution Break-Even Modeling and Sensitivity Analysis

Problem:
A manufacturing enterprise provides the following budgeted operational estimates:

- Budgeted Output Capacity = 80,000 units
- Fixed Operating Expenses = Rs. 4,00,000
- Variable Expenses per unit = Rs. 10
- Selling Price per unit = Rs. 20

You are required to: (a) Calculate Contribution per unit and P/V Ratio, (b) Determine Break-Even Point in units and in sales value, (c) Compute Operating Profit at budgeted capacity, (d) Construct the Contribution Break-Even Chart, and (e) Conduct a Sensitivity Analysis for input cost and price fluctuations.

Step 1: Fundamental Mathematical Computations:

1. **Unit Contribution & P/V Ratio:**

$$C_{\text{unit}} = s - v = 20 - 10 = \text{Rs. } \mathbf{10} \text{ per unit}$$
$$\text{P/V Ratio} = \frac{C_{\text{unit}}}{s} \times 100 = \frac{10}{20} \times 100 = \mathbf{50}\%$$

2. **Break-Even Point (BEP):**

$$\text{BEP (in Units)} = \frac{\text{Fixed Expenses}}{C_{\text{unit}}} = \frac{4{,}00{,}000}{10} = \mathbf{40{,}000} \text{ units}$$
$$\text{BEP (in Sales Value)} = \frac{\text{Fixed Expenses}}{\text{P/V Ratio}} = \frac{4{,}00{,}000}{0.50} = \text{Rs. } \mathbf{8{,}00{,}000}$$

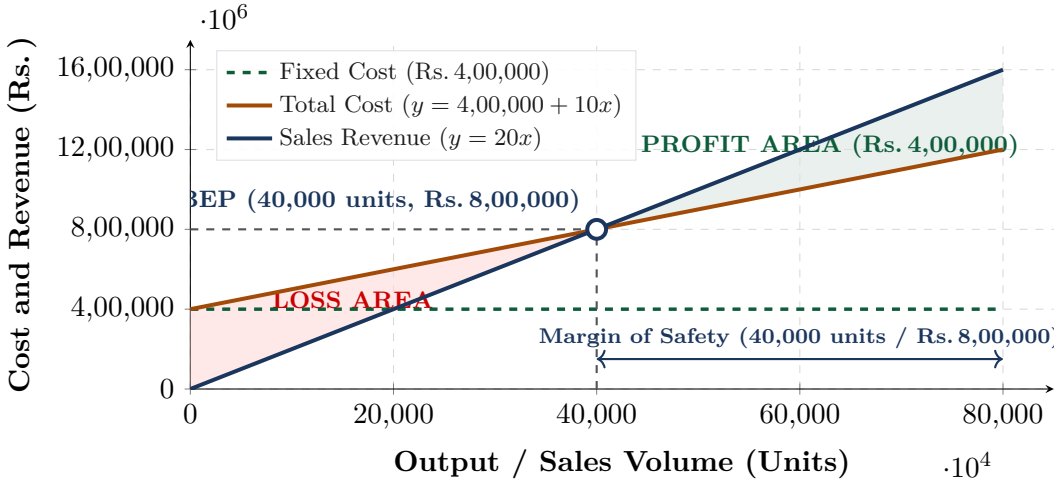
3. **Operating Performance at Budgeted Output (80,000 units):**

$$\begin{aligned} \text{Gross Revenue } (80{,}000 \times \text{Rs. } 20) &= \text{Rs. } 16{,}00{,}000 \\ \text{Less: Variable Expenses } (80{,}000 \times \text{Rs. } 10) &= (\text{Rs. } 8{,}00{,}000) \\ \text{Total Contribution Margin} &= \text{Rs. } 8{,}00{,}000 \\ \text{Less: Fixed Operating Expenses} &= (\text{Rs. } 4{,}00{,}000) \\ \hline \mathbf{Budgeted\ Operating\ Profit} &= \text{Rs. } \mathbf{4{,}00{,}000} \end{aligned}$$

4. **Margin of Safety at Budgeted Output:**

$$\text{MOS(Value)} = 16{,}00{,}000 - 8{,}00{,}000 = \text{Rs. } \mathbf{8{,}00{,}000} \quad (40{,}000 \text{ units})$$
$$\text{MOS Ratio} = \frac{8{,}00{,}000}{16{,}00{,}000} \times 100 = \mathbf{50}\%$$

Step 2: Graphic Vector Contribution Break-Even Chart:



Step 3: Sensitivity Analysis of Operational Variables:
Sensitivity Analysis evaluates the quantitative responsiveness of profitability and break-even equilibrium to independent shifts in key operational parameters:

1. **Scenario A: 10% Increase in Selling Price ($s \to$ Rs. 22):**
   $C_{\text{unit}} = 22 - 10 = \text{Rs. } 12 \implies \text{BEP} = \frac{4{,}00{,}000}{12} = \mathbf{33{,}333}$ units ($\downarrow 16.7\%$).
   Operating profit at 80,000 units surges to $(80{,}000 \times 12) - 4{,}00{,}000 = \text{Rs. } \mathbf{5{,}60{,}000}$ ($\uparrow 40\%$).
2. **Scenario B: 10% Increase in Variable Cost ($v \to$ Rs. 11):**
   $C_{\text{unit}} = 20 - 11 = \text{Rs. } 9 \implies \text{BEP} = \frac{4{,}00{,}000}{9} = \mathbf{44{,}444}$ units ($\uparrow 11.1\%$).
   Operating profit at 80,000 units declines to $(80{,}000 \times 9) - 4{,}00{,}000 = \text{Rs. } \mathbf{3{,}20{,}000}$ ($\downarrow 20\%$).
3. **Scenario C: 10% Increase in Fixed Overhead ($F \to$ Rs. 4,40,000):**
   $\text{BEP} = \frac{4{,}40{,}000}{10} = \mathbf{44{,}000}$ units ($\uparrow 10\%$).
   Operating profit declines to $8{,}00{,}000 - 4{,}40{,}000 = \text{Rs. } \mathbf{3{,}60{,}000}$ ($\downarrow 10\%$).

Conclusion: Profitability is most elastic with respect to unit selling price adjustments, followed by variable input cost control.

Chapter 4

Risk, Return, and Portfolio Mathematics

This chapter establishes the statistical, probabilistic, and quantitative foundations of investment analysis, asset pricing, and portfolio theory in accordance with Unit IV of the syllabus. We examine the fundamental trade-off between financial risk and return, explore classical measures of central tendency (arithmetic, geometric, and harmonic means), formalize absolute and relative dispersion techniques (variance, standard deviation, and coefficient of variation), analyze bivariate market co-movement through Pearson correlation, and introduce the quantitative foundations of foreign exchange exposure.

4.1 Financial Risk and Return Axioms & Measures of Central Tendency

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Definition 4.1.1: The Risk–Return Spectrum in Quantitative Finance

In financial mathematics and portfolio theory, investment decision-making is governed by two complementary economic dimensions:

1. **Financial Risk (Uncertainty and Dispersion):** The possibility that the realized return on an investment deviates adversely from its anticipated or expected value. It encompasses capital impairment, cash flow volatility, default risk, and market downside.
2. **Financial Return (Potential Profit):** The net economic reward, yield, or capital gain generated by committing economic capital over a designated temporal horizon.

The foundational axiom of modern financial economics asserts that risk and expected return are positively correlated: rational, risk-averse economic agents demand higher expected compensation for absorbing elevated levels of variance or downside uncertainty.

4.1.1 Measures of Central Tendency for Financial Return Series

An **average** is a single representative central value summarizing the location of an entire distribution of financial data points. In portfolio analysis and performance attribution, five primary measures of central tendency are employed:

Mathematical Measures of Central Tendency

- **Simple Arithmetic Mean ($\bar{X}$):** The arithmetic mean sums all observed outcomes

and divides by the sample cardinality N:

$$\bar{X} = \frac{1}{N}\sum_{i=1}^{N} X_i = \frac{X_1 + X_2 + \cdots + X_N}{N} \tag{4.1}$$

- **Weighted Arithmetic Mean ($\bar{X}_w$):** Accounts for differential economic importance, asset market capitalization, or portfolio allocation weights w_i:

$$\bar{X}_w = \frac{\sum_{i=1}^{N} w_i X_i}{\sum_{i=1}^{N} w_i} \qquad \text{with} \quad \sum_{i=1}^{N} w_i = 1 \implies \bar{X}_w = \sum_{i=1}^{N} w_i X_i \tag{4.2}$$

- **Mode (Modal Value):** The observation that occurs with the highest empirical frequency in a data series:

$$\text{Mode} = \arg\max_{x} f(x) \tag{4.3}$$

- **Geometric Mean (GM):** The n-th root of the product of n positive financial observations or growth multipliers:

$$\text{GM} = \sqrt[n]{\prod_{i=1}^{n} X_i} = (X_1 \cdot X_2 \cdots X_n)^{1/n} = \exp\left(\frac{1}{n}\sum_{i=1}^{n} \ln X_i\right) \tag{4.4}$$

  The geometric mean is the strictly appropriate metric for compounding multi-period investment returns and evaluating the Compound Annual Growth Rate (CAGR).
- **Harmonic Mean (HM):** The reciprocal of the arithmetic mean of the reciprocals of the data elements:

$$\text{HM} = \frac{n}{\sum_{i=1}^{n} \frac{1}{X_i}} = \frac{n}{\frac{1}{X_1} + \frac{1}{X_2} + \cdots + \frac{1}{X_n}} \tag{4.5}$$

  The harmonic mean is utilized in finance when averaging rates, price-to-earnings (P/E) multiples, and executing fixed-rupee cost averaging (SIP investment strategies).

Fundamental Classical Inequality: $\text{AM} \geq \text{GM} \geq \text{HM}$ (for all positive $X_i > 0$)

Equality holds if and only if all observed elements are identical ($X_1 = X_2 = \cdots = X_n$).

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Example 4.1.1: Arithmetic Mean of Monthly Employee Incomes

Problem:
The payroll audit records of a commercial financial enterprise disclose the following monthly incomes (in Rs.) for 10 corporate employees:

14,780,	15,760,	26,690,	27,750,	24,840,
24,920,	16,100,	17,810,	27,050,	26,950

Calculate the simple arithmetic mean of the monthly income distribution.

Solution:
Let X_i denote the monthly income of employee i ($i = 1, 2, \ldots, 10$), with sample size $N = 10$.

Employee (i)	**Monthly Income** (X_i **in Rs.**)
1	14,780
2	15,760
3	26,690
4	27,750
5	24,840
6	24,920
7	16,100
8	17,810
9	27,050
10	26,950
Aggregate Sum ($\sum X_i$)	**Rs. 2,22,650**

Applying the simple arithmetic mean formulation:

$$\bar{X} = \frac{\sum_{i=1}^{N} X_i}{N} = \frac{2{,}22{,}650}{10} = \text{Rs. } \mathbf{22{,}265}$$

Economic Interpretation: The representative central earning baseline across the 10 corporate employees is Rs. **22,265** per month.

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Example 4.1.2: Modal and Harmonic Mean Valuation of Asset Series

Problem:
Solve the following two foundational central tendency evaluations:

(a) Determine the mode for the discrete transaction price series:

$$3, \quad 5, \quad 8, \quad 5, \quad 4, \quad 5, \quad 9, \quad 3$$

(b) Compute the harmonic mean (HM) for the following series of 8 operational asset rates:

$$5, \quad 7, \quad 10, \quad 13, \quad 15, \quad 20, \quad 25, \quad 28$$

Solution:
Part (a): Modal Value Determination: Tabulating the empirical occurrence frequencies of the observation elements:

Transaction Value (X)	3	4	5	8	9
Frequency (f)	2	1	**3**	1	1

The numerical value 5 appears with the maximum frequency ($f = 3$). Therefore:

$$\text{Mode} = \mathbf{5}$$

Part (b): Harmonic Mean Computation: Let $n = 8$. We construct the schedule of reciprocal components $\frac{1}{X_i}$:

Observation (i)	**Asset Rate** (X_i)	**Reciprocal Value** ($1/X_i$)
1	5	$1/5 = 0.20000$
2	7	$1/7 \approx 0.14286$
3	10	$1/10 = 0.10000$
4	13	$1/13 \approx 0.07692$
5	15	$1/15 \approx 0.06667$
6	20	$1/20 = 0.05000$
7	25	$1/25 = 0.04000$
8	28	$1/28 \approx 0.03571$
Total ($n = 8$)	—	$\sum_{i=1}^{8} \frac{1}{X_i} \approx \mathbf{0.71216}$

Applying the harmonic mean formula:

$$\text{HM} = \frac{n}{\sum_{i=1}^{n} \frac{1}{X_i}} = \frac{8}{0.71216} \approx \mathbf{11.233}$$

Mathematical Verification: For the identical dataset, AM $= \frac{123}{8} = 15.375$. Since AM(15.375) > HM(11.233), the classical mean inequality AM $\geq$ HM is rigorously confirmed.

4.1.2 Theoretical Foundations of Portfolio Return and Diversification

In asset allocation and modern portfolio theory (Markowitz framework), the performance of a multi-asset investment portfolio is structured as follows:

- **Portfolio Expected Return ($E[R_p]$):** The expected return of a portfolio is the strictly linear weighted average of the expected returns of its constituent security holdings:

$$E[R_p] = \sum_{i=1}^{N} w_i E[R_i] \qquad \text{where} \quad \sum_{i=1}^{N} w_i = 1 \tag{4.6}$$

  where w_i represents the fraction of total investment capital committed to security i.

- **Diversification Mechanics and Non-Linear Risk Reduction:** While portfolio expected return is strictly additive, **portfolio risk (variance) is sub-additive** whenever the constituent assets are not perfectly positively correlated ($\rho_{ij} < +1$):

$$\sigma_p^2 = \sum_{i=1}^{N} w_i^2 \sigma_i^2 + 2 \sum_{i=1}^{N} \sum_{j>i} w_i w_j \, \text{Cov}(R_i, R_j) < \sum_{i=1}^{N} w_i \sigma_i \quad (\text{if } \rho_{ij} < 1) \tag{4.7}$$

  Diversification pools non-systematic (unsystematic / idiosyncratic) firm-specific shocks, eliminating uncompensated risk without sacrificing expected portfolio yield.

4.2 Measures of Dispersion, Absolute Risk, and Comparative Stability

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Definition 4.2.1: Absolute versus Relative Dispersion

In financial risk analysis, **dispersion** quantifies the extent to which asset returns, market prices, or cash flow outcomes scatter or diverge from their central expected mean.

1. **Absolute Dispersion:** Expressed in the identical physical or monetary units as the original dataset (e.g. rupees, percentage return points). Primary metrics include Variance (σ^2) and Standard Deviation (σ).
2. **Relative Dispersion:** Dimensionless, normalized percentage ratios that facilitate direct cross-sectional risk and stability comparisons across assets trading at fundamentally disparate price scales or denominated in different currencies. The primary metric is the Coefficient of Variation (CV).

4.2.1 Formulations of Absolute Dispersion: Variance and Standard Deviation

Direct and Shortcut Formulations for Standard Deviation

- **Population Variance (σ^2) and Standard Deviation (σ):**

$$\sigma^2 = \frac{\sum_{i=1}^{N}(X_i - \bar{X})^2}{N}, \qquad \sigma = \sqrt{\frac{\sum_{i=1}^{N}(X_i - \bar{X})^2}{N}} \tag{4.8}$$

- **Sample Variance (s^2) and Sample Standard Deviation (s):** Incorporate Bessel's degrees-of-freedom correction ($n-1$) to eliminate small-sample downward estimation bias:

$$s^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}, \qquad s = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}} \tag{4.9}$$

- **Assumed Mean Method (Shortcut Technique):** To eliminate cumbersome decimal arithmetic when the true mean $\bar{X}$ is non-integer, an arbitrary round constant A is selected as the **assumed mean**. Defining deviations $d_i = X_i - A$:

$$\bar{X} = A + \frac{\sum d_i}{n} \tag{4.10}$$

$$\sigma = \sqrt{\frac{\sum d_i^2}{n} - \left(\frac{\sum d_i}{n}\right)^2}, \qquad s = \sqrt{\frac{\sum d_i^2 - \frac{(\sum d_i)^2}{n}}{n-1}} \tag{4.11}$$

- **Shift of Origin and Scaling Method:** When financial observations X_i are tightly clustered around a large base value with small fractional increments, we define a transformed variable $y_i = X_i - A$ or $u_i = \frac{X_i - A}{h}$:

$$\sigma_X = h \cdot \sigma_u = h\sqrt{\frac{\sum u_i^2}{n} - \left(\frac{\sum u_i}{n}\right)^2} \tag{4.12}$$

Variance is mathematically invariant to shifts in origin ($X_i \to X_i - A$), but scales quadratically with changes of scale ($X_i \to hX_i \implies \sigma^2 \to h^2\sigma^2$).

4.2.2 Relative Dispersion and Stability: The Coefficient of Variation (CV)

To assess which financial asset exhibits greater price stability or lower risk relative to its earnings level, analysts utilize the **Coefficient of Variation** (CV):

$$\text{CV} = \frac{\text{Standard Deviation}}{\text{Mean}} \times 100 = \frac{\sigma}{\bar{X}} \times 100 \tag{4.13}$$

Decision Rule: Lower CV $\implies$ Superior Price Stability / Lower Relative Risk

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Example 4.2.1: Standard Deviation Computation via Assumed Mean and Origin Shift

Problem:
Solve the following two rigorous standard deviation evaluations:

(a) The blood serum cholesterol readings of 10 persons were recorded as follows:

240, 260, 290, 245, 255, 288, 272, 263, 277, 251 (in mg/dL)

Compute the standard deviation using the **Assumed Mean Method** with $A = 260$.

(b) Compute the standard deviation for the following 10 tightly clustered decimal security prices:

240.12, 240.13, 240.15, 240.12, 240.17,
240.16, 240.22, 240.21, 240.18, 240.17

Solution:
Part (a): Assumed Mean Method ($A = 260$):
Let $d_i = X_i - 260$ with $n = 10$. We construct the deviation table:

Index (i)	**Reading** (X_i)	**Deviation** ($d_i = X_i - 260$)	**Squared** (d_i^2)
1	240	-20	400
2	260	0	0
3	290	$+30$	900
4	245	-15	225
5	255	-5	25
6	288	$+28$	784
7	272	$+12$	144
8	263	$+3$	9
9	277	$+17$	289
10	251	-9	81
Total	—	$\sum d_i = \mathbf{41}$	$\sum d_i^2 = \mathbf{2{,}857}$

1. Arithmetic Mean:

$$\bar{X} = A + \frac{\sum d_i}{n} = 260 + \frac{41}{10} = \mathbf{264.1} \text{ mg/dL}$$

2. Population Standard Deviation (σ):

$$\sigma = \sqrt{\frac{\sum d_i^2}{n} - \left(\frac{\sum d_i}{n}\right)^2} = \sqrt{\frac{2{,}857}{10} - (4.1)^2}$$
$$= \sqrt{285.7 - 16.81} = \sqrt{268.89} \approx \mathbf{16.40} \text{ mg/dL}$$

3. Sample Standard Deviation (s):

$$s = \sqrt{\frac{\sum d_i^2 - \frac{(\sum d_i)^2}{n}}{n-1}} = \sqrt{\frac{2{,}857 - 168.1}{9}} = \sqrt{\frac{2{,}688.9}{9}} \approx \mathbf{17.29} \text{ mg/dL}$$

Part (b): Shift of Origin Method for Decimal Prices:
Let $y_i = X_i - 240$. The transformed observations become:

$$0.12, \quad 0.13, \quad 0.15, \quad 0.12, \quad 0.17, \quad 0.16, \quad 0.22, \quad 0.21, \quad 0.18, \quad 0.17$$

Sum of transformed variables:

$$\sum y_i = 1.63, \qquad \sum y_i^2 = 0.2765$$

1. Arithmetic Mean:

$$\bar{X} = 240 + \frac{1.63}{10} = \mathbf{240.163}$$

2. Sum of Squared Deviations:

$$\sum (X_i - \bar{X})^2 = \sum y_i^2 - \frac{(\sum y_i)^2}{n} = 0.2765 - \frac{(1.63)^2}{10} = 0.2765 - 0.26569 = \mathbf{0.01081}$$

3. Standard Deviation:

$$\text{Population:} \quad \sigma = \sqrt{\frac{0.01081}{10}} = \sqrt{0.001081} \approx \mathbf{0.0329}$$
$$\text{Sample:} \quad s = \sqrt{\frac{0.01081}{9}} \approx \mathbf{0.0347}$$

Example 4.2.2: Comparative Asset Stability: Reliance vs. JIO Share Prices

Problem:
From the consecutive trading price records of Reliance and JIO equity shares furnished below, determine which share is **more stable in value**:

Trading Day	1	2	3	4	5	6	7	8	9	10
Reliance (Rs.)	55	54	52	53	56	58	52	50	51	49
JIO (Rs.)	108	107	105	105	106	107	104	103	104	101

Solution:
To compare relative price stability across two distinct nominal price ranges, we compute the **Coefficient of Variation** (CV). A lower CV signifies smaller relative volatility and superior price stability.

Step 1: Statistical Computations for Reliance Shares (X):

Period (i)	**Price** (X_i)	**Deviation** ($X_i - 53$)	**Squared** ($(X_i - 53)^2$)
1	55	+2	4
2	54	+1	1
3	52	−1	1
4	53	0	0
5	56	+3	9
6	58	+5	25
7	52	−1	1
8	50	−3	9
9	51	−2	4
10	49	−4	16
Total ($N = 10$)	$\sum X_i = \mathbf{530}$	$\sum(X_i - \bar{X}) = 0$	$\sum(X_i - \bar{X})^2 = \mathbf{70}$

$$\bar{X} = \frac{\sum X_i}{N} = \frac{530}{10} = \mathbf{53.00}$$

$$\sigma_{\text{Reliance}} = \sqrt{\frac{\sum(X_i - \bar{X})^2}{N}} = \sqrt{\frac{70}{10}} = \sqrt{7} \approx \mathbf{2.646}$$

$$\text{CV}_{\text{Reliance}} = \frac{\sigma_{\text{Reliance}}}{\bar{X}} \times 100 = \frac{2.646}{53} \times 100 \approx \mathbf{4.99}\%$$

Step 2: Statistical Computations for JIO Shares (Y):

Period (i)	**Price** (Y_i)	**Deviation** ($Y_i - 105$)	**Squared** ($(Y_i - 105)^2$)
1	108	+3	9
2	107	+2	4
3	105	0	0
4	105	0	0
5	106	+1	1
6	107	+2	4
7	104	−1	1
8	103	−2	4
9	104	−1	1
10	101	−4	16
Total ($N = 10$)	$\sum Y_i = \mathbf{1{,}050}$	$\sum(Y_i - \bar{Y}) = 0$	$\sum(Y_i - \bar{Y})^2 = \mathbf{40}$

$$\bar{Y} = \frac{\sum Y_i}{N} = \frac{1{,}050}{10} = \mathbf{105.00}$$

$$\sigma_{\text{JIO}} = \sqrt{\frac{\sum(Y_i - \bar{Y})^2}{N}} = \sqrt{\frac{40}{10}} = \sqrt{4} = \mathbf{2.000}$$

$$\text{CV}_{\text{JIO}} = \frac{\sigma_{\text{JIO}}}{\bar{Y}} \times 100 = \frac{2.000}{105} \times 100 \approx \mathbf{1.90}\%$$

Step 3: Comparative Synthesis and Investment Conclusion:

Security	**Mean Price (Rs.)**	**Standard Deviation** (σ)	**Coefficient of Variation** (CV)
Reliance Shares	53.00	2.646	4.99%
JIO Shares	105.00	2.000	**1.90%**

$$\text{CV}_{\text{JIO}}\,(1.90\%) < \text{CV}_{\text{Reliance}}\,(4.99\%)$$

Conclusion: JIO shares exhibit a substantially lower coefficient of variation (1.90% vs. 4.99%). Although JIO trades at twice the nominal price of Reliance, its normalized dispersion is over 2.6× smaller. Therefore, **JIO shares are significantly more stable in value**.

4.3 Bivariate Risk, Market Co-movement, and Correlation Mechanics

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Definition 4.3.1: Covariance and Karl Pearson Correlation Coefficient

In bivariate financial time series, **covariance** and **correlation** quantify the directional tendency and linear strength with which two asset return series move in tandem:

1. **Covariance ($\mathrm{Cov}(X,Y)$):** The expected value of the cross-product of deviations of two random variables from their respective means:

$$\mathrm{Cov}(X,Y) = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y}) = \frac{\sum X_i Y_i}{n} - \bar{X}\bar{Y} \tag{4.14}$$

   While covariance establishes the direction of co-movement ($\mathrm{Cov} > 0 \implies$ positive co-movement; $\mathrm{Cov} < 0 \implies$ inverse co-movement), its magnitude is sensitive to measurement units.

2. **Karl Pearson Correlation Coefficient (r_{XY}):** The standardized, dimensionless normalization of covariance bounded strictly within $[-1, +1]$:

$$r_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2 \sum_{i=1}^{n}(Y_i - \bar{Y})^2}} \tag{4.15}$$

3. **Direct Raw-Score Product-Moment Formula:**

$$r_{XY} = \frac{n\sum X_i Y_i - \left(\sum X_i\right)\left(\sum Y_i\right)}{\sqrt{\left[n\sum X_i^2 - \left(\sum X_i\right)^2\right]\left[n\sum Y_i^2 - \left(\sum Y_i\right)^2\right]}} \tag{4.16}$$

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Example 4.3.1: Bivariate Co-movement: DLF Stock Returns vs. BSE National Index

Problem:
The monthly percentage return data (R_m and R_i) for **DLF Stock** (X) and the **BSE National Index** (Y) across 12 consecutive calendar months are recorded as follows:

Month (i)	1	2	3	4	5	6
DLF Return (X, %)	9.43	0.00	−4.31	−18.92	−6.67	26.57
BSE Return (Y, %)	7.41	−5.33	−7.35	−14.64	1.58	15.19
Month (i)	7	8	9	10	11	12
DLF Return (X, %)	20.00	2.93	5.25	21.45	23.13	32.83
BSE Return (Y, %)	5.11	0.76	−0.97	10.44	17.47	20.15

Calculate Karl Pearson's correlation coefficient (r_{XY}) and interpret the co-movement profile.

Solution:
Let X denote the monthly return of DLF Stock and Y denote the monthly return of the BSE National Index, with sample cardinality $n = 12$.
Step 1: Construction of the Product-Moment Calculation Schedule:

Month	DLF (X)	BSE (Y)	X^2	Y^2	XY
1	9.43	7.41	88.9249	54.9081	69.8763
2	0.00	−5.33	0.0000	28.4089	0.0000
3	−4.31	−7.35	18.5761	54.0225	31.6785
4	−18.92	−14.64	357.9664	214.3296	276.9888
5	−6.67	1.58	44.4889	2.4964	−10.5386
6	26.57	15.19	705.9649	230.7361	403.5983
7	20.00	5.11	400.0000	26.1121	102.2000
8	2.93	0.76	8.5849	0.5776	2.2268
9	5.25	−0.97	27.5625	0.9409	−5.0925
10	21.45	10.44	460.1025	108.9936	223.9380
11	23.13	17.47	534.9969	305.2009	404.0811
12	32.83	20.15	1077.8089	406.0225	661.5245
Total	**111.69**	**49.82**	**3,724.9769**	**1,432.7492**	**2,160.4812**

Step 2: Substitution into the Direct Pearson Formulation:

$$\begin{aligned} n\sum XY - \left(\sum X\right)\left(\sum Y\right) &= 12(2{,}160.4812) - (111.69)(49.82) \\ &= 25{,}925.7744 - 5{,}564.3958 = \mathbf{20{,}361.3786} \end{aligned}$$

$$\begin{aligned} n\sum X^2 - \left(\sum X\right)^2 &= 12(3{,}724.9769) - (111.69)^2 \\ &= 44{,}699.7228 - 12{,}474.6561 = \mathbf{32{,}225.0667} \end{aligned}$$

$$\begin{aligned} n\sum Y^2 - \left(\sum Y\right)^2 &= 12(1{,}432.7492) - (49.82)^2 \\ &= 17{,}192.9904 - 2{,}482.0324 = \mathbf{14{,}710.9580} \end{aligned}$$

Evaluating the denominator product and its square root:

$$\sqrt{32{,}225.0667 \times 14{,}710.9580} = \sqrt{474{,}061{,}699.7} \approx \mathbf{21{,}772.9558}$$

Computing the correlation coefficient r_{XY}:

$$r_{XY} = \frac{20{,}361.3786}{21{,}772.9558} \approx \mathbf{+0.9352} \quad (\approx \mathbf{+0.935})$$

Step 3: Financial Econometric Interpretation:

- **Strength of Association:** The estimated correlation $r \approx +0.935$ reveals an exceptionally strong, statistically robust positive linear association between DLF stock returns and the benchmark BSE National Index.
- **Systematic Market Risk Exposure:** DLF shares are driven overwhelmingly by macroeconomic market factors ($\approx 87.5\%$ of its variance, $R^2 = r^2 = 0.8746$, is explained by broad market index co-movements).
- **Portfolio Beta Implication:** Because $r_{XY} > 0.90$, holding DLF stock provides minimal diversification benefit against broad market pullbacks, demonstrating pronounced systematic market beta risk.

4.4 Sharpe's Single-Index Optimal Portfolio Construction

Lecture
30/10/2026

4.4.1 Theoretical Foundations of the Single-Index Model

In Harry Markowitz's classical mean-variance portfolio theory, optimizing an N-asset portfolio requires estimating N expected returns, N asset variances, and $N(N-1)/2$ pairwise covariances—creating severe computational intractability and parameter error for large investment universes ($N = 100 \implies$ 4,950 covariances).

To overcome this curse of dimensionality, William Sharpe (1963) developed the **Single-Index Model (SIM)**. The SIM posits that the co-movement between individual securities arises entirely from a common macroeconomic factor—represented by the broad market index return (R_m):

$$R_i = \alpha_i + \beta_i R_m + e_i \tag{4.17}$$

subject to the standard Gauss–Markov linear index assumptions:

$$E[e_i] = 0 \quad \text{for all } i, \tag{4.18}$$

$$\text{Var}(e_i) = \sigma_{ei}^2 \quad \text{(constant unsystematic / residual variance)}, \tag{4.19}$$

$$\text{Cov}(e_i, e_j) = 0 \quad \text{for all } i \neq j \quad \text{(zero residual cross-correlation)}, \tag{4.20}$$

$$\text{Cov}(e_i, R_m) = 0 \quad \text{for all } i \quad \text{(residuals orthogonal to market return)}. \tag{4.21}$$

Under these assumptions, the total risk of any security i decomposes into two mutually independent, orthogonal components:

$$\sigma_i^2 = \underbrace{\beta_i^2 \sigma_m^2}_{\text{Systematic Risk}} + \underbrace{\sigma_{ei}^2}_{\text{Unsystematic (Firm-Specific) Risk}} \tag{4.22}$$

and the covariance between any two distinct assets i and j ($i \neq j$) depends solely on their joint systematic exposure to the market index:

$$\text{Cov}(R_i, R_j) = \beta_i \beta_j \sigma_m^2 \tag{4.23}$$

4.4.2 Sharpe Ratio Maximization and Optimal Cut-off Rate C^*

The investor's objective is to construct an optimal tangent portfolio of risky assets that maximizes the **Sharpe ratio** (θ_p), defined as the slope of the Capital Allocation Line (CAL):

$$\max_{\{w_i\}} \theta_p = \frac{E[R_p] - R_f}{\sigma_p} \quad \text{subject to} \quad \sum_{i=1}^{N} w_i = 1, \quad w_i \geq 0 \quad \text{(no short sales)} \tag{4.24}$$

Elton, Gruber, and Padberg (1976) proved that maximizing θ_p under the Single-Index Model yields a unique optimal cut-off acceptance threshold, denoted by C^*. A security qualifies for inclusion in the optimal portfolio if and only if its excess return-to-beta ratio exceeds this critical cut-off rate:

$$Z_i > C^* \iff \text{Include Security } i \text{ in the Optimal Portfolio} \tag{4.25}$$

Securities with $Z_i \leq C^*$ are rejected because their inclusion would dilute the portfolio's Sharpe ratio.

4.4.3 Algorithmic Optimization Pipeline

The mathematical process of constructing Sharpe's optimal portfolio follows a four-stage systematic pipeline, illustrated in Figure 4.1.

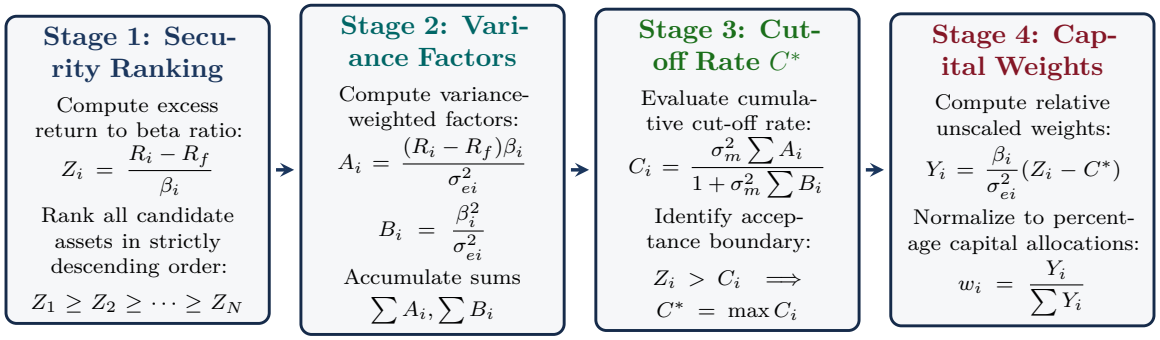


Figure 4.1: Systematic Four-Stage Architecture of Sharpe's Single-Index Optimal Portfolio Selection.

Definition 4.4.1: Sharpe's Optimal Portfolio Selection Formulations

Let R_f denote the riskless rate of return and σ_m^2 represent the variance of the macroeconomic market index. For an investment universe of N candidate risky securities:

1. **Excess Return-to-Beta Ratio (Ranking Metric):**

$$Z_i = \frac{R_i - R_f}{\beta_i} \tag{4.26}$$

   Securities are ranked from highest to lowest: $Z_1 \geq Z_2 \geq \cdots \geq Z_N$.

2. **Individual Variance-Weighted Factors:**

$$A_i = \frac{(R_i - R_f)\beta_i}{\sigma_{ei}^2}, \qquad B_i = \frac{\beta_i^2}{\sigma_{ei}^2} \tag{4.27}$$

3. **Cumulative Cut-off Rate (C_i):** For a portfolio containing the top i ranked securities:

$$C_i = \frac{\sigma_m^2 \sum_{j=1}^{i} A_j}{1 + \sigma_m^2 \sum_{j=1}^{i} B_j} = \frac{\sigma_m^2 \sum_{j=1}^{i} \frac{(R_j - R_f)\beta_j}{\sigma_{ej}^2}}{1 + \sigma_m^2 \sum_{j=1}^{i} \frac{\beta_j^2}{\sigma_{ej}^2}} \tag{4.28}$$

4. **Asset Selection Condition and Optimal Cut-off Point (C^*):** A security i is included in the optimal portfolio if and only if:

$$Z_i > C_i \tag{4.29}$$

   The optimal cut-off threshold is $C^* = C_k$, where k is the last asset index satisfying $Z_k > C_k$. At asset $k+1$, $Z_{k+1} \leq C_k$, terminating the search.

5. **Unscaled Weights and Normalized Portfolio Allocations:** For each selected security $i \in \{1, 2, \ldots, k\}$:

$$Y_i = \frac{\beta_i}{\sigma_{ei}^2}(Z_i - C^*), \qquad w_i = \frac{Y_i}{\sum_{j=1}^{k} Y_j} \quad \text{such that} \quad \sum_{i=1}^{k} w_i = 1 \tag{4.30}$$

Pedagogical Note on Notation Conventions

In the classroom lecture notes, the symbol Z_i denotes the excess return-to-beta ranking ratio ($\frac{R_i - R_f}{\beta_i}$) and Y_i denotes the unscaled relative asset weights. In standard textbook literature (e.g., Elton, Gruber, Brown, and Goetzmann, *Modern Portfolio Theory and Investment Analysis*), the excess return-to-beta ratio is written directly as $\frac{R_i - R_f}{\beta_i}$, while the unscaled weights are denoted by z_i. Both mathematical systems are identical in logic, intermediate values, and final capital allocations (w_i).

Example 4.4.1: Optimal Portfolio Construction via Single-Index Model (Jagan's Portfolio)

Problem:

Jagan is constructing an optimal investment portfolio from a candidate universe of five corporate stocks. The market variance is $\sigma_m^2 = 10\%$, and the prevailing riskless rate of return is $R_f = 5\%$. The expected returns, beta coefficients, and unsystematic residual risk levels of the five companies are presented in the initial portfolio data table below.

1. Rank the candidate stocks by their excess return-to-beta ratio (Z_i).
2. Construct the cumulative cut-off schedule (C_i) and determine the optimal cut-off point (C^*).
3. Identify the specific companies selected for inclusion in the optimum portfolio.
4. Calculate the relative unscaled weights (Y_i) and final percentage capital allocations (w_i).
5. Perform a post-optimization portfolio synthesis check: compute the portfolio's expected return $E[R_p]$, systematic beta β_p, unsystematic risk σ_{ep}^2, total variance σ_p^2, and Sharpe ratio θ_p.

Table 1: Given Initial Portfolio Inputs

Company (i)	Return (R_i, %)	Beta (β_i)	Unsystematic Risk (σ_{ei}^2)
1	20	1.0	20
2	22	1.5	10
3	10	0.5	30
4	24	2.0	20
5	12	1.0	30

Solution:

Step 1: Calculation of Excess Return to Beta (Z_i) and Security Ranking: Using the formula $Z_i = \frac{R_i - R_f}{\beta_i}$ with $R_f = 5\%$:

- Company 1: $Z_1 = \frac{20-5}{1.0} = \frac{15}{1.0} = 15.00$
- Company 2: $Z_2 = \frac{22-5}{1.5} = \frac{17}{1.5} = 11.33$
- Company 3: $Z_3 = \frac{10-5}{0.5} = \frac{5}{0.5} = 10.00$
- Company 4: $Z_4 = \frac{24-5}{2.0} = \frac{19}{2.0} = 9.50$
- Company 5: $Z_5 = \frac{12-5}{1.0} = \frac{7}{1.0} = 7.00$

Table 2: Step 1 — Excess Return to Beta (Z_i) Ranking Schedule

Company	Return (R_i)	$R_i - R_f$	Beta (β_i)	Z_i Ratio	Rank
1	20	15	1.0	**15.00**	**1**
2	22	17	1.5	**11.33**	**2**
3	10	5	0.5	**10.00**	**3**
4	24	19	2.0	**9.50**	**4**
5	12	7	1.0	**7.00**	**5**

Step 2: Calculation of the Cumulative Cut-off Schedule (C_i): For each ranked company, evaluate:

$$A_i = \frac{(R_i - R_f)\beta_i}{\sigma_{ei}^2}, \qquad B_i = \frac{\beta_i^2}{\sigma_{ei}^2}, \qquad C_i = \frac{\sigma_m^2 \sum_{j=1}^{i} A_j}{1 + \sigma_m^2 \sum_{j=1}^{i} B_j} \quad (\text{with } \sigma_m^2 = 10)$$

Table 3: Step 2 — Cumulative Cut-off Rate (C_i) Computation Schedule

Rank	Co.	Z_i	$A_i = \frac{(R_i - R_f)\beta_i}{\sigma_{ei}^2}$	$\sum A_i$	$B_i = \frac{\beta_i^2}{\sigma_{ei}^2}$	$\sum B_i$	C_i Calculation	C_i
1	1	15.00	$\frac{15\times1.0}{20} = 0.750$	0.750	$\frac{1.0^2}{20} = 0.050$	0.050	$\frac{10\times0.750}{1+10\times0.050} = \frac{7.500}{1.500}$	**5.000**
2	2	11.33	$\frac{17\times1.5}{10} = 2.550$	3.300	$\frac{1.5^2}{10} = 0.225$	0.275	$\frac{10\times3.300}{1+10\times0.275} = \frac{33.000}{3.750}$	**8.800**
3	3	10.00	$\frac{5\times0.5}{30} = 0.083$	3.383	$\frac{0.5^2}{30} = 0.008$	0.283	$\frac{10\times3.383}{1+10\times0.283} = \frac{33.833}{3.833}$	**8.826**
4	4	9.50	$\frac{19\times2.0}{20} = 1.900$	5.283	$\frac{2.0^2}{20} = 0.200$	0.483	$\frac{10\times5.283}{1+10\times0.483} = \frac{52.833}{5.833}$	**9.057**
5	5	7.00	$\frac{7\times1.0}{30} = 0.233$	5.516	$\frac{1.0^2}{30} = 0.033$	0.516	$\frac{10\times5.516}{1+10\times0.516} = \frac{55.167}{6.167}$	*8.946*

Determining the Optimal Cut-off Point (C^*): A security is included in the portfolio as long as its excess return-to-beta ratio exceeds the cumulative cut-off rate ($Z_i > C_i$):

- Company 1: $Z_1(15.00) > C_1(5.000) \implies$ Include
- Company 2: $Z_2(11.33) > C_2(8.800) \implies$ Include
- Company 3: $Z_3(10.00) > C_3(8.826) \implies$ Include
- Company 4: $Z_4(9.50) > C_4(9.057) \implies$ Include
- Company 5: $Z_5(7.00) < C_4(9.057) \implies$ **Exclude / Stop**

Because $Z_5 = 7.00$ falls below the prevailing cut-off rate $C_4 = 9.057$, Company 5 is excluded, and the optimization process terminates.

Conclusion on Security Selection:

$$\text{Optimal Cut-off Point: } \mathbf{C^* = 9.057}$$

The optimum portfolio consists strictly of **Companies 1, 2, 3, and 4**.

Step 3: Calculation of Optimum Portfolio Weights (w_i): For each selected asset $i \in \{1, 2, 3, 4\}$, the unscaled relative weight Y_i is computed as:

$$Y_i = \frac{\beta_i}{\sigma_{ei}^2}(Z_i - C^*) = \frac{\beta_i}{\sigma_{ei}^2}(Z_i - 9.057)$$

and the final percentage capital allocation is obtained via normalization: $w_i = \frac{Y_i}{\sum_{j=1}^{4} Y_j}$.

Table 4: Step 3 — Optimum Asset Weights and Percentage Capital Allocation

Company	Beta (β_i)	σ_{ei}^2	Z_i	$Z_i - C^*$	Relative Weight (Y_i)	Final Weight (w_i)
1	1.0	20	15.00	$15.00 - 9.057 = 5.943$	$\frac{1.0}{20} \times 5.943 = 0.29715$	$\frac{0.29715}{0.69857} \approx \mathbf{42.54}\%$
2	1.5	10	11.33	$11.33 - 9.057 = 2.276$	$\frac{1.5}{10} \times 2.276 = 0.34140$	$\frac{0.34140}{0.69857} \approx \mathbf{48.87}\%$
3	0.5	30	10.00	$10.00 - 9.057 = 0.943$	$\frac{0.5}{30} \times 0.943 = 0.01572$	$\frac{0.01572}{0.69857} \approx \mathbf{2.25}\%$
4	2.0	20	9.50	$9.50 - 9.057 = 0.443$	$\frac{2.0}{20} \times 0.443 = 0.04430$	$\frac{0.04430}{0.69857} \approx \mathbf{6.34}\%$
Total	—	—	—	—	$\sum Y_i = \mathbf{0.69857}$	**100.00**%

Step 4: Post-Optimization Portfolio Synthesis Check:

1. **Expected Portfolio Return ($E[R_p]$):**

$$E[R_p] = \sum_{i=1}^{4} w_i R_i = 0.4254(20) + 0.4887(22) + 0.0225(10) + 0.0634(24)$$
$$= 8.508\% + 10.751\% + 0.225\% + 1.522\% = \mathbf{21.01}\%$$

2. **Portfolio Beta Sensitivity (β_p):**

$$\beta_p = \sum_{i=1}^{4} w_i \beta_i = 0.4254(1.0) + 0.4887(1.5) + 0.0225(0.5) + 0.0634(2.0)$$
$$= 0.4254 + 0.7331 + 0.0113 + 0.1268 = \mathbf{1.2966} \approx \mathbf{1.30}$$

3. **Portfolio Unsystematic Variance (σ_{ep}^2):**

$$\sigma_{ep}^2 = \sum_{i=1}^{4} w_i^2 \sigma_{ei}^2 = (0.4254)^2(20) + (0.4887)^2(10) + (0.0225)^2(30) + (0.0634)^2(20)$$
$$= 3.6193 + 2.3883 + 0.0152 + 0.0804 = \mathbf{6.1032}$$

4. **Total Optimal Portfolio Variance and Standard Deviation (σ_p^2, σ_p):**

$$\sigma_p^2 = \beta_p^2 \sigma_m^2 + \sigma_{ep}^2 = (1.2966)^2(10) + 6.1032 = 16.8117 + 6.1032 = \mathbf{22.9149}$$
$$\sigma_p = \sqrt{22.9149} \approx \mathbf{4.787}\%$$

5. **Optimal Tangency Portfolio Sharpe Ratio (θ_p):**

$$\theta_p = \frac{E[R_p] - R_f}{\sigma_p} = \frac{21.006\% - 5.000\%}{4.787\%} = \frac{16.006}{4.787} \approx \mathbf{3.3436}$$

The assembled portfolio achieves a high Sharpe ratio of 3.34, demonstrating that Sharpe's single-index cut-off methodology successfully balances high expected return against systematic and residual risk.

4.5 Foreign Exchange Exposure and Currency Risk Foundations

Lecture
30/10/2026

Definition 4.5.1: Currency Exposure versus Exchange Rate Risk

In multinational financial management and quantitative corporate finance, international cash flows and firm valuation are subject to foreign exchange rate volatility:

1. **Foreign Exchange Exposure (Sensitivity):** The degree of structural sensitivity or responsiveness with which a corporation's operating cash flows, domestic asset values, or stock returns are affected by unpredictable exchange rate fluctuations. Econometrically, currency exposure is estimated via linear regression modeling:

$$R_{i,t} = \alpha_i + \beta_{\text{FX}} \left(\frac{\Delta S_t}{S_{t-1}} \right) + \epsilon_t \tag{4.31}$$

   where $\frac{\Delta S_t}{S_{t-1}}$ represents the percentage change in the spot exchange rate, and the regression slope $\beta_{\text{FX}} = \frac{\text{Cov}(R_i, \Delta S/S)}{\text{Var}(\Delta S/S)}$ represents the firm's **foreign exchange exposure coefficient**.

2. **Exchange Rate Risk (Variance / Volatility):** The actual variance, dispersion, or standard deviation in the firm's equity value or cash flows attributable directly to exchange rate movements:

$$\text{Var}(R_i) = \beta_{\text{FX}}^2 \sigma_{\text{FX}}^2 + \sigma_\epsilon^2 \tag{4.32}$$

   While exposure reflects the structural slope (β_{FX}), risk combines exposure with the underlying market volatility (σ_{FX}^2).

3. **FOREX Exposure (Cross-Border Capital Activities):** The overall enterprise vulnerability arising from commercial trade, offshore investments, cross-border liabilities, or debt financing denominated in currencies other than the home domestic reporting currency.

4.5.1 Tripartite Taxonomy of Foreign Exchange Exposure

Multinational enterprises classify currency exposures into three operational categories:

Exposure Type	Temporal Horizon	Economic Impact / Focus	Primary Hedging Tools
Transaction Exposure	Short-term (≤ 1 year)	Contracted foreign currency receivables, payables, and debt obligations with fixed settlement dates.	Forward contracts, futures, money market hedges, currency options.
Translation Exposure	Periodic (Quarterly / Annual reporting)	Paper accounting volatility arising when consolidating foreign subsidiary balance sheets into parent currency.	Balance sheet hedging, matching local currency assets with liabilities.
Economic / Operating Exposure	Long-term strategic horizon (Multi-year)	Uncontracted future operating cash flows and competitive market pricing power altered by structural exchange rate realignments.	Operational diversification, production relocation, multi-currency financing.

Comprehensive Formula Reference Sheet

Chapter 1: Interest Theory and Mathematical Foundations

1. Simple Interest and Total Accumulated Amount

$$\text{SI} = \frac{P \cdot N \cdot R}{100}, \qquad A = P + \text{SI} = P\left(1 + \frac{N \cdot R}{100}\right)$$

Where:

- P = Principal sum borrowed or invested
- N = Time period in years
- R = Annual rate of interest (%)
- SI = Simple interest earned
- A = Total accumulated amount (Principal + Interest)

2. Compound Interest and Multiple Conversion Periods

$$\text{Annual Compounding Amount:} \quad A = P\left(1 + \frac{R}{100}\right)^N$$

$$\text{Compound Interest Earned:} \quad \text{CI} = A - P = P\left[\left(1 + \frac{R}{100}\right)^N - 1\right]$$

$$\text{Sub-Annual Compounding } (m \text{ times/year}): \quad A = P\left(1 + \frac{R}{m \cdot 100}\right)^{m \cdot N}$$

$$\text{Continuous Compounding:} \quad A = P \cdot e^{r \cdot N} \qquad \left(r = \frac{R}{100}\right)$$

Where:

- P = Initial principal sum
- R = Nominal annual rate of interest (%)
- r = Nominal interest rate in decimal form ($r = R/100$)
- N = Total investment duration in years
- m = Compounding frequency per annum (2 semi-annual, 4 quarterly, 12 monthly)
- e = Base of the natural logarithm (≈ 2.71828)
- A = Compound terminal amount
- CI = Compound interest earned

3. Effective Annual Rate of Interest (ERI / EAR) and Depreciation

$$\text{Effective Annual Rate:} \quad \text{ERI} = \left(1 + \frac{R}{m \cdot 100}\right)^{m} - 1$$

$$\text{Reducing Balance Depreciation:} \quad A = P\left(1 - \frac{R}{100}\right)^{N}$$

Where:

- ERI = Effective rate of interest (true annualized yield in decimal)
- m = Number of compounding conversion periods per year
- P = Original historical cost of the asset
- R = Annual rate of depreciation (%)
- N = Operational lifespan / age in years
- A = Written Down Value (WDV) or residual scrap value

4. Doubling Period Approximations

$$\text{Exact Limit (Continuous):} \quad T_{\text{exact}} = \frac{\ln 2}{r} \approx \frac{0.6931}{r} = \frac{69.3}{R}$$

$$\text{Rule of 72:} \quad T_{\text{Rule of 72}} \approx \frac{72}{R}$$

$$\text{Rule of 69:} \quad T_{\text{Rule of 69}} \approx 0.35 + \frac{69}{R}$$

Where:

- T = Number of years required for principal to double
- R = Annual percentage interest rate ($R\%$)
- r = Annual interest rate in decimal ($r = R/100$)

5. Annuity Cash Flows: Future Value and Present Value

A. Closed-Form Factor Formulations:

$$\text{Ordinary Annuity Future Value:} \quad \text{FV} = A \cdot \text{CVAF}(r,n) = A\left[\frac{(1+r)^n - 1}{r}\right]$$

$$\text{Annuity Due Future Value:} \quad \text{FV}_{\text{due}} = \text{FV}_{\text{ord}} \times (1+r) = A\left[\frac{(1+r)^n - 1}{r}\right](1+r)$$

$$\text{Present Value of a Lump Sum:} \quad \text{PV} = \frac{A}{(1+r)^n} = A \cdot \text{PVF}(r,n)$$

$$\text{Ordinary Annuity Present Value:} \quad \text{PV} = A \cdot \text{PVAF}(r,n) = A\left[\frac{1-(1+r)^{-n}}{r}\right]$$

$$\text{Annuity Due Present Value:} \quad \text{PV}_{\text{due}} = \text{PV}_{\text{ord}} \times (1+r) = A\left[\frac{1-(1+r)^{-n}}{r}\right](1+r)$$

B. Year-by-Year Summation Formulas ($\sum$):

$$\text{Ordinary Annuity FV:} \quad \text{FV} = \sum_{t=1}^{n} A \times (1+r)^{n-t}$$

$$\text{Annuity Due FV:} \quad \text{FV}_{\text{due}} = \sum_{t=0}^{n-1} A \times (1+r)^{n-t}$$

$$\text{Ordinary Annuity PV:} \quad \text{PV} = \sum_{t=1}^{n} \frac{A}{(1+r)^t} = \frac{A}{(1+r)^1} + \frac{A}{(1+r)^2} + \cdots + \frac{A}{(1+r)^n}$$

$$\text{Annuity Due PV:} \quad \text{PV}_{\text{due}} = \sum_{t=0}^{n-1} \frac{A}{(1+r)^t} = A + \frac{A}{(1+r)^1} + \cdots + \frac{A}{(1+r)^{n-1}}$$

Where:

- A = Equal periodic cash payment / installment
- r = Periodic interest or discount rate in decimal ($R/100$)
- n = Total number of cash flow installments / conversion periods
- t = Payment / discounting period index
- $\text{CVAF}(r,n)$ = Compound Value Annuity Factor
- $\text{PVF}(r,n)$ = Present Value Factor ($(1+r)^{-n}$)
- $\text{PVAF}(r,n)$ = Present Value Annuity Factor
- $(1+r)$ = Advance compounding factor for payments at beginning of period (annuity due)

6. Perpetuities, Finite Growing Annuity, and Sinking Funds

A. Closed-Form Formulations:

$$\text{Constant Perpetuity:} \quad \text{PV} = \frac{C}{r}$$

$$\text{Growing Perpetuity:} \quad \text{PV} = \frac{C_1}{r-g} \qquad (r > g)$$

$$\text{Finite Growing Annuity:} \quad \text{PV} = \frac{C_1}{r-g}\left[1 - \left(\frac{1+g}{1+r}\right)^n\right] \qquad (r \neq g)$$

$$\text{Ordinary Sinking Fund Deposit:} \quad A = \frac{\text{FV}}{\text{CVAF}(r,n)} = \text{FV}\left[\frac{r}{(1+r)^n - 1}\right]$$

$$\text{Sinking Fund Annuity Due Deposit:} \quad A_{\text{due}} = \frac{A_{\text{ord}}}{1+r} = \frac{\text{FV}}{\text{CVAF}(r,n) \times (1+r)}$$

B. Year-by-Year Summation Formulation ($\sum$):

$$\text{Finite Growing Annuity PV:} \quad \text{PV} = \sum_{t=1}^{n} \frac{C_1 \times (1+g)^{t-1}}{(1+r)^t}$$

Where:

- C = Constant periodic perpetual cash inflow
- C_1 = Cash inflow generated at the end of Period 1
- r = Periodic discount rate in decimal ($R/100$)
- g = Constant compound growth rate of periodic cash flows ($G/100$)
- n = Total number of payment periods
- t = Cash flow period index ($t = 1, 2, \ldots, n$)
- FV = Target reserve accumulation sum required at maturity
- A = Periodic sinking fund contribution
- $\text{CVAF}(r,n)$ = Compound Value Annuity Factor

7. Continuous Exponential Growth and Decay Models

$$\text{Continuous Growth Model:} \quad n(t) = n_0\, e^{kt} \qquad (k > 0)$$

$$\text{Continuous Decay Model:} \quad n(t) = n_0\, e^{-kt} \qquad (k > 0)$$

$$\text{Rate Constant Calibration:} \quad k = \frac{1}{T} \ln\left(\frac{n_{\max}}{n_{\min}}\right)$$

Where:

- $n(t)$ = Modeled quantity / price impact level at elapsed time t
- n_0 = Initial quantity at baseline time $t = 0$
- k = Continuous rate constant ($k > 0$)
- T = Calibration interval between boundary values $[n_{\min}, n_{\max}]$

Chapter 2: Valuation of Financial Instruments

8. Security Yield Metrics and Holding Period Return

$$\text{Current Yield} = \frac{\text{Annual Dividend / Coupon}}{P_0} \times 100$$

$$\text{Dividend Yield } (dY) = \frac{\text{DPS}}{P_0} \times 100$$

$$\text{Holding Period Return (HPR)} = \frac{D_t + \Delta P}{P_1} = \frac{D_t + (P_t - P_1)}{P_1}$$

$$\text{Earnings Yield} = \frac{\text{EPS}}{P_0} \times 100$$

Where:

- P_0 = Current prevailing market purchase price of the share/security
- DPS = Annual dividend per share
- HPR = Total rate of return realized over the investment horizon
- D_t = Cash dividend received during the planned holding period
- P_1 = Initial purchase price paid at acquisition
- P_t = Terminal market price realized at liquidation
- ΔP = Capital appreciation or depreciation $(P_t - P_1)$
- EPS = Earnings per share

9. General Dividend Discount Valuation Models (DDM)

$$\text{Zero-Growth Model (Perpetual):} \quad P_0 = \frac{D}{K_e}$$

$$\text{Finite 2-Year Holding Horizon:} \quad P_0 = \frac{D_1}{1 + K_e} + \frac{D_2 + P_2}{(1 + K_e)^2}$$

$$\text{General Infinite DDM:} \quad P_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1 + K_e)^t}$$

$$\text{Gordon Constant Growth DDM:} \quad P_0 = \frac{D_1}{K_e - g} = \frac{D_0(1 + g)}{K_e - g} \qquad (K_e > g)$$

Where:

- P_0 = Theoretical intrinsic market price per share today
- D = Constant perpetual annual dividend per share
- D_1, D_2, D_t = Expected future dividend per share at the end of year 1, 2, and t
- D_0 = Baseline dividend per share currently paid
- P_2 = Projected terminal market price at the end of Year 2
- K_e = Required rate of return of equity investors / cost of equity capital
- g = Constant perpetual annual compound growth rate in dividends $(g < K_e)$

10. Gordon's Dividend Capitalization Model with Endogenous Growth

$$\text{Valuation Equation:} \quad P_0 = \frac{\text{EPS}_1(1-b)}{K_e - b \cdot r} = \frac{D_1}{K_e - g}$$

$$\text{Endogenous Growth Rate:} \quad g = b \times r$$

Where:

- P_0 = Theoretical market price per share (MPS)
- EPS_1 = Expected earnings per share for the upcoming period
- b = Retention ratio (fraction of earnings retained within the firm, $0 \leq b \leq 1$)
- $(1-b)$ = Dividend payout ratio (D_1/EPS_1)
- K_e = Cost of equity / capitalization rate
- r = Internal rate of return on reinvested corporate earnings (ROE)
- $g = b \cdot r$ = Endogenous growth rate of dividends and earnings

Payout Optimization Policy Rules:

Firm Category	Condition	Optimal Payout	Theoretical Dynamic
Growth Firm	$r > K_e$	NIL (0%)	Maximum retention ($b = 1$) maximizes MPS
Normal Firm	$r = K_e$	Indifferent	MPS is identical across all payout levels
Declining Firm	$r < K_e$	100%	Full distribution ($b = 0$) maximizes MPS

11. Walter's Valuation Formula and Firm Capitalization

$$\text{Market Price Per Share:} \quad P = \frac{D + \frac{r}{k}(E - D)}{k}$$

$$\text{Total Market Value of Firm:} \quad V = N \times P$$

Where:

- P = Theoretical market price per share (MPS)
- D = Dividend per share paid (DPS = EPS $\times$Payout Ratio)
- E = Earnings per share (EPS)
- $(E - D)$ = Retained earnings per share
- r = Internal rate of return on investment earned by the company
- k = Cost of capital / capitalization rate
- $\frac{r}{k}(E - D)$ = Capitalized market value of the earnings retained
- N = Total number of equity shares outstanding
- V = Aggregate market valuation of the firm

Payout Optimization Policy Rules:

Firm Category	Condition	Optimum Payout	Share Price Dynamic
Growth Firm	$r > k$	0%	MPS decreases as dividend payout increases
Normal Firm	$r = k$	Indifferent	MPS is invariant to payout choice
Declining Firm	$r < k$	100%	MPS increases as dividend payout increases

12. Modigliani–Miller (MM) Dividend Irrelevance Model

$$\text{Current Share Valuation:} \quad P_0 = \frac{D_1 + P_1}{1 + K_e}$$

$$\text{Period-End Share Price:} \quad P_1 = P_0(1 + K_e) - D_1$$

$$\text{External Financing Deficit:} \quad \Delta M = I - (E - n \cdot D_1)$$

$$\text{New Shares to be Issued:} \quad m = \Delta n = \frac{\Delta M}{P_1} = \frac{I - (E - n \cdot D_1)}{P_1}$$

$$\text{Total Market Value Invariant:} \quad V_1 = (n + m)P_1 = (n + \Delta n)P_1$$
$$= nP_0(1 + K_e) + I - E$$

Where:

- P_0 = Current prevailing market price per share at beginning of period
- P_1 = Market price per share at the end of the period
- D_1 = Dividend per share declared/distributed at the end of the period
- K_e = Rate of capitalization / cost of equity capital appropriate to risk class
- I = Total capital investment budget required for the period
- E = Total net income (earnings) of the firm during the period
- n = Number of existing equity shares outstanding at beginning of period
- nD_1 = Aggregate dividend distribution to existing shareholders
- ΔM = External equity financing deficit to be raised
- $m = \Delta n$ = Number of newly issued equity shares
- V_1 = Total aggregate market valuation of the firm at year end

13. Bond Valuation, Yield to Maturity (YTM), and Yield to Call (YTC)

A. Theoretical Present Value Valuation:

$$V_0 = \sum_{t=1}^{n} \frac{C}{(1+k_d)^t} + \frac{M}{(1+k_d)^n} = C \cdot \text{PVAF}(k_d, n) + M \cdot \text{PVF}(k_d, n)$$

$$V_{\text{call}} = \sum_{t=1}^{N_c} \frac{C}{(1+k_c)^t} + \frac{CP}{(1+k_c)^{N_c}} = C \cdot \text{PVAF}(k_c, N_c) + CP \cdot \text{PVF}(k_c, N_c)$$

B. Approximation Formulas:

$$\text{Approximate YTM} = \frac{C + \dfrac{M - P_0}{n}}{\dfrac{M + P_0}{2}}$$

$$\text{Approximate YTC} = \frac{C + \dfrac{CP - P_0}{N_c}}{\dfrac{CP + P_0}{2}}$$

Where:

- V_0 = Theoretical intrinsic economic value of the bond
- P_0 = Current prevailing market purchase price of the bond
- C = Annual coupon interest payment ($C = M \times$ Coupon Rate)
- M = Par or face value of the bond payable at final maturity
- n = Total number of years until final stated maturity
- N_c = Number of years remaining until the earliest call date
- CP = Call price payable upon redemption ($CP = M +$ Call Premium)
- k_d = Periodic required rate of return / discount rate of bondholders
- k_c = True Yield to Call rate (YTC)
- $\text{PVAF}(k, n)$ = Present Value Annuity Factor at rate k for n periods
- $\text{PVF}(k, n)$ = Present Value Factor at rate k for n periods

14. Loan Amortization and Equal Periodic Installment (EMI)

$$\text{Periodic Loan Installment (EMI):} \quad E = \frac{P \cdot r \cdot (1+r)^n}{(1+r)^n - 1} = \frac{P}{\text{PVAF}(r,n)}$$

$$\text{Interest Component for Period } t: \quad I_t = B_{t-1} \times r$$

$$\text{Principal Amortization for Period } t: \quad \Delta P_t = E - I_t$$

$$\text{Closing Principal Balance for Period } t: \quad B_t = B_{t-1} - \Delta P_t = B_{t-1} - (E - I_t)$$

Where:

- E = Equal periodic loan installment / repayment (EMI)
- $P = B_0$ = Initial principal loan amount borrowed
- r = Periodic interest rate in decimal $(R/(m \cdot 100))$
- n = Total number of periodic repayment installments $(N \times m)$
- t = Amortization period index $(t = 1, 2, \ldots, n)$
- B_{t-1} = Outstanding principal balance at the beginning of period t
- B_t = Outstanding principal balance remaining at the end of period t
- I_t = Interest charge incurred during period t
- ΔP_t = Principal reduction portion of installment E in period t
- $\text{PVAF}(r,n)$ = Present Value Annuity Factor for n periods at rate r

15. Price–Earnings (P/E) Valuation Model and Earnings Per Share

$$\text{Earnings Per Share (EPS):} \quad \text{EPS} = \frac{\text{EAT} - D_p}{N}$$

$$\text{Price–Earnings Ratio (P/E):} \quad \text{P/E Ratio} = \frac{P_0}{\text{EPS}}$$

$$\text{Implied Stock Valuation:} \quad P_0 = \text{EPS} \times (\text{P/E Ratio})$$

$$\text{Theoretical P/E Multiple (from Gordon DDM):} \quad \frac{P_0}{\text{EPS}_1} = \frac{1-b}{K_e - g} = \frac{\text{Payout Ratio}}{K_e - g}$$

Where:

- EPS = Earnings per share
- EAT = Corporate earnings after tax (net income)
- D_p = Total preference dividend obligations payable (EAT $- D_p$ = Earnings to Common Shareholders)
- N = Weighted average number of common equity shares outstanding
- P_0 = Current market price per share (MPS)
- P/E Ratio = Price-to-earnings valuation multiple
- EPS_1 = Projected forward earnings per share for the next annual period
- b = Retention ratio of earnings ($0 \leq b \leq 1$)
- $(1-b)$ = Dividend payout ratio (D_1/EPS_1)
- K_e = Cost of equity / required rate of return
- g = Sustainable constant annual growth rate in earnings and dividends ($g = b \cdot r$)

Chapter 3: Financial Ratios and Cash Flow Analysis

16. Profitability Ratios and Operational Margin Identities

$$\text{Gross Profit Ratio} = \frac{\text{Gross Profit}}{\text{Net Sales}} \times 100$$

$$\text{Operating Profit Ratio} = \frac{\text{Operating Profit}}{\text{Net Sales}} \times 100$$

$$\text{Operating Ratio} = \frac{\text{COGS} + \text{Operating Expenses}}{\text{Net Sales}} \times 100$$

$$\text{Invariance Identity:} \quad \text{Operating Ratio} + \text{Operating Profit Ratio} = 100\%$$

$$\text{Net Profit Ratio} = \frac{\text{NPAT}}{\text{Net Sales}} \times 100$$

$$\text{Return on Investment (ROI):} \quad \text{ROI} = \frac{\text{Operating Profit (EBIT)}}{\text{Capital Employed}} \times 100$$

Where:

- Net Sales = Total gross sales revenue less sales returns and allowances
- COGS = Cost of Goods Sold (Opening Stock + Purchases + Direct Expenses − Closing Stock)
- Gross Profit = Net revenue remaining after direct manufacturing/trading costs (Net Sales − COGS)
- Operating Expenses = Administrative, office, selling, distribution, and depreciation expenses
- Operating Profit = Core operational earnings before non-operating income, interest, and tax (EBIT)
- NPAT = Final net income available after non-operating gains/losses and corporate income taxes
- Capital Employed = Total long-term funds deployed (Shareholders' Equity + Long-Term Debt)

17. Activity and Turnover Ratios with Operational Velocity Periods

$$\text{Inventory Turnover Ratio (ITR):} \quad \text{ITR} = \frac{\text{COGS}}{\text{Average Inventory}} \quad \text{or} \quad \frac{\text{Net Sales}}{\text{Average Inventory}}$$

$$\text{Inventory Holding Velocity:} \quad T_{\text{inventory}} = \frac{365}{\text{ITR}} \text{ days} = \frac{12}{\text{ITR}} \text{ months}$$

$$\text{Debtors Turnover Ratio (DTR):} \quad \text{DTR} = \frac{\text{Net Credit Sales}}{\text{Average Accounts Receivable}}$$

$$\text{Debtors Collection Velocity:} \quad T_{\text{debtors}} = \frac{365}{\text{DTR}} \text{ days} = \frac{12}{\text{DTR}} \text{ months}$$

$$\text{Creditors Turnover Ratio (CTR):} \quad \text{CTR} = \frac{\text{Net Credit Purchases}}{\text{Average Accounts Payable}}$$

$$\text{Creditors Payment Velocity:} \quad T_{\text{creditors}} = \frac{365}{\text{CTR}} \text{ days} = \frac{12}{\text{CTR}} \text{ months}$$

$$\text{Fixed Assets Turnover:} \quad \text{FATR} = \frac{\text{COGS (or Net Sales)}}{\text{Net Fixed Assets}}$$

$$\text{Working Capital Turnover:} \quad \text{WCTR} = \frac{\text{Cost of Sales (or Net Sales)}}{\text{Net Working Capital}}$$

Where:

- Average Inventory $= \frac{1}{2}$(Opening Inventory + Closing Inventory)
- Net Credit Sales = Total gross sales less cash sales and sales returns
- Average Accounts Receivable = Sundry Debtors + Bills Receivable (gross before doubtful debt reserves)
- Net Credit Purchases = Total gross purchases less cash purchases and purchase returns
- Average Accounts Payable = Sundry Creditors + Bills Payable
- Net Fixed Assets = Book value of property, plant, and equipment less accumulated depreciation
- Net Working Capital = Current Assets − Current Liabilities
- $T_{\text{inventory}}, T_{\text{debtors}}, T_{\text{creditors}}$ = Operational velocity turnover periods in calendar days or months

18. Liquidity and Solvency Ratios with Conventional Benchmark Norms

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}} \qquad [\text{Ideal Norm: } 2:1]$$

$$\text{Quick / Acid-Test Ratio} = \frac{\text{QA}}{\text{CL}} = \frac{\text{CA} - \text{Stock} - \text{Prepaids}}{\text{CL}} \qquad [\text{Ideal Norm: } 1:1]$$

$$\text{Cash Position Ratio} = \frac{\text{Cash} + \text{Bank} + \text{Marketable Securities}}{\text{Current Liabilities}} \qquad [\text{Norm: } 0.75:1]$$

$$\text{Debt-to-Equity Ratio} = \frac{\text{Total Long-Term Debt}}{\text{Shareholders' Equity}}$$

$$\text{Proprietary Ratio} = \frac{\text{Proprietors' Funds}}{\text{Total Assets}}$$

Where:

- Current Assets = Short-term resources realizable within one annual operating cycle (Inventories + Debtors + Cash + Bank + Bills Receivable + Prepaids)
- Current Liabilities = Operational obligations maturing within one year (Creditors + Bills Payable + Overdraft + Provisions)
- Quick Assets = Current assets convertible to cash immediately without market loss (Current Assets − Stock − Prepaid Expenses)
- Cash Position = Most stringent liquid reserves (Cash in Hand + Cash at Bank + Marketable Securities)
- Shareholders' Equity = Paid-up equity capital + retained earnings + accumulated reserves

19. Fund Flow and Cash Flow Accounting Mechanics

$$\Delta\,\text{NWC} = \Delta\text{Current Assets} - \Delta\text{Current Liabilities}$$

$$\begin{aligned}\text{FFO} = {} & \text{Net Operating Profit} + \text{Depreciation \& Amortization} \\ & + \text{Non-Operating Losses} - \text{Non-Operating Gains}\end{aligned}$$

$$\begin{aligned}\text{CFO} = {} & \text{FFO} + \sum(\uparrow \text{CL}) + \sum(\downarrow \text{CA}) \\ & - \sum(\downarrow \text{CL}) - \sum(\uparrow \text{CA})\end{aligned}$$

$$\text{Funds Identity:} \quad \text{Total Sources} - \text{Total Applications} = \Delta\,\text{NWC}$$

Where:

- $\Delta\,\text{NWC}$ = Net change in corporate working capital between consecutive balance sheet dates
- FFO = Funds Generated from Operations (operational working capital surplus)
- CFO = Net cash flow generated directly from core operating activities
- $\sum(\uparrow \text{CL})$ = Cash inflows arising from short-term trade and credit expansion
- $\sum(\downarrow \text{CA})$ = Cash inflows released from liquidating inventories or collecting receivables
- $\sum(\uparrow \text{CA})$ = Cash outflows absorbed by increasing receivables, inventories, or prepaids
- $\sum(\downarrow \text{CL})$ = Cash outflows applied to retire trade creditors or short-term notes
- Sources = Long-term equity/debt capital raised, asset sales, and operating funds
- Applications = Fixed asset capital expenditures, long-term debt redemption, and proprietor drawings

20. Cost-Volume-Profit (CVP) and Break-Even Analysis Formulations

$$\text{Contribution Margin:} \quad C = S - V = F + P, \qquad C_{\text{unit}} = s - v$$

$$\text{Profit-Volume Ratio (P/V):} \quad \text{P/V Ratio} = \frac{C}{S} \times 100 = \frac{s - v}{s} \times 100 = \frac{\Delta\text{Profit}}{\Delta\text{Sales}} \times 100$$

$$\text{Break-Even Point (Units):} \quad \text{BEP}_{\text{units}} = \frac{\text{Fixed Cost}}{C_{\text{unit}}} = \frac{F}{s - v}$$

$$\text{Break-Even Point (Value):} \quad \text{BEP}_{\text{sales}} = \frac{\text{Fixed Cost}}{\text{P/V Ratio}} = \text{BEP}_{\text{units}} \times s$$

$$\text{Break-Even Ratio:} \quad \text{BE Ratio} = \frac{\text{BEP}_{\text{sales}}}{\text{Actual / Budgeted Sales}} \times 100$$

$$\text{Margin of Safety (MOS):} \quad \text{MOS} = \text{Actual Sales} - \text{BEP}_{\text{sales}} = \frac{\text{Profit}}{\text{P/V Ratio}}$$

$$\text{Margin of Safety Ratio:} \quad \text{MOS Ratio} = \frac{\text{MOS}}{\text{Actual Sales}} \times 100 = 100\% - \text{BE Ratio}$$

$$\text{Target Volume for Profit } P_{\text{target}}: \quad S_{\text{target}} = \frac{F + P_{\text{target}}}{\text{P/V Ratio}}$$

Where:

- S, s = Total sales revenue and unit selling price
- V, v = Total variable production costs and unit variable cost
- F = Total fixed operating overheads incurred over the period
- P = Net operating profit generated ($P = C - F$)
- C, C_{unit} = Aggregate contribution margin and unit contribution ($s - v$)
- P/V Ratio = Contribution margin percentage per rupee of sales
- BEP = Break-even operating volume yielding zero profit and zero loss
- MOS = Buffer of operational sales above the break-even volume threshold
- S_{target} = Required sales turnover to achieve a designated profit milestone

Chapter 4: Risk, Return, and Portfolio Mathematics

21. Statistical Measures of Central Tendency and Financial Return

$$\text{Simple Arithmetic Mean:} \quad \bar{X} = \frac{1}{N}\sum_{i=1}^{N} X_i$$

$$\text{Weighted Arithmetic Mean:} \quad \bar{X}_w = \frac{\sum_{i=1}^{N} w_i X_i}{\sum_{i=1}^{N} w_i} = \sum_{i=1}^{N} w_i X_i \qquad \left[\sum w_i = 1\right]$$

$$\text{Geometric Mean (Compounding):} \quad \text{GM} = \sqrt[n]{\prod_{i=1}^{n} X_i} = \exp\left(\frac{1}{n}\sum_{i=1}^{n} \ln X_i\right)$$

$$\text{Harmonic Mean (Rates / Multiples):} \quad \text{HM} = \frac{n}{\sum_{i=1}^{n} \frac{1}{X_i}} = \frac{n}{\frac{1}{X_1} + \frac{1}{X_2} + \cdots + \frac{1}{X_n}}$$

$$\text{Classical Mean Inequality:} \quad \text{AM} \geq \text{GM} \geq \text{HM} \qquad (\text{for all } X_i > 0)$$

$$\text{Portfolio Expected Return:} \quad E[R_p] = \sum_{i=1}^{N} w_i E[R_i]$$

Where:

- X_i, N, n = Asset observations, discrete return realizations, and sample size
- w_i = Portfolio capital allocation weights ($\sum_{i=1}^{N} w_i = 1$)
- $\bar{X}, \bar{X}_w$ = Simple and capitalization-weighted arithmetic mean returns
- GM = Multi-period geometric growth rate / Compound Annual Growth Rate (CAGR)
- HM = Harmonic mean asset multiple (appropriate for P/E and rupee-cost averaging)
- $E[R_p], E[R_i]$ = Expected return of the composite portfolio and asset i

22. Measures of Absolute Risk, Relative Dispersion, and Bivariate Co-movement

$$\text{Population Variance \& SD:} \quad \sigma^2 = \frac{\sum (X_i - \bar{X})^2}{N}, \qquad \sigma = \sqrt{\sigma^2}$$

$$\text{Sample Variance \& SD:} \quad s^2 = \frac{\sum (X_i - \bar{X})^2}{n-1}, \qquad s = \sqrt{s^2}$$

$$\text{Assumed Mean Shortcut:} \quad \sigma = \sqrt{\frac{\sum d_i^2}{n} - \left(\frac{\sum d_i}{n}\right)^2} \qquad [d_i = X_i - A]$$

$$\text{Origin / Scale Transformation:} \quad \sigma_X = h \cdot \sigma_u \qquad \left[u_i = \frac{X_i - A}{h}\right]$$

$$\text{Coefficient of Variation:} \quad \text{CV} = \frac{\sigma}{\bar{X}} \times 100 \qquad [\text{Lower CV} \implies \text{More Stable}]$$

$$\text{Covariance:} \quad \text{Cov}(X,Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n} = \frac{\sum X_i Y_i}{n} - \bar{X}\bar{Y}$$

$$\text{Karl Pearson Correlation:} \quad r_{XY} = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

$$= \frac{n \sum XY - (\sum X)(\sum Y)}{\sqrt{[n \sum X^2 - (\sum X)^2][n \sum Y^2 - (\sum Y)^2]}}$$

$$\text{FOREX Exposure Coefficient:} \quad \beta_{\text{FX}} = \frac{\text{Cov}(R_i, \Delta S/S)}{\text{Var}(\Delta S/S)}$$

Where:

- σ^2, σ = Population variance and standard deviation (total absolute risk)
- s^2, s = Unbiased sample variance and sample standard deviation (Bessel corrected)
- A, d_i, h, u_i = Assumed mean origin, deviations $(X_i - A)$, class width, and scaled scores
- CV = Coefficient of Variation (percentage relative risk per unit of expected return)
- $\text{Cov}(X,Y)$ = Covariance measuring joint directional co-movement between assets X and Y
- r_{XY} = Standardized Karl Pearson correlation coefficient $(-1 \le r_{XY} \le +1)$
- β_{FX} = Foreign exchange exposure sensitivity coefficient via market regression

23. Sharpe's Single-Index Optimal Portfolio Selection and Asset Weighting

$$\text{Excess Return-to-Beta Ratio:} \quad Z_i = \frac{R_i - R_f}{\beta_i} \qquad [Z_1 \geq Z_2 \geq \cdots \geq Z_N]$$

$$\text{Individual Variance Factors:} \quad A_i = \frac{(R_i - R_f)\beta_i}{\sigma_{ei}^2}, \qquad B_i = \frac{\beta_i^2}{\sigma_{ei}^2}$$

$$\text{Cumulative Cut-off Rate:} \quad C_i = \frac{\sigma_m^2 \sum_{j=1}^{i} A_j}{1 + \sigma_m^2 \sum_{j=1}^{i} B_j} = \frac{\sigma_m^2 \sum_{j=1}^{i} \frac{(R_j - R_f)\beta_j}{\sigma_{ej}^2}}{1 + \sigma_m^2 \sum_{j=1}^{i} \frac{\beta_j^2}{\sigma_{ej}^2}}$$

$$\text{Asset Inclusion Condition:} \quad \text{Include Security } i \iff Z_i > C^* \qquad \left[C^* = \max_k C_k\right]$$

$$\text{Relative Unscaled Weights:} \quad Y_i = \frac{\beta_i}{\sigma_{ei}^2}(Z_i - C^*) \qquad (\text{for all } i \in S^*)$$

$$\text{Normalized Capital Allocation:} \quad w_i = \frac{Y_i}{\sum_{k \in S^*} Y_k} \qquad \left[\sum_{k \in S^*} w_k = 1\right]$$

$$\text{Optimal Portfolio Variance:} \quad \sigma_p^2 = \beta_p^2 \sigma_m^2 + \sum_{i \in S^*} w_i^2 \sigma_{ei}^2 \qquad \left[\beta_p = \sum_{i \in S^*} w_i \beta_i\right]$$

Where:

- R_i, R_f = Expected rate of return on security i and riskless interest rate
- β_i, σ_{ei}^2 = Systematic market beta sensitivity and residual unsystematic variance of asset i
- σ_m^2 = Expected macroeconomic variance of the broad market index
- Z_i = Excess return-to-beta ratio (Treynor ranking metric)
- A_i, B_i = Cumulative excess return and beta scaling factors weighted by unique residual risk
- C_i, C^* = Cumulative acceptance rate and optimal cut-off threshold ($C^* = \max C_i$)
- Y_i, w_i = Unscaled relative weight factor and final normalized percentage portfolio allocation
- S^*, β_p, σ_p^2 = Set of selected assets satisfying $Z_i > C^*$, optimal portfolio beta, and total portfolio variance